

Does Artificial Intelligence Democratize Science?

Evidence from AlphaFold*

Francois Bares[†]

Anantha Divakaruni[‡]

Ludovic Phalippou[§]

September 2026

Abstract

Scientific discovery has traditionally concentrated among elite universities. This paper studies whether the diffusion of an open-access artificial intelligence tool can attenuate that concentration in a resource-intensive scientific field. We examine AlphaFold, the deep-learning system used for protein structure prediction, across its major public releases. Using publication data from top journals and universities worldwide, we show that lower-ranked institutions increase their share of protein research publications in top journals within two years of AlphaFold's public release. The gains are specific to protein research and are strongest after the release of the open AlphaFold 2 protein structure database and among universities lacking high-performance computing facilities. We find no such gain outside protein research or in lower-tier journals. Lower-ranked institutions also exhibit higher research novelty, directional pivoting, and citation impact, alongside reduced reliance on collaborations with top-ranked scientists. The pattern is consistent with open-access artificial intelligence lowering the human and financial resources required to produce state of the art protein research.

JEL classifications: O33, O31, I23, L17, O34

Keywords: artificial intelligence; science; inequality; structural biology; AlphaFold

*We are grateful to Pierre Azoulay, Kyle Myers, Jacqueline Lane, Fabrizio Dell'Acqua, Alex Chan, Hans Hvide, Savvas Savvides, Rob van der Kant, Emiel Michiels, and Lucrezia Vittoria Viti for their invaluable feedback and support. We also thank seminar participants at Harvard University, University of Bergen, and University of Illinois for useful comments. An earlier version of the paper was circulated under the title "AI-qualizing Science".

[†]Saïd Business School, University of Oxford, UK. Email: francois.bares@dphil.sbs.ox.ac.uk

[‡]Department of Economics, University of Bergen, Norway. Email: anantha.divakaruni@uib.no

[§]Saïd Business School, University of Oxford, UK. Email: ludovic.phalippou@sbs.ox.ac.uk

1 Introduction

Although several articles explore the influence of artificial intelligence (AI) on low and medium-skill activities such as image recognition ([Caplin et al., 2026](#)), programming ([Cui et al., 2024](#)), writing ([Noy and Zhang, 2023](#)), customer support ([Brynjolfsson et al., 2025](#)) and more general tasks ([Fügener et al., 2026](#); [Krakowski et al., 2026](#)), there is limited evidence on the impact of AI on productivity within highly specialized sectors. Filling this gap is crucial due to AI rapidly advancing from automating relatively straightforward tasks (e.g., chatbots, code generators) to engaging in complex, high-skill areas that demand substantial expertise and research infrastructure.

The most profound impact of AI in a specialized field has arguably been observed recently within the life sciences. An AI tool called AlphaFold, developed by Google DeepMind, has achieved very high accuracy levels in predicting three-dimensional (3D) protein structures that closely match those of experimental methods ([Abramson et al., 2024](#); [AlQuraishi, 2019](#); [Jumper et al., 2021](#)). As precise protein structures are crucial for understanding protein functions, designing therapeutics, and investigating biological pathways, AlphaFold is widely recognized as a significant advance in protein research. Consequently, its creators were honored with the 2023 Breakthrough Prize in Life Sciences and the 2024 Nobel Prize in Chemistry.

In protein research, where groundbreaking discoveries often depend on access to state-of-the-art resources and specialized knowledge, the *Matthew* effect is particularly evident: resources, recognition, and influence accrue to those who already have them ([Azoulay et al., 2014](#); [Bol et al., 2018](#); [Brogaard et al., 2024](#); [Merton, 1968](#)). A small minority of researchers and institutions receive the majority of public funding and account for the majority of top-journal publications ([Ma et al., 2015](#); [Madsen and Aagaard, 2020](#)). Top scientists often serve as editors of journals and grant reviewers, giving them an outsized influence on the direction of science ([Azoulay et al., 2019](#); [Nicholson and Ioannidis, 2012](#); [Packalen and Bhattacharya, 2018](#)).

In this context, a tool as powerful as AlphaFold might widen existing disparities, enabling elite institutions to exploit technical advantages more rapidly and thoroughly due to their superior human capital, infrastructure, and entrenched influence within the publication ecosystem. Alternatively, by sharply lowering the technical and computational demands of protein structure prediction, AlphaFold may help level the playing field.

This study investigates whether AlphaFold and other similar AI tools have modified the existing stratification of top-journal publications across institutions in protein research. Universities are classified into quartiles according to their overall publication shares in top

journals prior to the release of AlphaFold. By examining the changes in each quartile's proportion of top-journal publications, we assess whether the innovative features and open access associated with AlphaFold have broadened access to top journals or have perpetuated existing disparities.

Our analysis shows that following the introduction of AlphaFold, bottom-quartile universities (ranked 137th or lower pre-AlphaFold) experienced notable increases in top-journal publications. Similar, though smaller, gains are observed for other institutions in the second and third quartiles. In contrast, top-quartile universities saw a decline in their market share of top-journal publications after AlphaFold's release.

These gains in market share are most pronounced after the July 2021 release of AlphaFold 2, when DeepMind made both the trained AI model and a large public database of predicted protein structures openly available. While AlphaFold 1 marked a major technical breakthrough in predicting protein structures, its adoption relied largely on institutions' ability to reimplement the methodology independently. The stronger effects observed after AlphaFold 2 therefore point to the importance of broad accessibility, beyond predictive accuracy alone.

These patterns are absent in other scientific fields such as microbiology (excluding protein research), materials science, economics and finance, and social science, indicating that the redistribution we document reflects AlphaFold's specific impact on protein research rather than concurrent AI developments or broader trends in top-journal academic publishing.

Furthermore, we find that the post-AlphaFold increase in top-journal publications is concentrated among universities that lacked access to high-performance computing (HPC) prior to AlphaFold's introduction. This pattern suggests that AlphaFold partially substitutes for expensive computational resources, reducing a key barrier to structural biology that historically benefited leading institutions.

Computational infrastructure, however, is only one of many resources that protein research depends on. Many fundamental questions about a protein require knowing its three-dimensional structure, which, prior to AlphaFold , relied on laboratory experiments costing up to \$1 million each ([Stevens, 2003](#)). These experiments demand expensive infrastructure, specialized facilities, and trained structural biologists, creating sharp resource disparities across universities ([Cheng et al., 2015](#); [Garman, 2014](#); [Nogales, 2016](#)). AlphaFold eases these constraints by making predicted structures publicly available, providing equal access to universities regardless of their existing resources. Whether a predicted structure can fully substitute for an experimental one, however, remains contested. [Qian \(2026\)](#) finds that experimental work declines for proteins that AlphaFold

predicts well, suggesting direct substitution. In contrast, [Hill and Stein \(2026\)](#) and [Gans and Mohnen \(2026\)](#) report no such decline, observing instead that predictions serve as inputs to new experimental work. While these studies demonstrate how predictions enter experimental research at the protein level, we examine whether institutional focus among university ranks shifted across different classes of proteins post-AlphaFold.

Since papers do not report studied proteins in a structured format, we extract this information from publication text and curated biological records. Our novel approach combines three independent methods to map each paper directly to its underlying proteins. Using this dataset, we test whether post-AlphaFold research shifts were uniform across university ranks, or diverged between proteins with established experimental templates and those lacking structural precedent. A shift toward proteins with established templates would point to expansion into work where predictions substitute for experimental structure determination. Conversely, a shift toward proteins without structural precedent indicates movement into research that continues to require laboratory validation. We further assess whether these patterns vary between structural biology and non-structural protein research, which uses structural information to investigate protein function, disease mechanisms, and drug targets.

The protein-level results align with our main findings, demonstrating that post-AlphaFold 2 publication gains among lower-ranked universities concentrated specifically in research on proteins backed by pre-existing structural templates. Following AlphaFold 1, publication gains across university ranks were statistically indistinguishable between proteins with and without prior structural templates. Following AlphaFold 2, point estimates moved systematically toward template-backed proteins, attaining statistical significance within non-structural protein research and strengthening under more stringent template classification thresholds. Functional substitution offers a plausible interpretation of this pattern: where an experimental structure or a close template already existed, a prediction can be checked against that evidence and used without new laboratory work, bypassing the capital-intensive experimental step that lower-ranked institutions are least able to fund. Where no such precedent exists, the prediction remains a hypothesis requiring experimental confirmation ([Akdel et al., 2022](#); [Terwilliger et al., 2024](#)), and institutional resource barriers remain binding. Consistent with this account, the difference is largest at universities with no structural biologist among the authors of their pre-AlphaFold top-journal protein papers. A separate test of team composition points the same way: post-AlphaFold 2, lower-ranked papers are 10.2 percentage points less likely to include a structural biologist as a co-author. Within structural biology itself, where predictions do not replace physical structure determination as the field's core empirical contribution, no

corresponding pattern emerges, and no estimated difference attains statistical significance.

Finally, we present evidence suggesting that changes in research practices may have accompanied these shifts in publication patterns. In particular, protein researchers at lower-ranked institutions exhibit greater pivots in research direction, engage more frequently with less commonly used concepts (more novelty), and experience increases in forward citations. While these patterns are most pronounced in structural biology, we note similar trends in non-structural protein domains. Collaboration dynamics also evolve: institutions with stronger structural biology capacity become less reliant on higher-ranked partners, while those with limited in-house expertise increase such collaborations.

Taken together, these results indicate that, in protein research, an openly accessible AI tool weakened entrenched inequalities by lowering dependence on costly infrastructure, specialized expertise, and elite collaboration networks. Whether the same redistribution arises in other resource-intensive domains is an open question.

2 Sample Construction

We examine the proportion of top-journal publications in protein science for the top 500 universities over time. Publications are attributed to universities on the basis of the institutional affiliation of the authors. To identify top journals in protein research, we compile a list of 2,170 “biochemistry, genetics, and molecular biology” journals and 175 interdisciplinary journals publishing articles related to protein research. We rank the journals by their average SCImago Journal Rank (SJR), a prestige-weighted journal metric that accounts for citation counts of articles published in a given journal and the prestige of the journals where these citations appear. We use SJR rather than the Journal Impact Factor because it is field-normalized, prestige-weighted (so less sensitive to citation-volume inflation), and openly documented; the resulting top set is unchanged under standard alternatives (see Section B.2). To account for annual fluctuations in SJR, we use the average SJR over the six-year period (2013–18) preceding AlphaFold’s release. There are 17 journals with an average SJR of 10 or above (see Table A2 for details), and we classify these as the top journals for protein research.

We obtain data on all articles published in these 17 journals from OpenAlex (the successor to Microsoft Academic Graph). OpenAlex provides detailed article metadata including titles, abstracts, covered topics, citations, author affiliations and the publishing journal. Each article is assigned to one of three mutually exclusive categories, reflecting a nested hierarchy in which structural biology is a subset of protein research, which itself is a subset of microbiology:

- *Structural biology*: if at least one author is a structural biologist or the paper covers any of 238 identified structural biology topics. This is the most specialized subset, focused on determining three-dimensional protein structures.
- *Non-structural protein*: if none of the authors is a structural biologist and the paper's topics fall under one or more of 1,066 protein research topics outside structural biology. This category captures the broader domain of protein research that does not involve structural determination.
- *Non-protein*: for all other publications in these journals that do not involve protein research.

We rank universities based on their average total normalized citation scores (TNCS) between 2013 and 2018 in the category "biomedical and health sciences" (computed by the CWTS Leiden Ranking at Leiden University). The CWTS Leiden Ranking is the standard subfield-normalized scientometric ranking; computing it over the pre-AlphaFold window yields a fixed, time-invariant institutional stratification that cannot be mechanically affected by AlphaFold itself (see Section B.3 for further detail and robustness to alternative rankings). The score for university u is defined as:

$$Score_u = \frac{1}{6} \sum_{t=2013}^{2018} TNCS_{u,t} \quad (1)$$

where $TNCS_{u,t}$ denotes the total normalized citation scores in year t . The top 12 universities are Harvard, Johns Hopkins, UCSF, Toronto, UPenn, Washington, UCL, Stanford, Duke, Michigan, UCLA and Oxford (Table A4).

Each author is assigned to a single university (the highest-ranked university for authors with multiple affiliations). The research output of each university u in year t is measured as the annual count of articles published in our list of top journals, where at least one author is affiliated with that university. We denote this output $P_{u,t}^{all}$ when we count all authors listed on a publication and $P_{u,t}^{main}$ when we count only the two main authors, listed first and last¹. The corresponding shares of top-journal publications attributed to university u in year t , which we call its "market share" is computed as follows:

$$MktS_{u,t}^{all} = \frac{P_{u,t}^{all}}{\sum_{i=1}^{500} P_{i,t}^{all}} \quad (2)$$

$$MktS_{u,t}^{main} = \frac{P_{u,t}^{main}}{\sum_{i=1}^{500} P_{i,t}^{main}} \quad (3)$$

¹See section B.4 for a formal definition of $P_{u,t}^{all}$ and $P_{u,t}^{main}$

Each measure uses full counting: a multi-university article contributes one count to every represented university. The denominator applies the same rule, so market shares sum to 100% in each year.

The cumulative density function (CDF) of $MktS_{u,t}^{all}$ is quasi-monotone and concave (Figure 1). The top 10 universities generate 18% of all the top-journal publications. The top 50 generate nearly half of all publications. The top 150 universities produce almost three times as many top-journal publications as the next 350 universities (Figure 1(A)). Figure 1(B) shows that the distribution is similar if we only take into account the main authors (first and last authors) of each publication. The monotonicity in Figure 1 shows that our ranking based on top-journal share closely mirrors a more traditional ranking based on citation, such as the CWTS ranking.

We classify universities by their market share of top-journal publications from 2013 to 2018 and assign them to quartiles, each representing 25% of total output in that period. The top quartile consists of the top 12 universities by TNCS score, the following 36 universities make up the second quartile, the next 88 make up the third, and 364 make up the fourth (lower) quartile. Geographically, top-quartile institutions are concentrated in the United States. The second quartile includes universities from the US and Europe, along with three in Australia. Half of third quartile institutions are in Europe, and a quarter in Asia (notably China, Japan, and South Korea). The fourth quartile is more globally dispersed, with concentrations in Europe and East Asia and representation from South America, the Middle East, and Africa. See Figure A1 for a visual summary.

3 Publication Rates around AlphaFold

3.1 Changes in Aggregate Market Shares

We plot the aggregate market shares of each university quartile over time in Figure 2. Panels 2(A) and 2(B) report structural-biology and non-structural protein publications, respectively. In both panels, each quartile’s market share is stable before the release of AlphaFold. The figures also highlight three milestones in AlphaFold’s development: (i) AlphaFold 1 (AF1) won the 13th Critical Assessment of Protein Structure Prediction (CASP13) in December 2018; (ii) AlphaFold 2 (AF2) won CASP14 in November 2020; and (iii) the open release of AlphaFold 2’s code and database in July 2021 included predictions for about 350,000 proteins, with the database subsequently expanding to over 214 million predicted structures. Section D.3 provides further details.

We observe a notable shift within structural biology publications after the December

2018 CASP13 result (Figure 2(A)). Universities in the bottom quartile expanded their market share at the expense of those in the top two quartiles. In just two years, the market share of bottom-quartile universities rose from slightly above 24% to nearly 29%. This upward trend persisted after the July 2021 open release of AlphaFold 2's code and database, and by 2023, the market share of the bottom quartile reached 30%, while the market share of the top two quartiles both fell to around 22%. A similar picture emerges when we compute market share only taking into account main (first and last) authors (Equation 3). Bottom-quartile universities increased their share from 25% to more than 30%, while the share of the top quartile fell from 25% to 20% (Figures 2(C) and 2(D)).

The pattern observed in non-structural protein publications is also remarkable (panel 2(B)). We observe no comparable shift after AlphaFold 1, but after AlphaFold 2 the share of top publications from bottom-quartile institutions rises sharply. Their market share increased from 26% to 30% within two years. Universities in the third quartile also gained share, although by less. In contrast, institutions in the top two quartiles experienced declining market shares, with the largest decline in the top quartile.

A potential concern is that the increase in lower-quartile publications reflects the broader wave of AI tools released during this period, including large language models and machine learning applications that may have benefited lower-quartile institutions across multiple fields. Specialized AI tools have also emerged in materials science (Cheng et al., 2026; Choudhary et al., 2022; Merchant et al., 2023) and other domains, raising the possibility that the observed effects are not unique to AlphaFold. We test this alternative explanation by examining market share changes in top-journal publications across microbiology (excluding protein research), economics and finance, materials science, and social sciences. The results are presented in Figure 3. We do not observe any changes in top-journal market shares in any of these fields. The same graphics that showed a large increase in lower-quartile publication and market share post-AlphaFold are remarkably flat outside protein research. When we take all the top journals related to microbiology, exclude those that are make our list of protein research top journals, and plot the market share for the remaining journals, we find no changes in market share before and after AlphaFold . The change in market share after AlphaFold unique to protein research specifically.

Alternatively, the gains by lower-quartile universities in top journals could reflect increased overall research output following AlphaFold, rather than improved research quality. We test this hypothesis by examining protein-related publications in lower-ranked journals (SJR between 1 and 10) for all four university quartiles. The corresponding yearly market shares are shown in Figure A2. We do not observe any pre-trend and no changes in market share after AlphaFold.

Overall, these results show that the increase in top-journal market share among lower-quartile universities after AlphaFold occurs exclusively in protein research. The gains reflect improved research quality rather than mere increases in output, as lower-quartile universities show no comparable changes in their share of publications in lower-ranked journals. Moreover, the absence of similar patterns in other fields indicates that the observed shifts are attributable to AlphaFold’s specific impact on protein science rather than broader trends in scientific publishing or concurrent AI developments affecting multiple fields.

3.2 Regression Analysis

To further test whether the changes in market shares observed in Figure 2 can be attributed to the introduction of AlphaFold, we employ a two-way fixed effects regression framework. Our analysis is conducted at the university level using a yearly panel of universities from 2013 to 2023. Each university is assigned to its pre-AlphaFold quartile as in Figure 2. This quartile-based approach allows us to estimate the average post-AlphaFold effect within each group, capturing shifts in publication dynamics while accounting for both time-invariant university characteristics and common yearly shocks. Specifically, for each university u in year t , we estimate the following equation:

$$y_{ut} = \beta_0 + \sum_{q=2}^4 Quartile_{q,u} \cdot PostAF1_t \cdot \beta_q + \mathbf{X}'_{ut} \delta + \alpha_u + \gamma_t + \epsilon_{ut} \quad (4)$$

where $Quartile_{q,u}$ is a binary variable equal to 1 if university u belongs to quartile q (zero otherwise). $PostAF1$ is a binary variable set to 1 for all years from 2019 onward, capturing the period after the December 2018 CASP13 result. The vector $\mathbf{X}_{u,t}$ includes university-level control variables, with δ representing the corresponding coefficients. Fixed effects α_u control for time-invariant university characteristics, while γ_t captures common shocks affecting all universities in a given year.

We estimate the impact of AlphaFold on university research output using three variations of the outcome variable y_{ut} . Each variation captures university publication activity in a different form. First, we define y_{ut} as the number of top-journal publications (in log) by all authors at university u in year t , to assess absolute changes in publication output. Second, we assess relative changes by defining y_{ut} as the university’s annual market share based on all authors, using equation (2). Third, we define y_{ut} as the university’s annual market

share based on main authors (first and last) only, using equation (3).

We first estimate these models on the full sample of protein research publications. Panel 1(A) of Table 1 shows estimates for the three outcome variables². In each of these specifications, the interaction terms for the second, third, and fourth quartiles (relative to the top quartile) are positive and highly statistically significant. Ordinary least squares (OLS) estimates in model (1) indicate that the average fourth-quartile university its number of yearly publications post-AlphaFold by 60% relative to the first quartile. The effect is monotonic. The second- and third-quartile universities exhibit the same upward trend, but with smaller magnitudes.

Given that market shares are bounded between zero and one, we estimate models (2) and (3) using a fractional logit specification (Papke and Wooldridge, 1996). This approach fits a logistic conditional mean via quasi-maximum likelihood, ensuring fitted values remain strictly within the unit interval. Consistency requires only that the conditional expectation function is correctly specified, placing no restrictive assumptions on the underlying distribution of the dependent variable. The estimated coefficients capture differences on the logit index rather than linear percentage-point changes in market share. For small individual university shares, however, the logit index closely approximates the natural logarithm of the share.³ A point estimate β therefore yields a proportional change of $e^\beta - 1$ in a university's expected market share.

The 0.21 estimate in model (2) implies a 23 percent relative increase in expected annual market share for a bottom-quartile university following AlphaFold, relative to top-quartile peers. The second and third quartiles exhibit relative gains of 6 and 14 percent, respectively. Operating from a pre-treatment baseline of 25 percent per quartile, these relative shifts imply the bottom quartile's absolute market share expanded to approximately 28 percent on average during the 2019–2023 period. By accounting identity, the top quartile's share correspondingly declined from 25 to roughly 22.5 percent. Restricting the share definition to first and last authors in model (3) yields analogous relative gains of 11, 15, and 26 percent for the second, third, and bottom quartiles.

We also conduct separate analyses for publications in structural and non-structural protein research (see Panels 1(B) and 1(C)). The estimated AlphaFold effect shows similar strength in both domains, indicating that its impact reaches beyond structural biology alone. Note that the results reported here reflect average outcomes before and after AlphaFold's release: annual trends are discussed in Section E.2. Table A7 places the baseline estimates alongside alternative estimators for each outcome. Column (1) applies

²See Table A5 and Table A6 show robustness for different combinations of controls and fixed effects.

³At the largest quartile mean share in Table A3, 0.37 percent, $\ln[p/(1-p)]$ and $\ln p$ differ by less than 0.004.

Poisson pseudo-maximum-likelihood estimation to the publication count itself rather than to its logarithm (Santos and Tenreyro, 2006). Column (2) applies the same estimator to the count of unique author–publication pairs. Columns (3) and (4) reproduce the fractional-logit estimates of models (2) and (3) (Papke and Wooldridge, 1996), adding average marginal effects beneath each coefficient. Column (5) estimates the outcome of model (2) by OLS in place of the fractional logit.⁴ The lower the quartile, the larger the increase in academic output post AlphaFold. The exception to this rule is when we regress the number of unique Author-Publications pairs (column 2). In this case the monotonicity breaks down, suggesting that the improvement in lower quartile output is not driven by an increase in the number of co-authors per paper, which could mechanically improve lower-quartile publication counts.

4 Mechanisms

4.1 The Role of Accessibility

Having established that AlphaFold increased the share of top-journal protein publications from lower-quartile universities, we now examine mechanisms that may explain this shift. AI-based protein-structure prediction has grown substantially since 2014, with the number of such tools submitted to CASP competitions rising steadily over the past decade (Figure A3). AlphaFold’s breakthrough performance at CASP13 in 2018 accelerated this trend. Despite the subsequent proliferation of competing AI approaches, AlphaFold has maintained a dominant position, with more than 60% of prediction models submitted to recent CASP competitions either directly using AlphaFold or building on its methodology. This pattern raises a key question: does the observed democratization in top-journal protein publications stem primarily from AlphaFold’s technical superiority, or does its distinctive open accessibility play an additional role?

While AlphaFold 1 demonstrated breakthrough prediction accuracy in CASP13, DeepMind published only its underlying methodology without releasing the model weights. This potentially limited adoption to institutions with structural biology expertise and computational resources to implement the tool independently. Its successor AlphaFold 2 won CASP14 in November 2020; the open release of its code and comprehensive public database followed in July 2021, initially containing predicted structures for nearly 1 million

⁴The results are also robust to COVID-era publishing dynamics. Excluding publications linked to COVID-19 topics yields virtually identical quartile estimates, while dropping the 2020–2021 period entirely leads to larger point estimates, indicating that our baseline results are conservative (see Figure A10).

proteins and subsequently expanding to over 214 million. This staggered sequence allows us to evaluate whether subsequent open access improved AlphaFold’s democratizing effect beyond what its breakthrough performance at CASP13 could alone achieve.

We therefore compare publication changes before and after the open release of AlphaFold, and assess whether adoption barriers under the closed-source model limited the democratizing effect of AlphaFold. We re-estimate our baseline specification, replacing the *PostAF1* indicator with two separate indicators: *PostAF1* (for publications after the December 2018 CASP13 result and before the July 2021 open release) and *PostAF2* (for publications from the July 2021 open release of AlphaFold 2’s code and database onward), with the pre-AlphaFold period as the reference category. The results reported in Table 2 show that the gains for lower-quartile universities are substantially larger following AlphaFold 2’s release. For all top-journal protein publications, bottom-quartile universities saw an average marginal rise in market share of 0.05 percentage points per university-year (64% relative increase) after AlphaFold 1, intensifying to 0.09 percentage points (110% relative increase) post-AlphaFold 2. Second- and third-quartile universities display analogous but smaller improvements in their respective publication shares. Model (3), restricting analysis to main authors, reinforces this pattern with increases of 0.05 percentage points (56% relative increase) post-AlphaFold 1 and 0.09 percentage points (95% relative increase) after AlphaFold 2. Similar patterns emerge in both structural biology and non-structural protein research, with the bottom quartile consistently showing the biggest gains.

The larger effects following AlphaFold 2 indicate that open-source code and data availability reduced adoption barriers beyond what AlphaFold 1’s breakthrough accuracy could achieve. Lower-quartile institutions benefited most from this increased accessibility, as they could use predicted structures directly without requiring significant structural biology expertise.⁵

4.2 The Role of Resource Constraints

We next examine whether AlphaFold reduces protein researchers’ traditional dependence on expensive computational infrastructure. Protein research has historically required substantial computing resources for tasks such as molecular dynamics simulations, fitting structural models to experimental data, drug discovery, and functional genomics (Cheng et al., 2015; Garman, 2014; McKeon and Boudou, 2024; Nogales, 2016). Table A3 shows that 51% of top-quartile universities had access to high-performance computing (HPC)

⁵These timing dynamics are not an artifact of pandemic-era publishing shocks. The concentration of gains in the post-AlphaFold 2 period remains robust when excluding COVID-19-related publications, and persists when omitting the 2020–2021 pandemic years from the sample entirely (Figure A11).

compared with only 9% of bottom-quartile institutions. If AlphaFold’s novel technology, publicly available source code, and predicted structures substitute for computationally intensive methods, then its democratizing effects should be most pronounced among institutions that traditionally lacked HPC infrastructure.

For this purpose, we run our main specification separately for universities with and without HPC access. In both sub-samples, we use the categorical time variable that distinguishes between the post AlphaFold 1 and AlphaFold 2 phases, with the pre-AlphaFold phase serving as the reference group. Table 3 reports the results, showing that the effects are consistently stronger for universities lacking HPC access. Specifically, among bottom-quartile universities with HPC, market shares rose on average by 0.02 percentage points (18% relative increase) in AlphaFold 1 and by 0.09 percentage points (60% relative increase) after AlphaFold 2. In contrast, bottom-quartile institutions without HPC experienced increases of 0.07 percentage points (236% relative increase) in AlphaFold 1 and 0.09 percentage points (181% relative increase) in AlphaFold 2. Second- and third-quartile universities exhibit analogous patterns, though with smaller magnitudes and generally weaker statistical significance compared to the bottom quartile, but the disparity between HPC-equipped and non-equipped institutions persists across quartiles.

These patterns indicate that AlphaFold substituted for expensive computational resources, with effects emerging after AlphaFold 1 and strengthening after AlphaFold 2. Institutions with HPC were already competitive in protein research using established computational methods, so AlphaFold 1 offered them smaller relative gains. By contrast, institutions lacking HPC benefited more as accurate structure predictions began circulating through collaborations and early adopters following AlphaFold 1’s demonstrated success. The larger effects following AlphaFold 2, however, underscore that direct access to open-source code and a comprehensive public database was critical for fully bypassing the resource dependence that previously conferred a competitive advantage to computationally well-equipped institutions.

4.3 The Role of Prior Protein Information

The preceding analyses infer mechanisms from differences in universities’ scientific expertise and computing resources. We now turn from research inputs to research content by identifying the proteins studied in each paper. Our results so far suggest that AlphaFold improved performance at lower-quartile universities by providing a substitute for costly experiments. If this is true, their gains should be concentrated in research where AlphaFold predictions can substitute for experimental work. This is not always the case. [Gans](#)

and Mohnen (2026) and Hill and Stein (2026) show that AlphaFold can complement experimental work by providing predicted structures that serve as starting hypotheses for experimental validation. AlphaFold is particularly useful when researchers explore unusual proteins with little available homology. Although AlphaFold can complement experimental work on unusual structures, it can also substitute for some experiments on more common structures, where predictions are more reliable and may not require experimental validation. We test whether lower-quartile universities gain more in the latter setting.

Implementing this test requires a publication–protein linkage that standard bibliometric data do not provide. Establishing this link is challenging because protein names are non-unique, species-dependent, and reported inconsistently. The most reliable linkage available, used in Hill and Stein (2026) among others, is curated in Swiss-Prot, where each link is individually validated by biocurators. These links are precise but cover only a small part of the protein space. We retain this channel and complement it with linkages based on PubTator3’s normalized gene identifiers and biomedical named-entity recognition over abstracts. Crucially, we link publications only to experimental structures and homology templates available before AlphaFold 2’s public database release on 22 July 2021. This ensures that our research classification is not mechanically affected by AlphaFold itself.

To identify the proteins studied in each paper, we match every publication against the reviewed section of UniProtKB, the reference catalog of over 570,000 curated protein entries. The difficulty is that papers can state the proteins they study in numerous ways in the text. The proteins must therefore be inferred from how the text refers to them. The inference is further complicated by how proteins are named. A single protein goes by several names: its gene symbol, its full recommended name, an abbreviation, or any of its recorded synonyms. A single name, in turn, can refer to several proteins, because most proteins have counterparts in other species and each counterpart is a separate entry in the catalog. We therefore build the link between papers and proteins ourselves, using three distinct and novel methods.

The first method exploits the literature citations that UniProt curators attach to each protein’s entry. Each entry cites the publications that document the protein’s properties. Read in reverse, this record links a paper to every protein whose entry cites it. Links of this kind are precise: each exists because a human curator judged that the paper reports experimental evidence on the given protein. Curation, however, is a slow process. Because entries are updated years after the underlying papers appear, the publications least likely to be linked are the most recent ones: Those are precisely the post-AlphaFold papers that identify our effect.

Our second method replaces the curator’s reading with PubTator3, an automated annotation system run by the National Library of Medicine. PubTator3 scans the PubMed literature weekly and tags each gene a paper mentions with its standard gene identifier. Every entry in the UniProt catalog records the identifier of its own gene, which turns the match into an exact lookup and puts a paper published last month on the same footing as one published a decade ago. The third method requires only the paper’s abstract. Models trained to recognize protein names mark each mention, each mention is searched in the catalog, and only the best match is retained. Because a link needs nothing beyond a name in the abstract, this method reaches papers that neither curators nor PubTator3 have processed.

A paper is linked to a protein when at least one of the three methods establishes a match. In our data, each successive channel links papers that the preceding channels leave unlinked, which is why we use all three rather than relying on curation alone.⁶ The union links at least one protein to 2,859 of the 10,581 protein research papers in the sample, yielding a coverage rate of 27.0% (Table A8). Coverage is broadly similar across university quartiles, reducing concern that the results are driven by differences in linkage coverage by university rank.

We then ask what was already known about each of these proteins. A protein has a known structure if, before the AlphaFold protein structure database was released on 22 July 2021, its own three-dimensional structure had been solved experimentally, or a usable homology template existed. The second case rests on a basic regularity of biology: proteins with similar amino-acid sequences fold into similar shapes. The solved structure of a close relative can therefore stand in for the target protein, with the share of sequence the two have in common measuring how far the substitute can be trusted. We treat a template as usable at 30% sequence identity or above, the conventional threshold below which a related structure no longer supports reliable modeling. Of the 48,314 linked proteins, 18.5% had their own experimental structure, 16.1% had only a homology template, and 65.4% had neither (Table A9). A paper counts as having known protein structure when at least half of its linked proteins had either their own experimental structure or a homology template before the database release.

Results and interpretation We test these differences more formally in the regression framework used so far, adding triple interaction terms that measure the difference in post-AlphaFold gains between papers with and without prior structures (Table 6). Post-AlphaFold 1, the third and fourth quartiles gain more on papers *without* prior structures,

⁶Appendix C sets out the three channels, the matching rules, and the limitations of the procedure.

but the differences are not statistically significant. Post-AlphaFold 2, the pattern reverses: lower-quartile universities gain more on papers *with* prior structures. This reversal is concentrated in non-structural protein research, where all three quartiles show larger and statistically significant estimates in every outcome except the third-quartile market share, which is positive but insignificant. In structural biology post-AlphaFold 2, no estimate is statistically significant, but the market share gains of the third and fourth quartiles remain larger for papers *without* prior structures.

Figure 4 and A14 illustrate this reversal between the two periods. The reversal is robust to alternative thresholds for classifying a paper as drawing on prior structures, as shown in Figure A15. These thresholds further highlight the dynamics: the stricter the threshold for classifying a paper as having a prior structure, the larger the post-AlphaFold 2 gains for lower-quartile universities.

The post-AlphaFold 2 gains on papers with prior structures in non-structural protein research point to the role of the AlphaFold database as a low-cost structural input. Researchers in these fields may use a structure as an input into functional or applied studies rather than as the principal contribution. When an existing experimental structure or homology template supports it, an AlphaFold prediction can be credible without experimental validation. Lower-quartile universities may benefit most because they have less experimental capacity. For proteins with no prior structural information, experimental confirmation retains more of its value, and top universities retain more of their advantage. This interpretation explains why lower-quartile gains are smaller for papers without prior structures and in structural biology, where the value of experimentation is greater.

Two additional results further support this interpretation. Table A11 splits universities by whether a structural biologist appeared on any of their pre-AlphaFold top-journal papers. After AlphaFold 2, the lower-quartile gains at universities without such an author are larger for papers with a prior structure. The difference in lower-quartile gains between papers with and without prior structures is also greater for universities without a structural biologist pre-AlphaFold. Team composition changed as well. Table A12 shows that, after AlphaFold 2, papers from bottom-quartile universities are 10.2 percentage points less likely to include a structural biologist as a co-author.

Mechanism summary Overall, the evidence presented in this section points in the same direction. AlphaFold 1 generated initial gains, including at universities without HPC, but those gains were limited by closed implementation. AlphaFold 2 widened access through a ready-to-use database. AlphaFold thus made lower-quartile universities more competitive in protein research by compensating for their relative inability to produce

structures experimentally, whether because of limited expertise or computing resources. This dynamic is made clearer when we investigate the studies that drive lower-quartile gains. The gains are concentrated in applied or functional studies of proteins with pre-existing structural information, where AlphaFold can be used credibly without experimental validation.

5 Research Reorientation and Team Formation

The preceding section shows that AlphaFold made lower-quartile universities more competitive by reducing their dependence on computing infrastructure, structural-biology expertise, and new experimental structure determination. We now ask whether this relaxation of constraints was accompanied by changes in the research portfolios and team formation of lower-quartile universities. We examine two dimensions of this adjustment. First, we assess research orientation using three metrics: *pivot behavior*, which captures shifts toward areas farther from an author’s prior work; *novelty*, which captures engagement with less common concepts; and *disruptiveness*, which captures changes in subsequent citation patterns. Second, we examine whether reliance on collaborations with top-quartile universities changed after AlphaFold.

5.1 Qualitative Measures of Research

We examine three qualitative metrics that reflect distinct dimensions of research activity at the author paper level. These indicators offer insight into how AlphaFold may be reshaping scientific inquiry, enabling researchers to pursue new directions, introduce more original ideas, and contribute more meaningfully to ongoing debates in the field.⁷ Analyzing these metrics across university quartiles helps assess whether the rising presence

⁷To motivate the analysis, we present some examples of how AlphaFold impacted various fields in protein research: AlphaFold has enabled the discovery of new 3D shapes (folds) in parts of viruses that bind to sugar molecules (glycan-binding domains), shifting research from studying rigid, crystal-like arrangements (paracrystalline) to more flexible, varied forms (polymorphic assembly in rotavirus studies (Hu et al., 2022)). In drug discovery, AlphaFold sped up the identification of mind-affecting drugs (psychotropic agonists) that target a brain protein involved in mood (TAAR1), achieving a higher success rate (hit rate) than predictions based on similar known structures (traditional homology models) (Díaz-Holguín et al., 2024). Additionally, AlphaFold supported predictions across all proteins in an organism (proteome-wide) for changes in DNA that alter protein building blocks (missense variant effects), improving understanding of their disease-causing potential (pathogenicity) and aiding gene-based medicine (clinical genomics) (Cheng et al., 2023). AlphaFold has also been instrumental to uncovering the structure of an immune signaling complex (IL-27), revealing new ways cells communicate during inflammation (Jin et al., 2022). Access to AlphaFold also accelerated the discovery of a small molecule inhibitor for CDK20, a protein linked to cancer, by generating accurate structures for rapid screening (Ren et al., 2023).

of lower-quartile institutions in top journals is accompanied by changes in the character and potential impact of their research output.

The three metrics are defined as follows:

- **Pivot score:** This measure quantifies the extent to which a given paper by a researcher differs from their prior work based on the journals cited. It is calculated as 1 minus the cosine similarity between the distribution of journals referenced in the focal paper and those in the author’s previous publications over the past three years ([Hill et al., 2025](#)) (see detailed description of this measure in section [B.7](#)). The pivot score ranges from 0 (no departure from prior work) to 1 (implying complete departure). A higher pivot score indicates a greater shift in research direction, which may be facilitated by AlphaFold’s provision of novel structural methodologies or predictions.
- **Novelty:** This metric evaluates originality in a paper by quantifying the rarity of concepts associated with its cited references, using OpenAlex topic classifications. As such, it follows an entity-based approach to estimating novelty ([Azoulay et al., 2011](#); [Hill and Stein, 2025](#); [Liu et al., 2022](#); [Luo et al., 2022](#)), leveraging OpenAlex’s concept classification. For each reference, concepts with relevance scores above 0.5 are included, where the relevance score reflects centrality to the reference content. Novelty is then calculated as the average inverse of the works count for these concepts, with works count indicating prevalence in the broader literature (refer to Section [B.7](#) for a detailed description of this metric.). This approach denotes novelty because reliance on less common concepts signals that the paper builds on underrepresented ideas or unusual knowledge combinations, setting it apart from incremental advances in mainstream subject areas. Novelty matters in the context of AlphaFold because predicted 3D structures reduce the need for experiments requiring substantial resources, allowing researchers to study proteins that were previously inaccessible.
- **Disruptiveness:** This variable measures the degree to which a paper affects subsequent research, based on citation patterns in which future works cite the focal paper but not its predecessors ([Funk and Owen-Smith, 2017](#); [Park et al., 2023](#)); see Section [B.7](#) for details. A positive disruptiveness score indicates that the paper draws citations primarily to itself, implying that it reduces the prominence of earlier work. Accurate protein structures supplied by AlphaFold may facilitate this form of research, particularly for scientists at lower-quartile institutions.

Importantly, these measures describe intellectual orientation and citation impact, not the structural novelty of the focal protein. A paper can cite unfamiliar journals or

rare concepts while studying a protein with prior structural information. Research reorientation can therefore coexist with the preceding result that lower-quartile gains concentrate where AlphaFold predictions can be used without new experimental validation.

Changes in pivot behavior

Analysis of pivot scores reveals significant shifts in research direction across university quartiles following the introduction of AlphaFold. Prior to AlphaFold, researchers at top-quartile universities consistently exhibited higher pivot scores, indicating a greater propensity to make substantial shifts in their research direction. This pattern likely stems from their access to superior resources, funding, and collaborative networks, which facilitate exploration across diverse subfields of protein research. However, post-AlphaFold, this disparity narrows, with lower-quartile universities experiencing larger increases in pivot scores.

In structural biology, pivot scores increased across all university quartiles following AlphaFold, with larger gains observed in lower quartiles (Figure 5(A)). We quantify this effect using the regression specification in Equation 4, which estimates the differential impact of AlphaFold on pivot scores across quartiles while controlling for university fixed effects and time trends. Results from Model (1) in Table 4 show that, relative to the top quartile, bottom-quartile universities experienced a 2.6% increase in pivot scores, third-quartile universities a 1.7% increase, and second-quartile universities a 1% increase, all highly statistically significant.

We observe similar trends in non-structural protein research. Pivot scores increased across all university quartiles following AlphaFold, but with a more pronounced pattern favoring lower-quartile institutions relative to those observed in structural biology (Figure 5(B)). We once again estimate this effect using Equation 4. Results from Model (4) in Table 4 show that, relative to the top quartile, second-quartile universities experienced an increase of 3.4% in pivot scores and fourth-quartile universities an increase of 3.3%, both highly statistically significant.

Taken together, these patterns suggest that AlphaFold, by offering a widely accessible tool for protein structure prediction, has helped lower barriers to entry in both structural and non-structural protein research. Once structural predictions became a low-cost input, lower-quartile researchers could pursue questions farther from their prior citation portfolios. These findings are consistent with prior work showing that technological shocks can broaden participation in science by reducing costs and enabling the exploration of ideas that are more novel or distant from researchers' existing trajectories (Agrawal and

Goldfarb, 2008; Furman and Teodoridis, 2020).

Changes in novelty

Analysis of novelty scores reveals distinct patterns across university quartiles following the introduction of AlphaFold. These patterns likely stem from AlphaFold's capacity to provide access to uncommon protein structures and predictions, allowing researchers to pursue hypotheses that were previously difficult to explore due to various constraints.

In structural biology, novelty scores increased across all university quartiles post-AlphaFold, with significant gains in lower quartiles (Figure 5(C)). Regression estimates using Equation 4 shown in Model (2) of Table 4 show that, relative to the top quartile, second-quartile universities experienced a 2% increase, while third- and fourth-quartile universities both saw a 1.5% rise in novelty. These shifts underscore AlphaFold's pivotal role in enabling researchers at resource-constrained institutions to address novel structural challenges, including the prediction of previously unresolved protein folds.

In the case of non-structural protein research, changes in novelty patterns are more pronounced, with novelty scores rising sharply for lower-quartile institutions (Figure 5(D)). The corresponding regression results presented in Model (5) of Table 4 indicate no significant effect for the second quartile but a substantial 1% (2.7%) increase for the third (bottom) quartile, relative to the top quartile. These results suggest that by leveraging AlphaFold-derived structural insights, researchers at lower-quartile institutions are able to integrate accurate and new protein structures into non-structural domains such as drug discovery and functional genomics despite lacking experimental resources.⁸

⁸For example, researchers at McMaster University (ranked 69 in our sample) and others used ai predictions to study how proteins interact based on their genetic sequences, showing that the sequence can be more important than the 3D structure for predicting these interactions in genomics (Hosseini et al., 2024). This approach allowed them to incorporate uncommon ideas about protein behavior that were hard to test previously, boosting the originality of their findings, and helped better understand how enzymes repair DNA in bacteria. Similarly, researchers at the University of Alberta (ranked 84 in our sample) incorporated AlphaFold predictions to identify molecular markers for better medical diagnosis and treatment, enabling exploration of rare protein functions in bacterial genomes by adding notes to predicted structures in their system, which enhanced the novelty of their research by uncovering new biological insights (Jackson et al., 2025). Additionally, researchers at Universidade de Lisboa (ranked 315 in our sample) used AlphaFold to model a protein structure (CDS2), revealing potential spots for drugs to target in cancer-related pathways, and to design small binders for a receptor involved in cell processes for targeted breakdown (Huang et al., 2025). Their approach introduced original concepts in combining structure with function for new therapeutic ideas. Lastly, researchers at University of Coimbra (ranked 372 in our sample) participated in a study that applied AlphaFold to predict a protein structure (TEX264), identifying parts involved in clearing damaged DNA, which brought fresh perspectives to DNA repair mechanisms and increased the work's originality by linking structure to rare cellular cleanup processes (Lascaux et al., 2024).

Changes in disruptiveness

Disruptiveness, a citation-based metric, evaluates whether a paper consolidates (negative values) or disrupts (positive values) the existing citation network by attracting citations independently of its references. Our findings show that protein papers published in top journals post-AlphaFold have modestly redirected future studies away from prior literature. Due to citation lags, the full impact of AlphaFold on citation patterns may take years to emerge.

In structural biology, disruptiveness scores exhibit small but statistically significant increases across all university quartiles, with no clear trend favoring lower quartiles (Figure 5(E)). Regression estimates from Model (3) in Table 4 indicate average annual increases of 1.3% for lower-quartile (fourth-quartile) universities, 1.2% for second-quartile, and 0.5% for third-quartile universities, relative to the top quartile. These modest gains suggest that AlphaFold supports incremental disruptions by providing accessible structural predictions that address challenges in traditional experimental methods, such as predicting complex protein folds previously difficult to resolve.

In non-structural protein research, disruptiveness effects are much weaker, with scores showing some overall increase (Figure 5(F)). Regression results in Model (6) of Table 4 reveal a small positive effect of 0.6% for lower-quartile (fourth-quartile) universities and a small negative effect of -0.6% for third-quartile universities, relative to the top quartile, both marginally significant at the 10% level. These subtle shifts indicate that AlphaFold’s structural insights have not yet significantly altered citation dynamics in non-structural protein domains. This may be due to the initial integration of structural predictions into studies aligning with existing research trajectories rather than immediately disrupting them. The limited impact could also reflect the time-lagged nature of citation-based metrics, where significant changes in research patterns often take years to fully emerge in the literature.

5.2 Collaboration Patterns

Collaboration represents another pathway through which AlphaFold, as a technological shock, may have reshaped the competitive dynamics of protein research publishing in top journals. Prior work shows that such shocks can influence collaboration patterns (Agrawal and Goldfarb, 2008; Murray et al., 2016; Teodoridis, 2018), which in turn have important implications for publication outcomes. High-impact research is increasingly produced through multi-university collaborations, which tend to involve institutions of comparable prestige (Jones et al., 2008; Milojević, 2014; Wu et al., 2019; Wuchty et al., 2007).

Overcoming this tendency toward institutional homophily can be particularly beneficial for researchers at lower-quartile universities, who often see improved publication outcomes when collaborating with peers from top-quartile institutions (Agrawal and Goldfarb, 2008; Khor and Yu, 2016; Kwiek, 2015; Li et al., 2019). Such collaborations offer access to advanced infrastructure, technical expertise, and greater visibility. Empirical evidence also indicates that the citation impact of multi-university teams tends to converge toward that of the more prestigious partner, effectively enhancing the scholarly reach of lower-quartile institutions (Jones et al., 2008).

Accordingly, we examine whether AlphaFold increases the likelihood that researchers from lower-quartile institutions collaborate with peers from top-quartile universities—a dynamic that could disproportionately benefit the former and help account for their observed gains in top-journal market share. We estimate equation 4, using as the outcome variable the fraction of a university’s top journal publications in protein science that include at least one co-author from a top-quartile institution in a given year. In addition to the standard two-way interaction term $Quartile_{q,u} \times PostAF1_t$, we include a triple interaction with the share of structural biologists in the university’s biology faculty for that year to distinguish changes in collaboration patterns among structural biologists and non-structural biologists.

Results are shown in Table 5. In the baseline specifications (models 1, 3, and 5), collaborations with top-quartile authors generally decline among lower-quartile universities following AlphaFold. The decline is most pronounced for fourth-quartile institutions and is also evident for second-quartile universities in all three samples, while the third-quartile estimate in the full protein sample is close to zero. Taken together, these estimates suggest a broad move toward greater self-reliance among lower-quartile universities, especially those with larger in-house structural biology capacity.

Models 2, 4, and 6 include the triple interaction with the share of structural biologists. In these specifications, the coefficient on $Quartile_{q,u} \times PostAF1_t$ is positive for the second, third, and fourth quartiles. Because the model includes the triple interaction, this coefficient measures the post-AlphaFold change for a university with no structural biologists. The coefficients on $Quartile_{q,u} \times PostAF1_t \times Structural\ Biologists(\%)$ are negative and statistically significant. This pattern points to substantial heterogeneity in collaboration responses. Collaborations with top-quartile institutions rise primarily among lower-quartile universities with limited in-house structural biology expertise, and this increase weakens as structural biology capacity rises. One possible interpretation is that AlphaFold improves the productivity of structural biologists, who are disproportionately concentrated at top-quartile institutions, allowing them to participate in a larger number of projects.

Lower-quartile researchers with fewer in-house structural biologists may therefore become more likely to collaborate with top-quartile structural biologists in the post-AlphaFold period.

6 Conclusions

The persistent concentration of scientific output within elite institutions has long been attributed to unequal access to resources, infrastructure, skill, and reputational capital. Our findings provide empirical support for the idea that AI can disrupt these structural advantages. The introduction of AlphaFold represents an exogenous technological shock that reduced the skill and resource intensity historically required for cutting-edge protein research.

The introduction of AlphaFold represents an exogenous technological shock that reduced the skill and resource intensity historically required for cutting-edge protein research, particularly in structural biology. This shock enabled researchers at lower-ranked institutions to increase their presence in top journals, suggesting that AI can function as a scientific equalizer. Notably, the effects are both quantitative and qualitative: lower-ranked universities not only increased their publication volumes and market shares in top journals but also exhibited significant improvements in research quality relative to elite institutions, thereby narrowing pre-existing institutional gaps in innovation and impact within protein science.

These results align with prior work on the redistributive effects of technological shocks in science. Prior studies have shown that when experimental costs fall or computational tools become more accessible, the institutional distribution of research can shift ([Furman and Stern, 2011](#); [Furman and Teodoridis, 2020](#); [Murray et al., 2016](#)). Whereas such shocks manifest gradually and are limited in scope, AlphaFold's effects have been rapid and large, possibly pointing to a different mechanism: one linked to the general-purpose and open-access nature of modern AI tools. Beyond AlphaFold, [Hao et al. \(2026\)](#) document how quickly and broadly other AI tools -from simple random forest to large language models- have been adopted across the natural sciences. They find that researchers using AI increased their publication output by a factor of three. We complement their study by showing that this increase in productivity benefited lower-ranked scientists.

Our findings also contribute to long-standing debates about the *Matthew* effect in science. Elite universities have historically leveraged cumulative advantages in funding, talent, editorial control, and collaboration networks to reinforce their dominance ([Azoulay et al., 2014](#); [Merton, 1968](#)). The changes we observe after AlphaFold suggest that such

feedback loops can be interrupted when new tools lower the resources needed to produce top-quality research. For instance, post-AlphaFold, lower-ranked universities with stronger structural-biology capacity exhibited reduced reliance on collaborations with top-ranked partners to publish protein research in top journals, as evidenced by an overall decrease in their propensity for such partnerships. This reduced dependence is consistent with AlphaFold broadening access to structural insights and allowing lower-ranked institutions to pursue projects more autonomously. Importantly, AlphaFold's open accessibility played a central role: unlike many scientific breakthroughs constrained by proprietary datasets or infrastructure, AlphaFold 2 provided a freely available computational model and a comprehensive, open-access protein structure database, effectively diluting traditional barriers to entry in structural biology research.

Whether future AI innovations will follow AlphaFold's trajectory remains to be seen. As AI tools become more complex and expensive to train, the risk increases that control over their development and deployment will be centralized within a small number of firms. AlphaFold itself was developed by Google DeepMind and made freely available, but future iterations of protein prediction models may be gated by proprietary data, computational scale, and commercial licensing. This raises the concern that AlphaFold's democratizing effects may be temporary, potentially giving way to new hierarchies rooted in unequal access to advanced AI infrastructure and expertise. Furthermore, while AlphaFold lowers the cost of protein structure determination, downstream activities such as experimental validation and drug discovery remain capital-intensive and skewed toward elite institutions, which disproportionately hold advanced facilities, proprietary datasets, and interdisciplinary expertise. As such, the increased publication volume and improved research quality we document for lower-ranked universities may not immediately lead to comparable advances in practical applications, such as faster development of new therapies or more equitable access to research funding and infrastructure.

Future research should examine whether the rebalancing of top publication shares observed post-AlphaFold represents a lasting shift, translating into more durable changes in scientific influence, collaboration networks, and funding allocation. It is also important to assess whether these democratizing effects extend beyond protein science to other domains such as genomics, materials science, and climate modelling, where AI tools may similarly empower under-represented institutions or, conversely, introduce new forms of exclusion. Beyond top-journal publications, other aspects of inequality should be studied. For example, [Hao et al. \(2026\)](#) finds that the adoption of AI leads to greater concentration of citations among the most influential papers.

Recent developments surrounding AlphaFold 3 highlight the fragility of such gains.

It was initially released with restricted access in May 2024, which prompted widespread calls from the scientific community for greater access. It was ultimately made open access in November 2024. This episode underscores the importance of longitudinal studies on the evolving role of AI in research ecosystems, particularly regarding how access and governance decisions shape equitable innovation.

As AI continues to transform scientific discovery, global policy responses are accelerating, and many initiatives, such as the U.S. AI Action Plan, have emphasized equitable access to AI technologies. In this context, a deeper understanding of AI's potential to either reinforce or reduce institutional hierarchies will be essential for designing inclusive science and innovation policies that broaden participation and enhance societal benefit.

Figure 1: Cumulative market share of top 500 universities before release of AlphaFold

Figure illustrates the cumulative distribution of top-journal publication for the top 500 universities. Panel (A) shows the distribution when publications are allocated based on the affiliation of any of the paper authors; Panel (B) shows the distribution when publications are allocated based on the affiliation of the first or last author. The university rankings are determined by the average *total normalized citation score* (TNCS) of each university in the "biomedical and health sciences" category between 2013 and 2018.

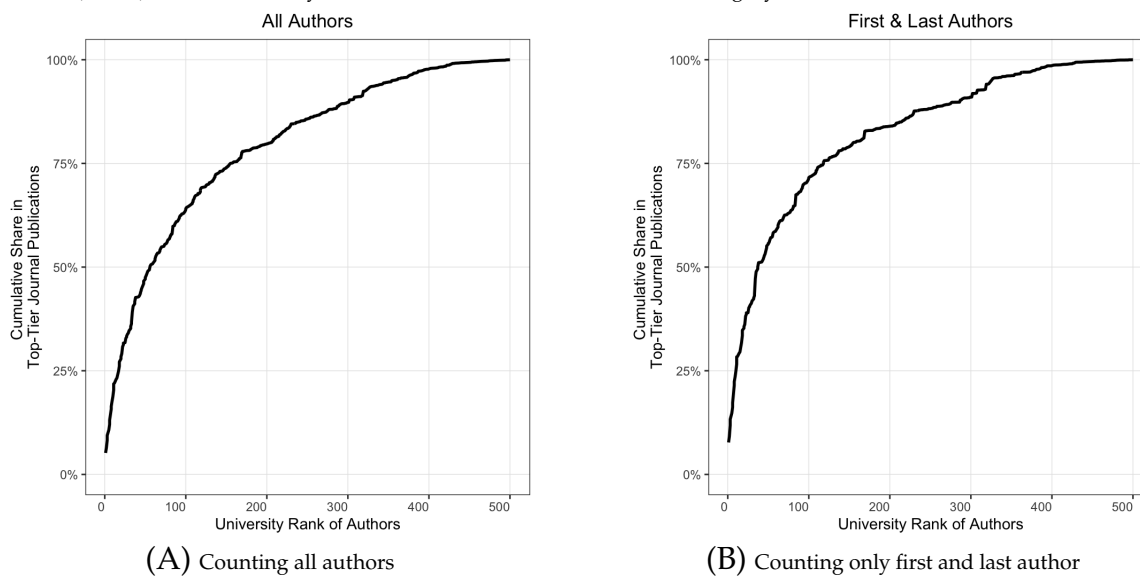


Figure 2: Protein publications in top journals by university quartiles

The figure reports annual market shares of protein research publications in top journals, defined as journals with an average SJR of at least 10 during 2013–2018. The sample is limited to authors affiliated with the top 500 universities. Universities are ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares. Panels A and B show market share computed with all authors and panels C and D show market share computed with first and last authors only. The top journal shares are computed for structural biology papers (panels A and C) and for protein research excluding structural biology (panels B and D).

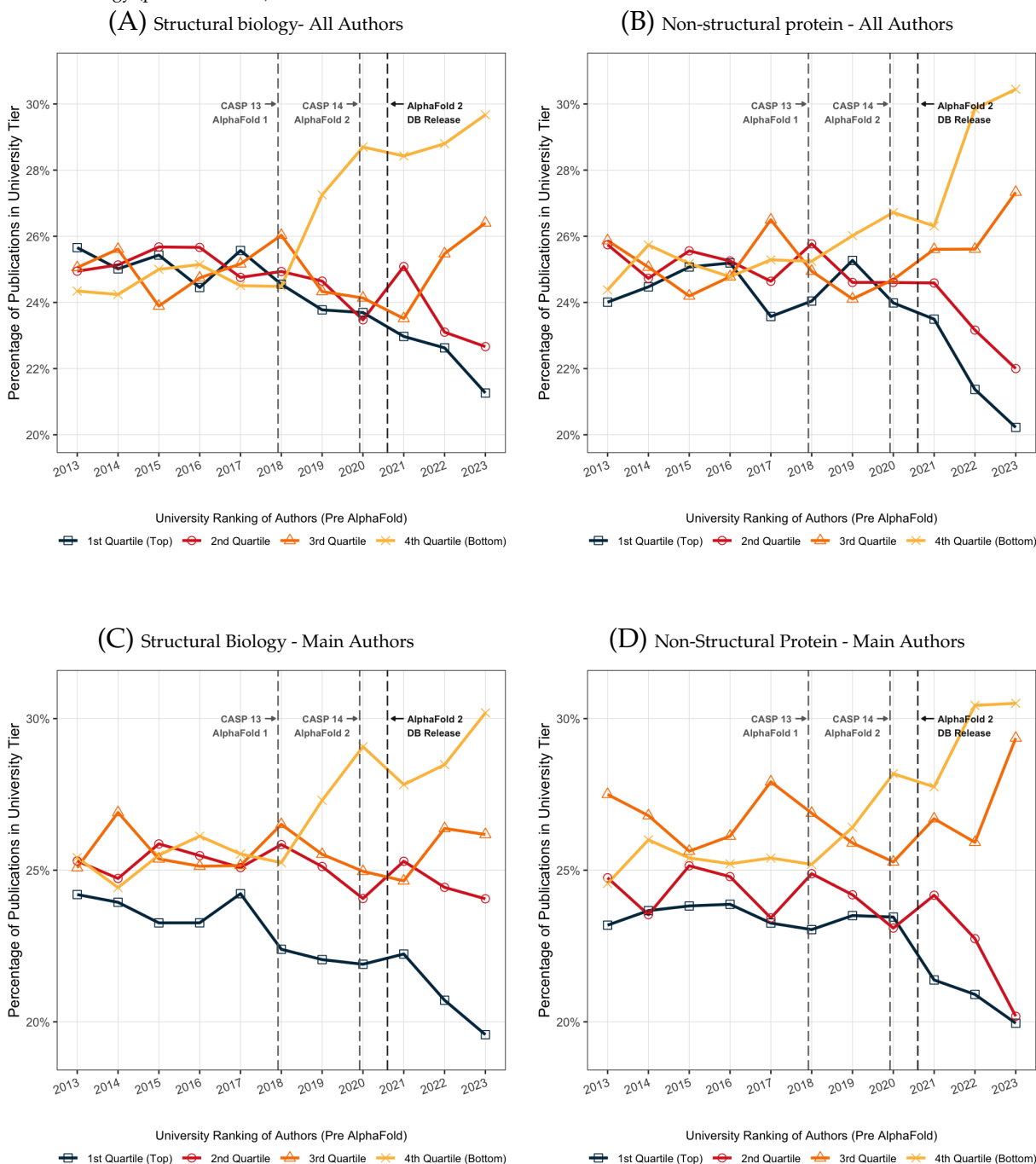
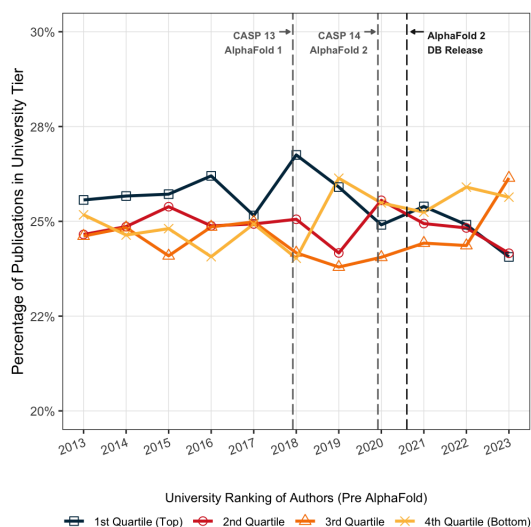


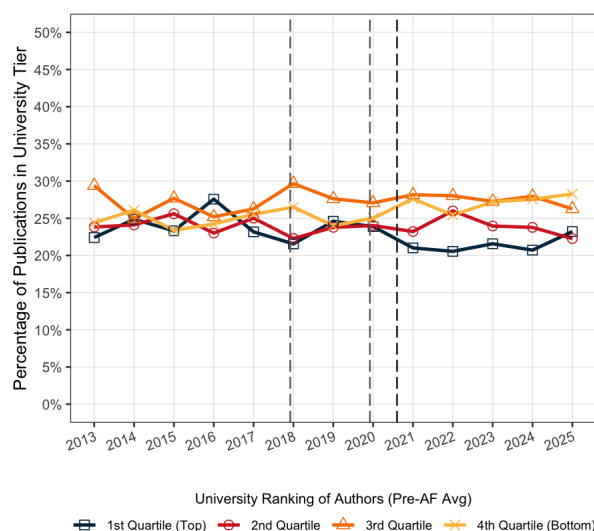
Figure 3: Publications in other fields in top journals by university quartiles

Figure illustrates the percentage of publications in top journals ($SJR \geq 10$) held by each university quartile over time. Universities are ranked based on their field-specific TNCS. As in protein research, quartiles are fixed based on each university's average rank in the pre-AlphaFold period (before 2019) rather than being redefined annually. Panel (A) shows trends for Microbiology (excluding protein research) publications. Panel (B) shows trends for Economics journals and Panel (C) for Materials Science journals and Panel (D) for the Social Sciences. University quartile composition is fixed within each panel but defined separately across panels. A top quartile university for Microbiology might not be in the top quartile for Materials Science. Unlike in protein research, there are no visible changes in top journal share after 2018.

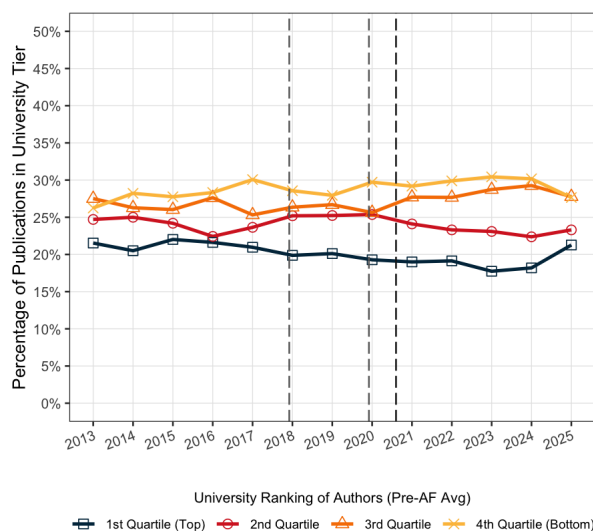
(A) Microbiology (no protein)



(B) Economics & Finance



(C) Materials Science



(D) Social Sciences

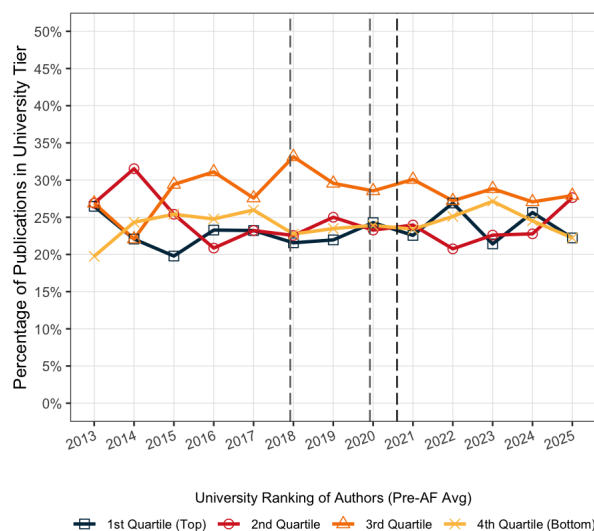


Figure 4: Bottom-quartile coefficients by prior protein information and period

Each point compares the bottom quartile with the top-quartile reference group. Bars are 95% confidence intervals.

Where the bottom-quartile gain lands, by period

Same specification, three panels, two dependent variables

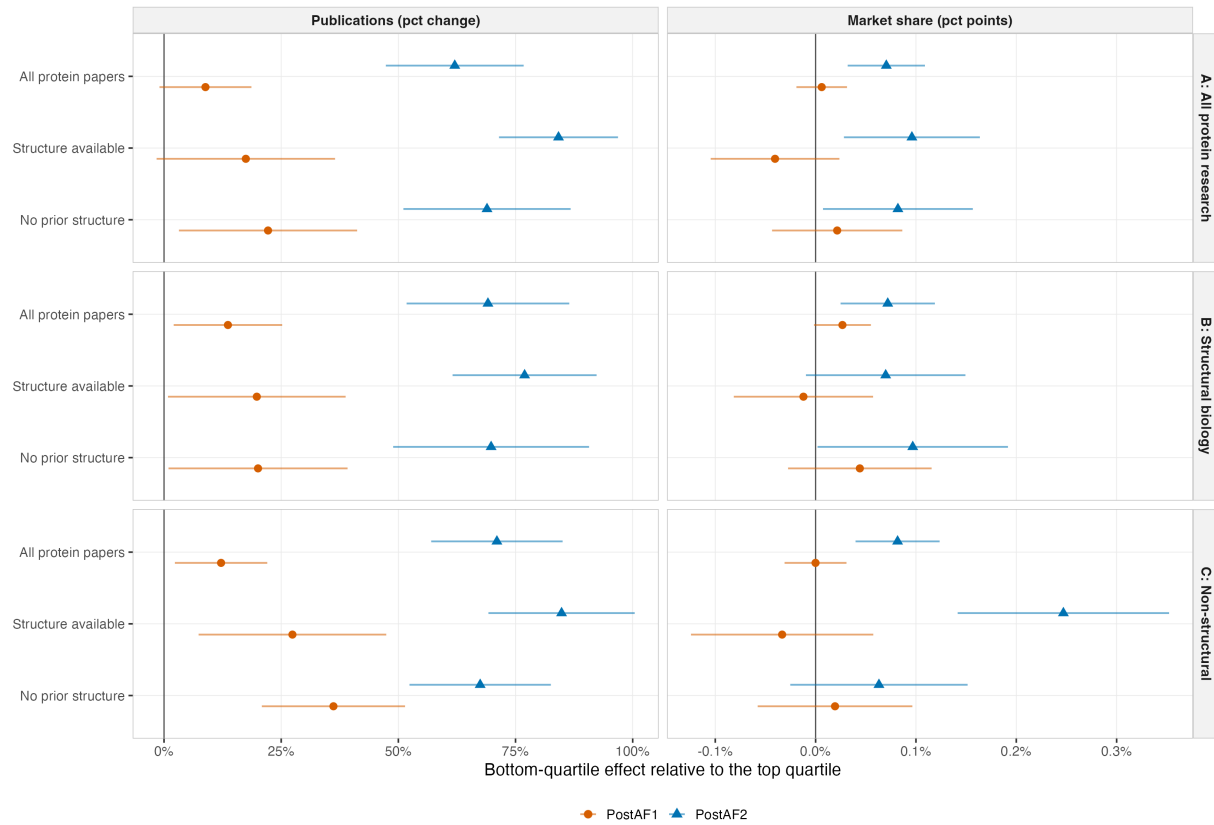


Figure 5: Research direction measures in top journal publications by university quartile

Figure shows pivot scores, novelty, and disruptiveness measures based on three-year publication histories of protein researchers that published in top journals (SJ_R ≥ 10). *p*-values shown indicate the statistical significance of pre- vs. post-AlphaFold differences within each quartile. Panels (A) and (B) show pivot scores for structural biology and non-structural protein publications. Panel (C) and (D) show novelty measures. Panel (E) and (F) show disruptiveness measures. Universities are divided into four quartiles based on their aggregate share of publications in top journals during 2013-18.

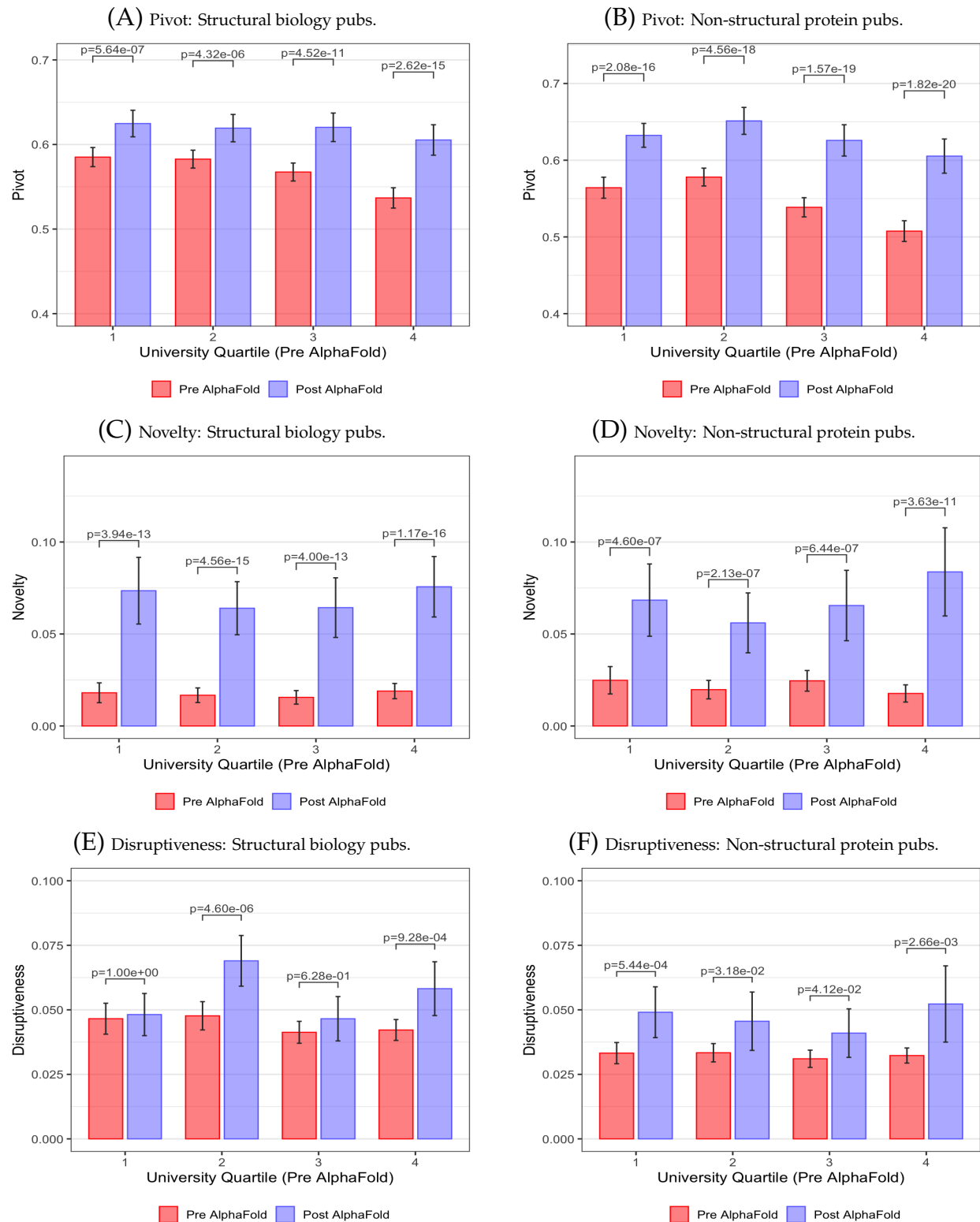


Table 1: Protein publications in top journals around AlphaFold

PostAF1 is a binary variable set to one for papers published after the December 2018 CASP13 result. The sample is limited to all protein research papers (both structural biology and non-structural protein research) published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 results. Universities are ranked based on their average TNCS in the period preceding the CASP13. The highest-ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile $\times$ PostAF1	0.19*** (0.02)	0.06*** (0.00)	0.10*** (0.00)
University Tier = 3rd Quartile $\times$ PostAF1	0.31*** (0.03)	0.13** (0.02)	0.14** (0.02)
University Tier = 4th Quartile $\times$ PostAF1	0.47*** (0.04)	0.21*** (0.02)	0.23*** (0.02)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
Adjusted/Pseudo R ²	0.82	0.92	0.93

(B) Structural biology publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile $\times$ PostAF1	0.23*** (0.02)	0.13*** (0.00)	0.15*** (0.00)
University Tier = 3rd Quartile $\times$ PostAF1	0.36*** (0.03)	0.16*** (0.02)	0.16*** (0.01)
University Tier = 4th Quartile $\times$ PostAF1	0.54*** (0.04)	0.25*** (0.03)	0.24*** (0.02)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
Adjusted/Pseudo R ²	0.81	0.91	0.92

(C) Non-structural protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile $\times$ PostAF1	0.28*** (0.03)	0.09*** (0.01)	0.12*** (0.01)
University Tier = 3rd Quartile $\times$ PostAF1	0.40*** (0.03)	0.17*** (0.03)	0.17*** (0.03)
University Tier = 4th Quartile $\times$ PostAF1	0.54*** (0.03)	0.25*** (0.02)	0.30*** (0.03)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
Adjusted/Pseudo R ²	0.79	0.90	0.91

Table 2: Protein publications in top journals around AlphaFold: AF1 versus AF2

PostAF1/AF2 is a categorical variable with the pre-period as the reference group, where *PostAF1* is set to one for papers published after the December 2018 CASP13 result but before the July 2021 open release of AlphaFold 2's code and database, and *PostAF2* is set to one for papers published after that open release. The sample includes all protein research papers (both structural and non-structural biology) published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 result. Universities are ranked based on their average TNCS in the period preceding CASP13. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile $\times$ PostAF1	0.3383*** (0.0885)	0.1166 (0.0777)	0.1243* (0.0750)
University Tier = 2nd Quartile $\times$ PostAF2	0.1853* (0.0963)	0.0828 (0.0926)	0.1222 (0.0978)
University Tier = 3rd Quartile $\times$ PostAF1	0.4729*** (0.0769)	0.1733** (0.0703)	0.1740*** (0.0651)
University Tier = 3rd Quartile $\times$ PostAF2	0.4223*** (0.0886)	0.2862*** (0.0952)	0.2591** (0.1033)
University Tier = 4th Quartile $\times$ PostAF1	0.8294*** (0.0648)	0.4937*** (0.0653)	0.4476*** (0.0647)
University Tier = 4th Quartile $\times$ PostAF2	0.9476*** (0.0844)	0.7451*** (0.1003)	0.6685*** (0.1049)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R ²	0.81	0.89	0.90

(B) Structural biology publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile $\times$ PostAF1	0.3702*** (0.1003)	0.2097** (0.0911)	0.1844** (0.0880)
University Tier = 2nd Quartile $\times$ PostAF2	0.2320** (0.1131)	0.1335 (0.1153)	0.1903 (0.1202)
University Tier = 3rd Quartile $\times$ PostAF1	0.5352*** (0.0828)	0.2914*** (0.0761)	0.3028*** (0.0761)
University Tier = 3rd Quartile $\times$ PostAF2	0.4918*** (0.1125)	0.2880** (0.1246)	0.2526* (0.1304)
University Tier = 4th Quartile $\times$ PostAF1	0.8934*** (0.0711)	0.7100*** (0.0741)	0.5966*** (0.0736)
University Tier = 4th Quartile $\times$ PostAF2	1.049*** (0.1072)	0.8544*** (0.1244)	0.7365*** (0.1289)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R ²	0.81	0.87	0.88

(C) Non-structural protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile $\times$ PostAF1	0.4239*** (0.1192)	0.1851* (0.0954)	0.1832* (0.0934)
University Tier = 2nd Quartile $\times$ PostAF2	0.3600*** (0.1180)	0.1415 (0.1131)	0.1562 (0.1152)
University Tier = 3rd Quartile $\times$ PostAF1	0.5298*** (0.1070)	0.2236** (0.0870)	0.1853** (0.0790)
University Tier = 3rd Quartile $\times$ PostAF2	0.6346*** (0.1085)	0.4181*** (0.1113)	0.3988*** (0.1138)
University Tier = 4th Quartile $\times$ PostAF1	0.8420*** (0.0984)	0.6054*** (0.0755)	0.5765*** (0.0735)
University Tier = 4th Quartile $\times$ PostAF2	1.102*** (0.1028)	0.9565*** (0.1138)	0.9067*** (0.1146)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R ²	0.79	0.84	0.85

Table 3: Protein publications in top journals around AlphaFold: differences by institutional access to high-performance computing

PostAF1/AF2 is a categorical variable with the pre-period as the reference group, where *PostAF1* is set to one for papers published after the December 2018 CASP13 result but before the July 2021 open release of AlphaFold 2's code and database, and *PostAF2* is set to one for papers published after that open release. The sample is limited to all protein research papers (both structural and non-structural biology) published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 result. Universities are ranked based on their average TNCs in the period preceding CASP13. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Dependent Variables: HPC Access at Institution	Log(1 + No. of Publications)		Market Share (% Publications, All Authors)		Market Share (% Publications, Main Authors)	
	Yes	No	Yes	No	Yes	No
Model:	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile × PostAF1	0.1500* (0.0817)	0.6671** (0.3207)	0.0350 (0.0667)	0.4463 (0.3023)	0.0664 (0.0670)	0.2349 (0.2960)
University Tier = 2nd Quartile × PostAF2	0.1664 (0.1023)	0.3563** (0.1725)	0.0815 (0.1123)	0.1603 (0.1853)	0.1277 (0.1198)	0.2900 (0.2368)
University Tier = 3rd Quartile × PostAF1	0.2473*** (0.0788)	0.9863*** (0.2690)	0.0705 (0.0694)	0.6465** (0.2856)	0.0808 (0.0652)	0.4347* (0.2621)
University Tier = 3rd Quartile × PostAF2	0.3391*** (0.0921)	0.6634*** (0.1255)	0.2391** (0.1163)	0.4189** (0.1678)	0.2071 (0.1258)	0.4463** (0.2226)
University Tier = 4th Quartile × PostAF1	0.3394*** (0.0732)	1.512** (0.2598)	0.1626** (0.0731)	1.213** (0.2794)	0.2004*** (0.0719)	0.9426*** (0.2605)
University Tier = 4th Quartile × PostAF2	0.6579*** (0.0932)	1.288*** (0.1171)	0.4704*** (0.1238)	1.035*** (0.1684)	0.4753*** (0.1302)	0.9703*** (0.2214)
Controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	1,925	3,575	1,925	3,575	1,925	3,575
R ²	0.87	0.72	0.94	0.73	0.94	0.72

(B) Structural biology publications

Dependent Variables: HPC Access at Institution	Log(1 + No. of Publications)		Market Share (% Publications, All Authors)		Market Share (% Publications, Main Authors)	
	Yes	No	Yes	No	Yes	No
Model:	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile × PostAF1	0.2660*** (0.1022)	0.4856** (0.2355)	0.1296* (0.0768)	0.2416 (0.2261)	0.1247 (0.0772)	0.0398 (0.1919)
University Tier = 2nd Quartile × PostAF2	0.2570** (0.1153)	0.4465*** (0.1680)	0.1818 (0.1271)	0.2068 (0.1942)	0.2617** (0.1303)	0.0933 (0.2001)
University Tier = 3rd Quartile × PostAF1	0.3457*** (0.0823)	0.7840*** (0.1819)	0.1868*** (0.0678)	0.4622*** (0.1646)	0.2065*** (0.0675)	0.3265*** (0.1218)
University Tier = 3rd Quartile × PostAF2	0.4662*** (0.1124)	0.6141*** (0.1458)	0.3077** (0.1404)	0.2689 (0.1844)	0.2766** (0.1374)	0.1738 (0.1995)
University Tier = 4th Quartile × PostAF1	0.4978*** (0.0795)	1.251*** (0.1690)	0.3308*** (0.0778)	1.072*** (0.1586)	0.3159*** (0.0750)	0.7767*** (0.1098)
University Tier = 4th Quartile × PostAF2	0.8074*** (0.1081)	1.272*** (0.1365)	0.5536*** (0.1355)	1.045*** (0.1829)	0.5898*** (0.1344)	0.7381*** (0.1931)
Controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	1,925	3,575	1,925	3,575	1,925	3,575
R ²	0.87	0.72	0.94	0.73	0.94	0.73

(C) Non-structural protein publications

Dependent Variables:	Log(1 + No. of Publications)		Market Share (% Publications, All Authors)		Market Share (% Publications, Main Authors)	
	Yes	No	Yes	No	Yes	No
Model:	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile × PostAF1	0.2331** (0.0928)	0.5927 (0.5801)	0.0752 (0.0783)	0.6682 (0.5592)	0.1003 (0.0775)	0.4856 (0.5128)
University Tier = 2nd Quartile × PostAF2	0.3703*** (0.1373)	0.1838 (0.3826)	0.1358 (0.1449)	0.0501 (0.2013)	0.2094 (0.1493)	0.1185 (0.2194)
University Tier = 3rd Quartile × PostAF1	0.3005*** (0.0891)	0.7914 (0.5679)	0.0589 (0.0862)	0.8381 (0.5535)	0.0634 (0.0782)	0.5085 (0.4965)
University Tier = 3rd Quartile × PostAF2	0.5396*** (0.1337)	0.3846 (0.3583)	0.3078** (0.1451)	0.2478 (0.1695)	0.2914** (0.1444)	0.2916* (0.1755)
University Tier = 4th Quartile × PostAF1	0.3588*** (0.0760)	1.212** (0.5653)	0.1808** (0.0808)	1.432*** (0.5479)	0.2378*** (0.0799)	1.152** (0.4933)
University Tier = 4th Quartile × PostAF2	0.8654*** (0.1293)	0.9276*** (0.3553)	0.6721*** (0.1439)	0.9119*** (0.1715)	0.7266*** (0.1442)	0.8281*** (0.1784)
Controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	1,925	3,575	1,925	3,575	1,925	3,575
R ²	0.85	0.71	0.92	0.68	0.92	0.67

Table 4: Research patterns in top-journal publications around AlphaFold

Table presents regression estimates examining qualitative changes in the research produced following the December 2018 CASP13 result. Dependent variables are pivot scores, novelty measures, and disruptiveness scores for publications in top journals ($\text{SJR} \geq 10$), averaged at the university year level. *PostAF1* is a binary indicator for papers published after the CASP13 result in December 2018. Sample includes structural biology publications (columns 1-3) and non-structural protein publications (columns 4-6) by authors affiliated with top 500 universities. University quartiles based on pre-AlphaFold publication shares in top journals (2013-2018), with top quartile as reference group. Novelty and disruptiveness are min-max normalized within each calendar quarter across all universities to the [0,1] range; the pivot score is naturally bounded on the same range and is not additionally scaled. Coefficients are therefore differences in scale units, not percentage changes in raw measures. Standard errors clustered by university in parentheses. ***, **, * denote significance at 1%, 5%, 10% levels.

Dependent Variables: Model:	Structural Biology Pubs.			Non-Structural Protein Pubs.		
	Pivot Score	Novelty	Disruptiveness	Pivot Score	Novelty	Disruptiveness
	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile $\times$ PostAF1	0.0104** (0.0025)	0.0196*** (0.0020)	0.0117*** (0.0003)	0.0345*** (0.0013)	0.0026 (0.0015)	-0.0010 (0.0012)
University Tier = 3rd Quartile $\times$ PostAF1	0.0166*** (0.0026)	0.0152*** (0.0020)	0.0046** (0.0010)	0.0028 (0.0025)	0.0112*** (0.0011)	-0.0056* (0.0022)
University Tier = 4th Quartile $\times$ PostAF1	0.0263*** (0.0021)	0.0154** (0.0032)	0.0126*** (0.0015)	0.0326*** (0.0034)	0.0268*** (0.0018)	0.0065* (0.0026)
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500	5,500
R ²	0.03	0.18	0.07	0.02	0.16	0.06

Table 5: Collaborations with top-quartile universities in top journal publications around AlphaFold

Table presents regression estimates examining changes in collaboration patterns following the December 2018 CASP13 result. The dependent variable is the share of top-journal publications produced by university u in year t that were co-authored with at least one author from a top-quartile university. For each sample, we run two specifications: One where we only include the usual double interaction term between *PostAF1* and *University Tier* (column 1,3,5) and one where we include the triple interaction term between *PostAF1*, *University Tier*, and the share of structural biologists at the university (column 2,4,6) in year t . University quartiles based on pre-AlphaFold publication shares in high-impact journals (2013-2018), with 1st quartile as reference group. *PostAF1* is a binary indicator for papers published after the CASP13 result in December 2018. Quartile 1 is the reference group and the controls are the covariates shown in Table A3. Standard errors clustered by university in parentheses. ***, **, * denote significance at 1%, 5%, 10% levels.

Dependent Variable:	Collaboration 1st Quartile (%)					
	All Protein Pubs.		Structural Biology Pubs.		Non-Structural Protein Pubs.	
	(1)	(2)	(3)	(4)	(5)	(6)
Model:						
University Tier = 2nd Quartile $\times$ PostAF1	-0.0124*** (0.0004)	0.0738** (0.0199)	-0.0491*** (0.0021)	0.1535*** (0.0210)	-0.0337** (0.0081)	0.1704*** (0.0163)
University Tier = 3rd Quartile $\times$ PostAF1	0.0006 (0.0091)	0.1093*** (0.0175)	-0.0657** (0.0118)	0.1678*** (0.0146)	-0.0506** (0.0151)	0.1544*** (0.0166)
University Tier = 4th Quartile $\times$ PostAF1	-0.0566** (0.0124)	0.0377* (0.0131)	-0.1295*** (0.0146)	0.0937*** (0.0102)	-0.1162*** (0.0174)	0.0665** (0.0158)
University Tier = 2nd Quartile $\times$ PostAF1 $\times$ Structural Biologists (%)		-1.240** (0.2747)		-2.750*** (0.3011)		-2.788*** (0.2796)
University Tier = 3rd Quartile $\times$ PostAF1 $\times$ Structural Biologists (%)		-1.432** (0.3628)		-3.030*** (0.3633)		-2.815*** (0.3428)
University Tier = 4th Quartile $\times$ PostAF1 $\times$ Structural Biologists (%)		-1.079** (0.3363)		-2.745*** (0.3140)		-2.377*** (0.3096)
Controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500	5,500
Adjusted R ²	0.19	0.20	0.29	0.29	0.30	0.30

Table 6: The gap between the two protein strata across periods

A protein counts as having prior known information if an experimental structure of its own, or a homology template of at least 30 pct sequence identity, is dated before the AlphaFold Protein Structure Database release on 22 July 2021. Papers are assigned to mutually exclusive strata at a 50 pct threshold: a paper is Structure available when at least half of its linked proteins meet that test and No prior structure otherwise. The stratum enters every interaction as an indicator for structure-available papers, so a positive coefficient means the lower-quartile differential is larger for papers with a pre-existing structure available. Publication-count coefficients are log-point differences for $\text{Log}(1 + \text{publications})$; market-share coefficients are fractional-logit index differences, not percentage-point effects.

Dependent Variables:	Log(1 + No. of publications)			Market share (% of publications)		
	All protein research	Structural biology	Non-structural	All protein research	Structural biology	Non-structural
Model:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
2nd Quartile $\times$ PostAF1	0.0356 (0.1232)	0.0262 (0.1299)	0.1089 (0.1033)	-0.0361 (0.1477)	-0.0413 (0.1684)	-0.0122 (0.1558)
3rd Quartile $\times$ PostAF1	0.1887* (0.1035)	0.1736* (0.1031)	0.2828*** (0.0833)	0.1475 (0.1232)	0.1578 (0.1318)	0.1400 (0.1349)
4th Quartile $\times$ PostAF1	0.2219** (0.0970)	0.2006** (0.0974)	0.3615*** (0.0779)	0.0926 (0.1425)	0.1792 (0.1483)	0.0750 (0.1524)
2nd Quartile $\times$ PostAF2	0.1087 (0.1139)	0.1875 (0.1312)	0.2438** (0.0981)	-0.2075 (0.1863)	-0.0303 (0.2110)	-0.2073 (0.1952)
3rd Quartile $\times$ PostAF2	0.4418*** (0.0970)	0.4380*** (0.1115)	0.4975*** (0.0825)	0.0890 (0.1744)	0.0291 (0.2082)	0.0357 (0.1910)
4th Quartile $\times$ PostAF2	0.6892*** (0.0910)	0.6979*** (0.1066)	0.6746*** (0.0769)	0.3534** (0.1639)	0.3936** (0.1966)	0.2448 (0.1747)
2nd Quartile $\times$ PostAF1 $\times$ Structure available	0.0948 (0.1348)	0.1383 (0.1443)	0.0028 (0.1340)	0.0029 (0.1811)	0.0940 (0.1996)	-0.1668 (0.1871)
3rd Quartile $\times$ PostAF1 $\times$ Structure available	-0.0860 (0.1175)	-0.0288 (0.1217)	-0.0834 (0.1276)	-0.2387 (0.1778)	-0.1680 (0.1959)	-0.1839 (0.2224)
4th Quartile $\times$ PostAF1 $\times$ Structure available	-0.0474 (0.1058)	-0.0026 (0.1086)	-0.0875 (0.1179)	-0.2600 (0.1873)	-0.2272 (0.1891)	-0.1953 (0.2049)
2nd Quartile $\times$ PostAF2 $\times$ Structure available	0.3517** (0.1371)	0.1601 (0.1371)	0.2999*** (0.0996)	0.4475** (0.2126)	0.1542 (0.2353)	0.7773*** (0.2228)
3rd Quartile $\times$ PostAF2 $\times$ Structure available	0.1338 (0.1134)	0.0642 (0.1144)	0.1609* (0.0835)	0.0073 (0.2007)	-0.0673 (0.2356)	0.3503 (0.2385)
4th Quartile $\times$ PostAF2 $\times$ Structure available	0.1525 (0.1049)	0.0714 (0.1074)	0.1737** (0.0759)	0.0439 (0.1818)	-0.1171 (0.2169)	0.6456*** (0.1966)
<i>Fixed-effects</i>						
University	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	11,154	10,890	10,802	10,340	9,900	9,504
Adjusted/Pseudo R ²	0.57575	0.55721	0.49510	0.64771	0.60925	0.53166

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Appendix

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A Supplementary Figures and Tables

A.1 Supplementary Figures

Figure A1: Geographical distribution of universities by quartile

The figure shows the location of the top 500 universities in our sample. Colors denote quartiles. Universities are ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares. Circle diameters

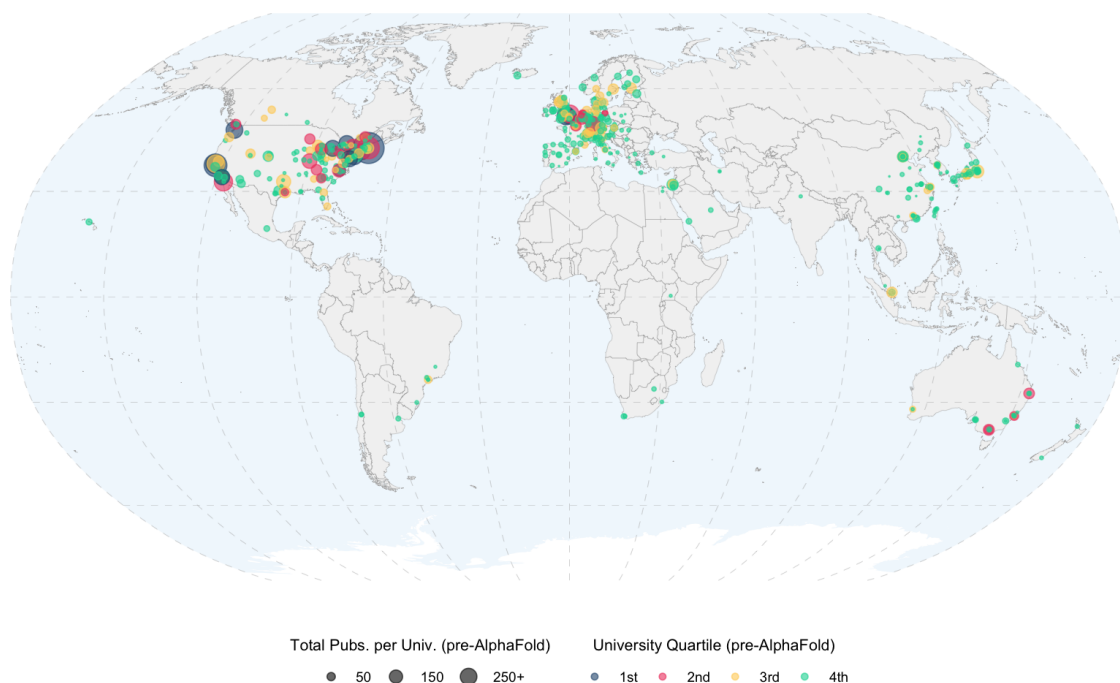
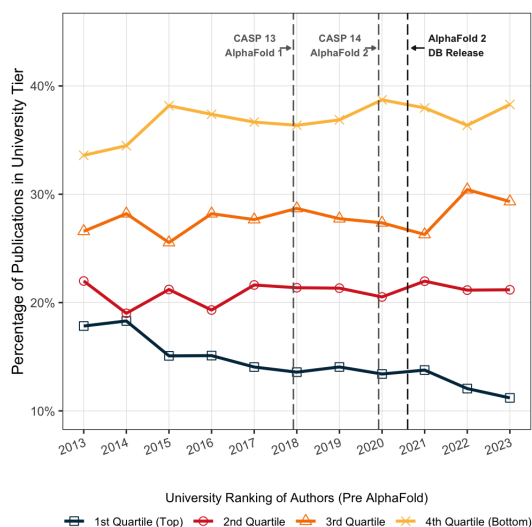


Figure A2: Publications in lower-ranked journals by university quartile

The figure reports the percentage of protein research publications in lower-ranked journals, defined as those with an average SJR between 1 and 10 during 2013–2018. The analysis is limited to authors affiliated with the top 500 universities, ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares.

(A) Structural biology publications



(B) Non-structural protein publications

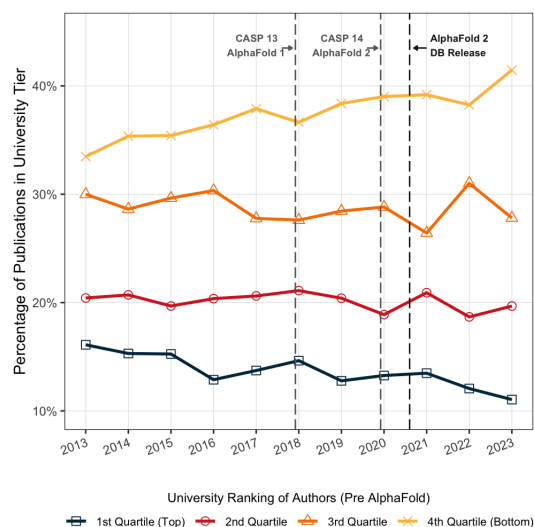


Figure A3: AI tools introduced during CASP competitions

The figure reports the number and share of prediction tools using AI at each CASP competition, together with the share of participating models that

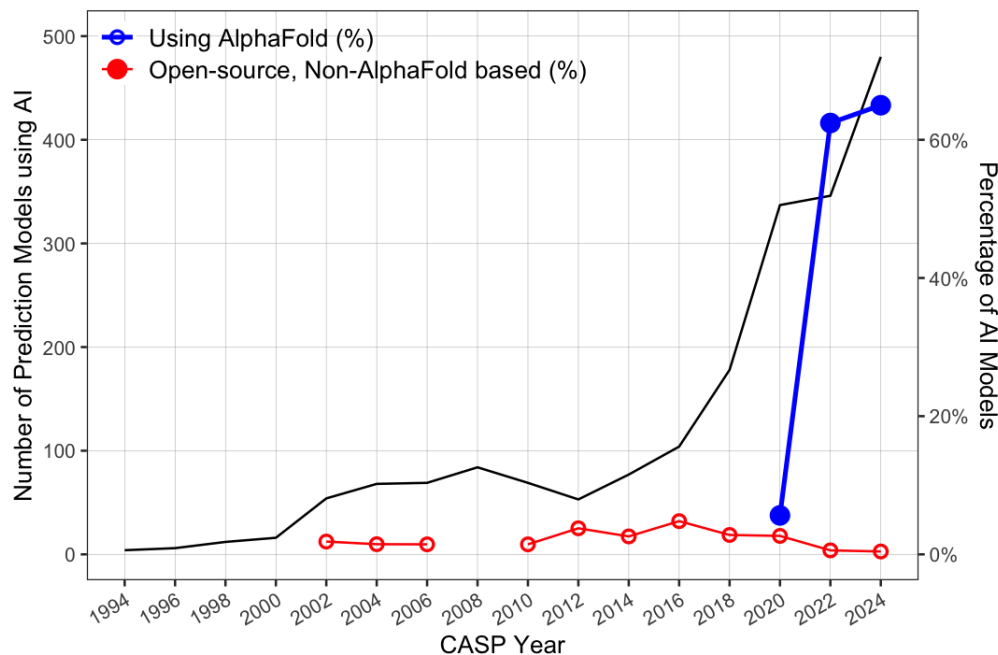


Figure A4: Distribution of top-journal publications over time

The figure reports publication volume over time for all top-journal publications, defined as those in journals with SJR ≥ 10 , in the sample.

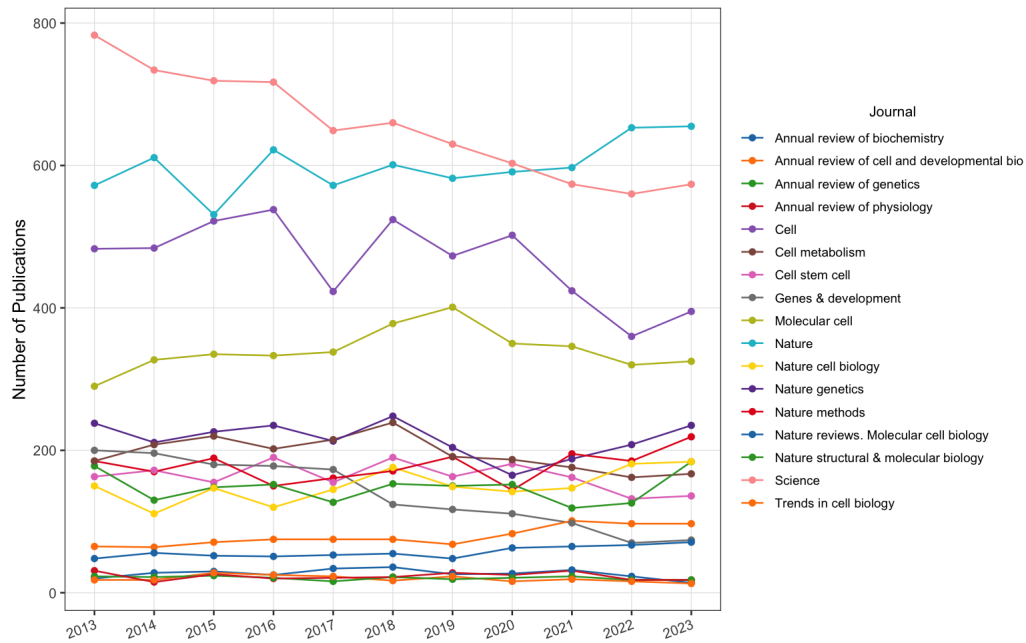


Figure A5: Timeline of AI development in structural biology

The development of AI systems in structural biology occurred in two distinct phases. In the early phase, following the December 2018 CASP13 announcement, adoption likely required substantial domain knowledge and ICT skills. In the later phase, the July 2021 open release of AlphaFold 2 (AF2) code and database made adoption easier and likely reached a wider group. A third version, AlphaFold 3 (AF3), was subsequently released.

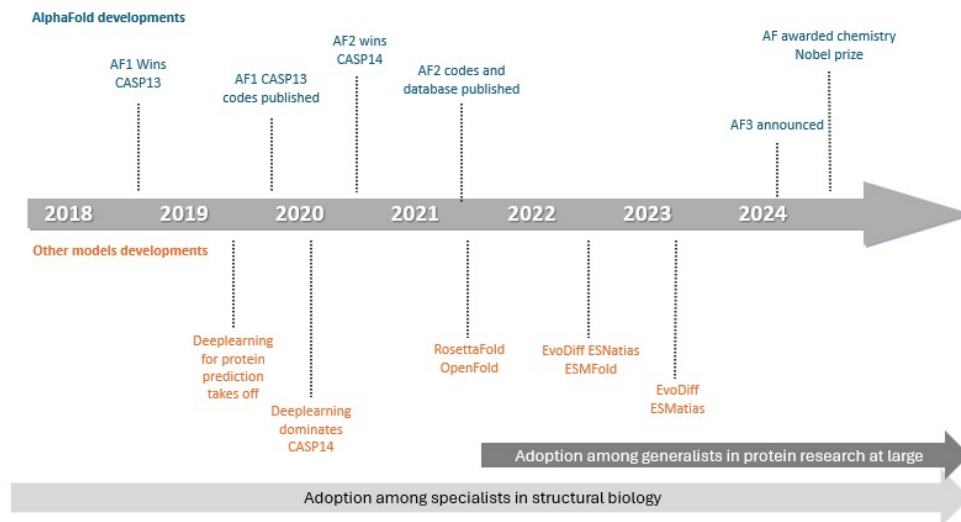
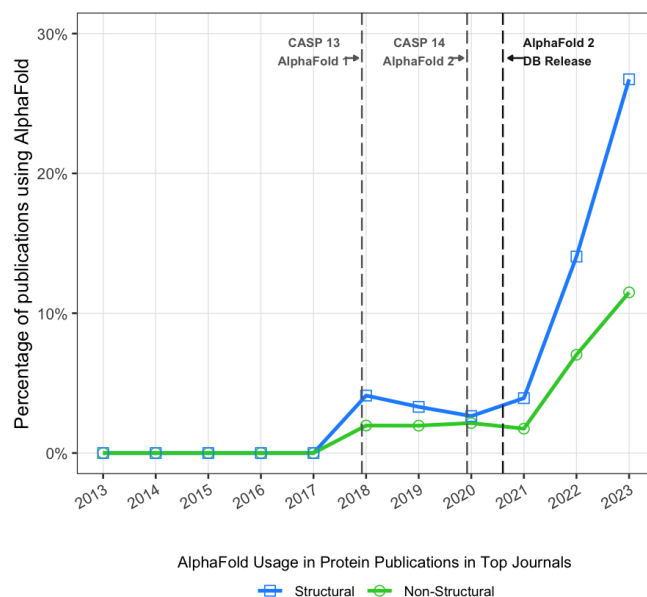


Figure A6: AlphaFold mentions in top protein publications

Figure shows details on AlphaFold adoption in top journals (SJR ≥ 10), specifically the share of protein research papers published that mention explicit use of AlphaFold in their abstract or related AI tools, such as RoseTTAFold. Panel (A) shows this share for structural and non-structural protein research. Panel (B) breaks down the adoption rate by subfield of structural biology.

(A) All protein publications



(B) AlphaFold mentions in top subfields within structural biology

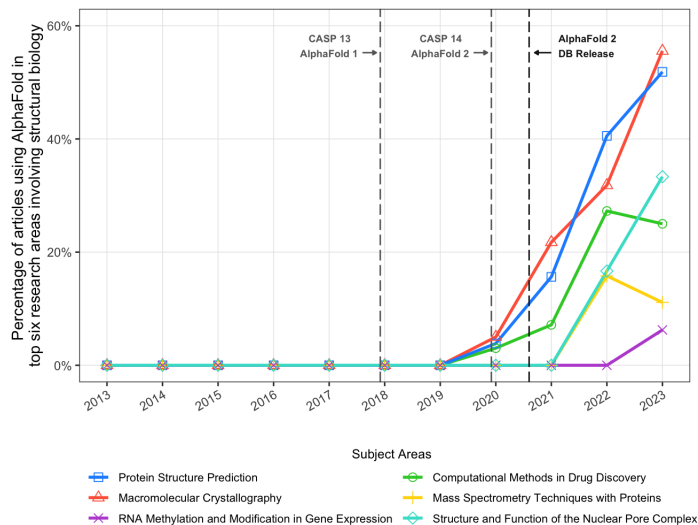
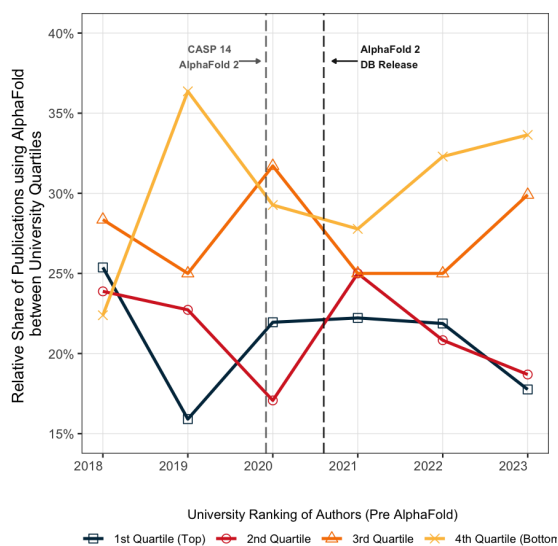


Figure A7: AlphaFold mentions in top protein publications by university quartile

The figure reports the share of publications using AlphaFold attributable to each of the four university quartiles. These shares are computed for structural-biology publications in Panel (A) and non-structural protein publications in Panel (B).

(A) Structural biology publications



(B) Non-structural protein publications

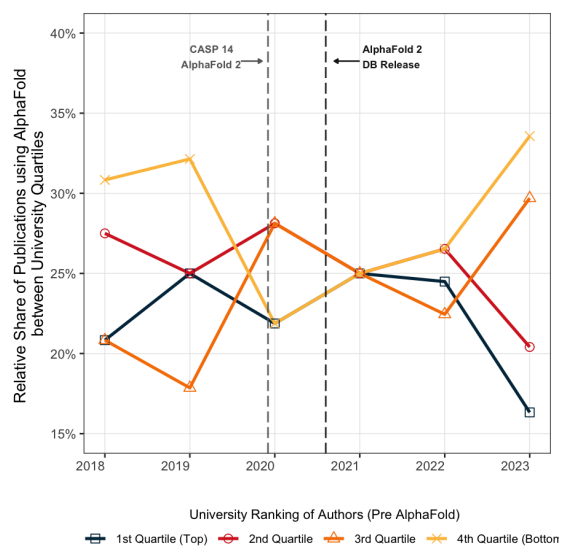
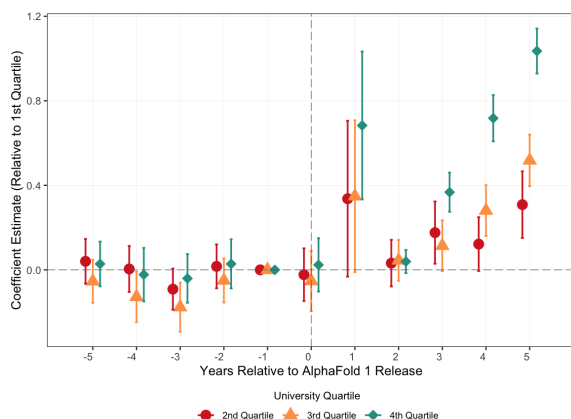


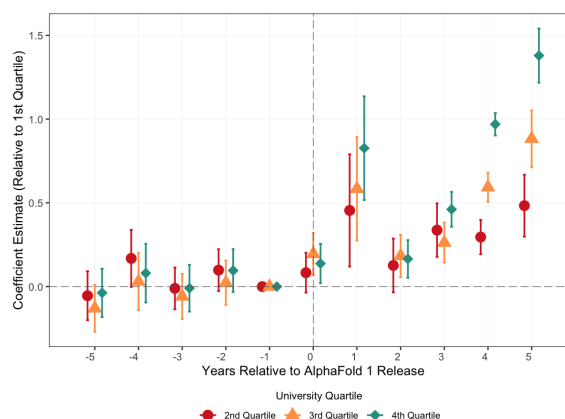
Figure A8: Protein publications by universities in top journals around AlphaFold : dynamic estimates

The figure reports annual estimates from regressions of the number of top-journal protein publications on university-quartile indicators and the controls in Equation 4. The plot shows point estimates with 95% confidence intervals. The dotted line marks the release of AlphaFold. The analysis includes all protein publications in Panel (A), structural-biology publications in Panel (B), and non-structural protein publications in Panel (C). Universities are ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares.

(A) All protein publications



(B) Structural biology publications



(C) Non-structural protein publications

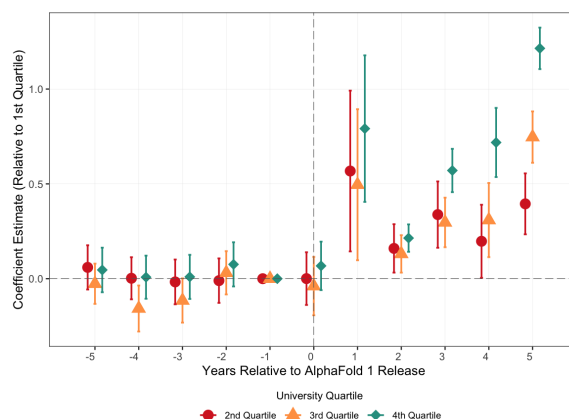
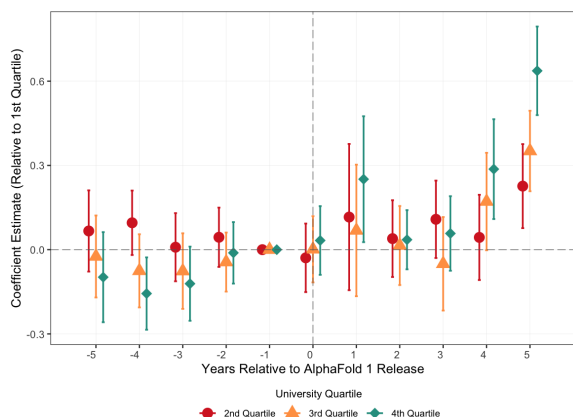


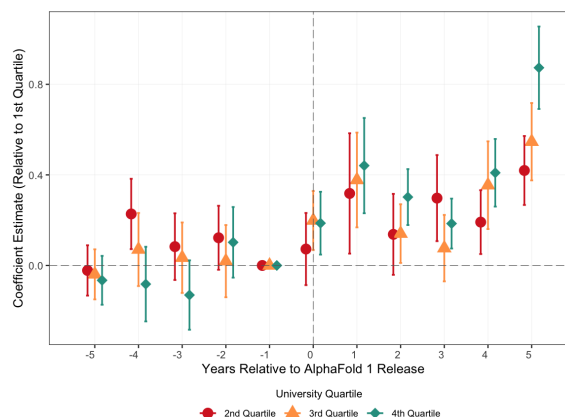
Figure A9: Market shares of universities in top-journal protein publications around AlphaFold : dynamic estimates

The figure reports annual estimates from regressions of top-journal market share on university-quartile indicators and the controls in the main specification. The plot shows point estimates with 95% confidence intervals. The dotted line marks the release of AlphaFold. The analysis includes all protein publications in Panel (A), structural-biology publications in Panel (B), and non-structural protein publications in Panel (C). Universities are ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares.

(A) All protein publications



(B) Structural biology publications



(C) Non-structural protein publications

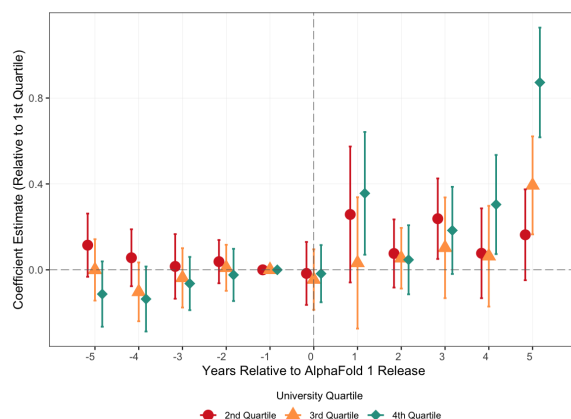
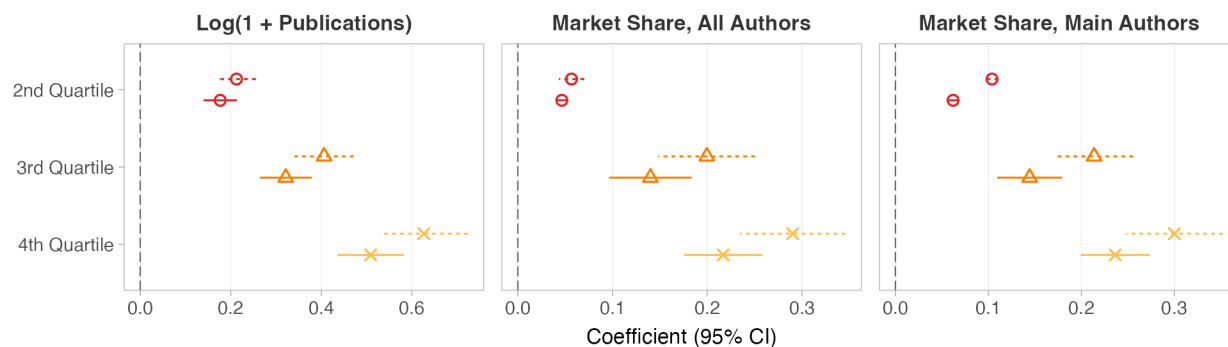


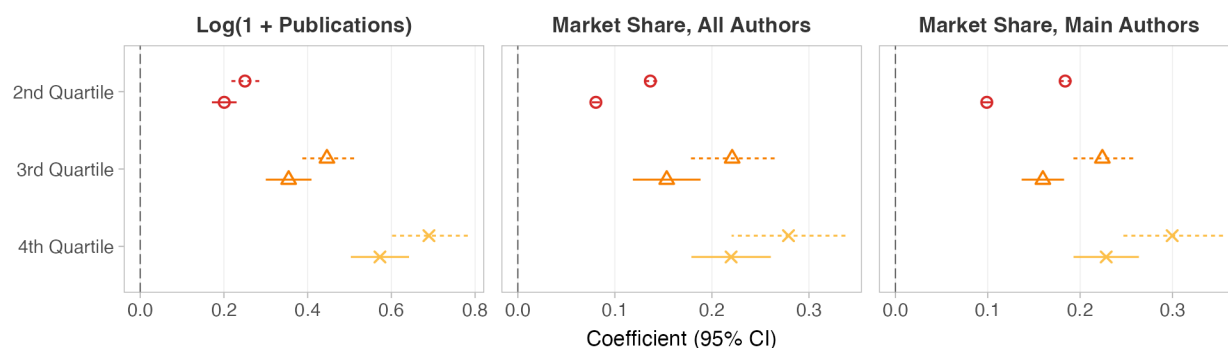
Figure A10: University-quartile gains in top-journal protein publications, excluding COVID-19 research and the pandemic years

The figure reports the quartile $\times$ PostAF1 estimates from Equation 4 for two restricted samples: one excluding publications assigned to COVID-19 topics, and one excluding the pandemic years 2020–2021. Point estimates are shown with 95% confidence intervals, with standard errors clustered by university. The first quartile is the reference group. Panel (A) covers all protein publications, Panel (B) structural-biology publications, and Panel (C) non-structural protein publications.

(A) All protein publications



(B) Structural biology publications



(C) Non-structural protein publications

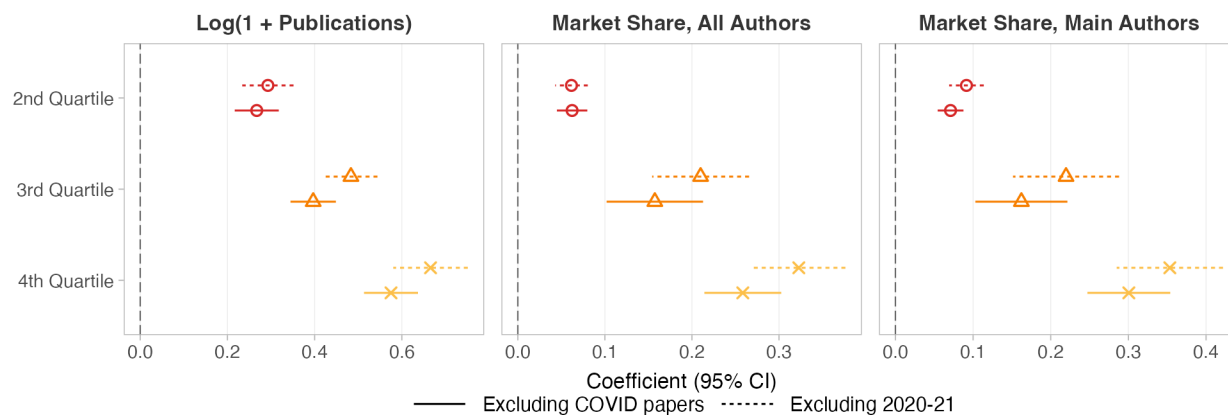
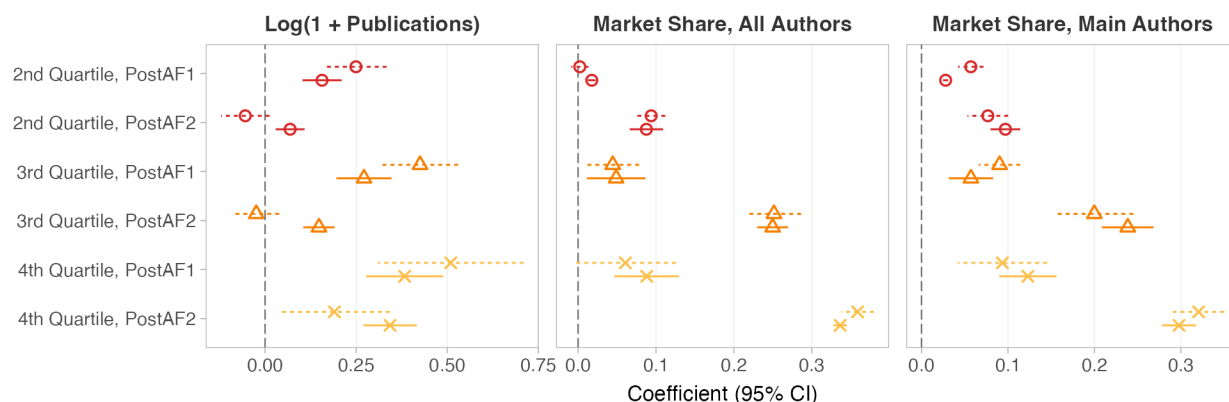


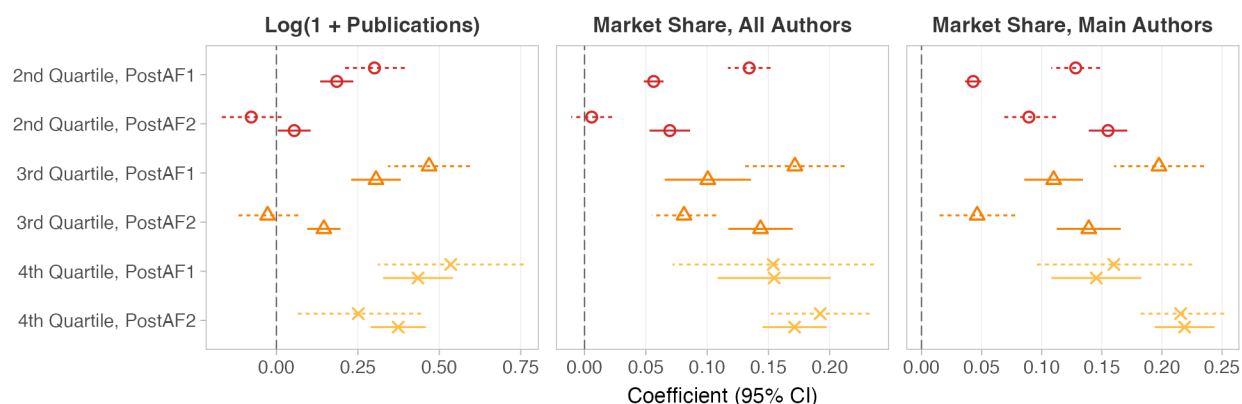
Figure A11: Timing of the quartile gains across the two AlphaFold releases, excluding COVID-19 research and the pandemic years

The figure separates the estimates in Figure A10 into the period following the release of AlphaFold 1 and the period following the release of the AlphaFold Protein Structure Database in July 2021. It reports the quartile $\times$ PostAF1 and quartile $\times$ PostAF2 estimates for two restricted samples: one excluding publications assigned to COVID-19 topics, and one excluding the pandemic years 2020–2021. Point estimates are shown with 95% confidence intervals, with standard errors clustered by university. The first quartile is the reference group. Panel (A) covers all protein publications, Panel (B) structural-biology publications, and Panel (C) non-structural protein publications.

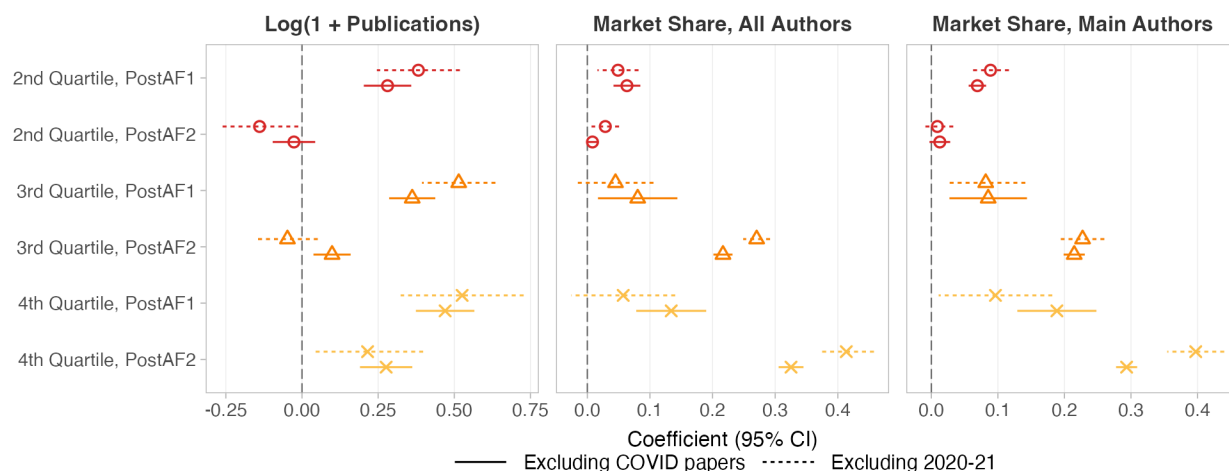
(A) All protein publications



(B) Structural biology publications



(C) Non-structural protein publications



— Excluding COVID papers - - - - - Excluding 2020-21

Figure A12: Distribution of top-journal publications in other fields over time

The figure reports the annual number of publications in each top journal in our sample, defined as journals with SJR ≥ 10 , for economics and finance in Panel (A), materials science in Panel (B), and social sciences in Panel (C).

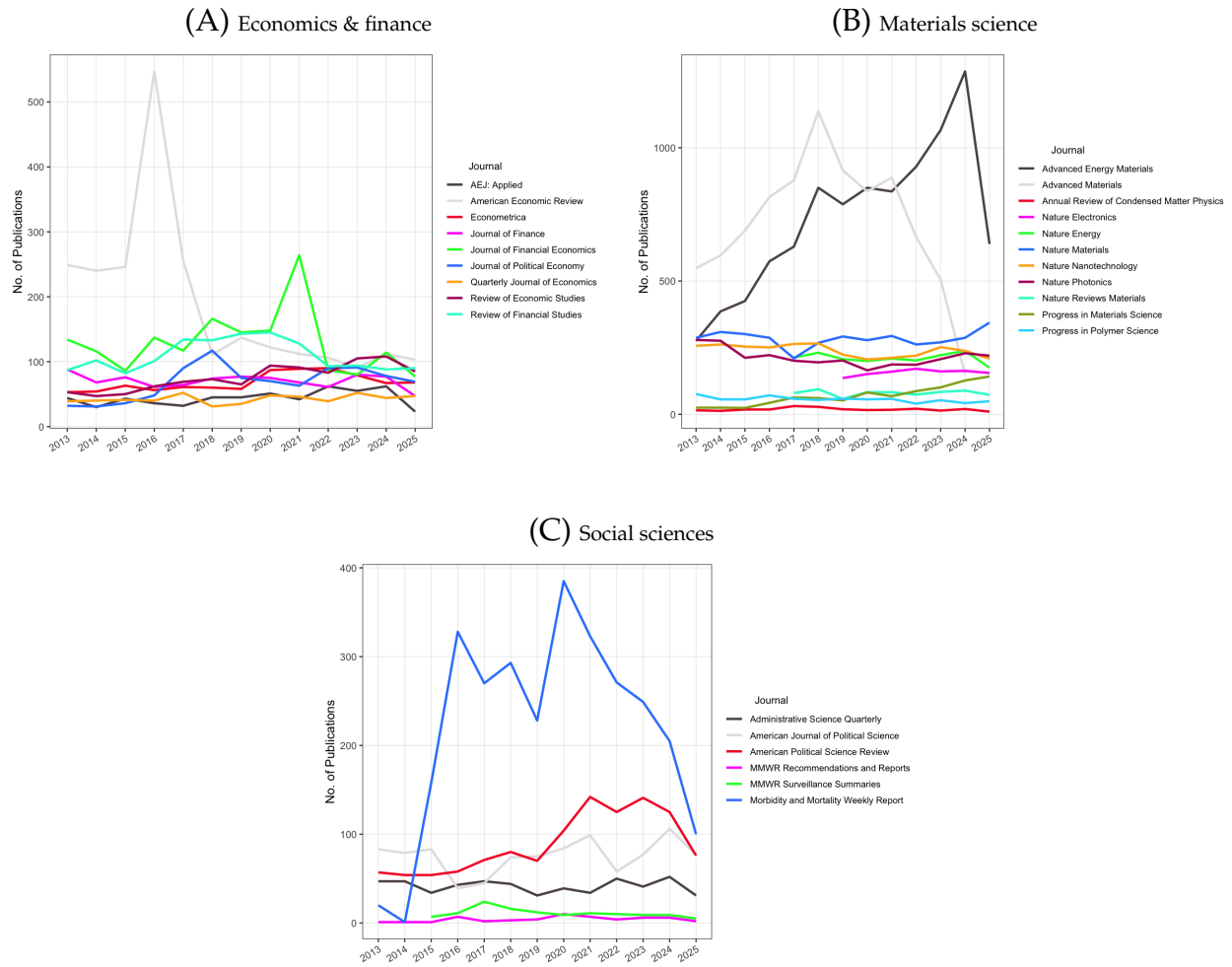


Figure A13: Market shares of top 500 universities in top-journal publications in other fields

The figure plots the cumulative distribution of top-journal publication shares across universities. University rankings are based on each university's Total Normalized Citation Score (TNCS) in a given year. For economics and finance in Panel (A) and social sciences in Panel (C), we use the Leiden Ranking category "social sciences and humanities." For materials science in Panel (B), we use "physical sciences and engineering." As in protein research, the top 500 universities capture nearly all top-journal publication counts. The distributions in all three fields show substantial concentration among top universities.

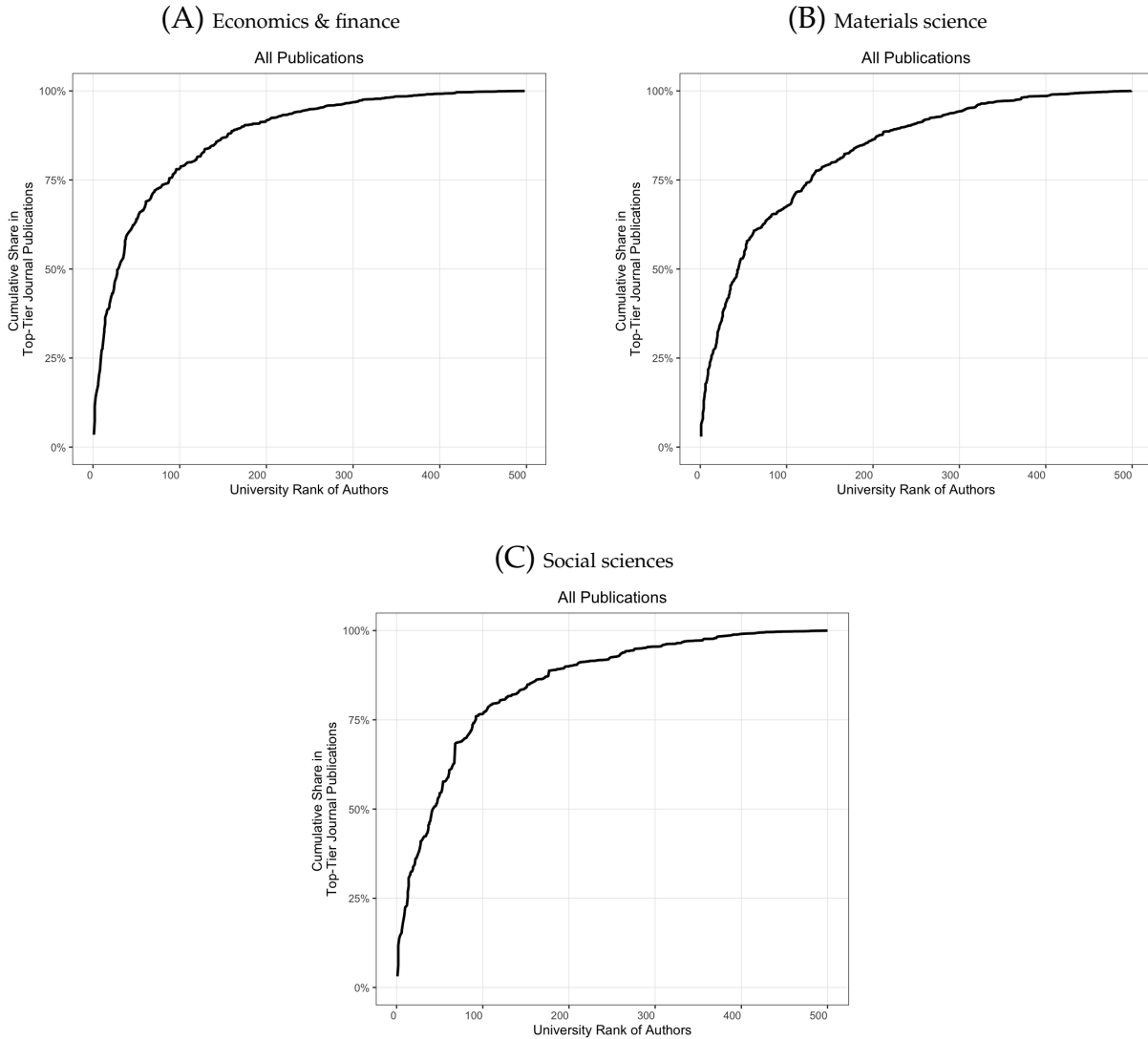


Figure A14: **Bottom-quartile changes in top-journal protein publications by protein stratum, period, and subfield**

Each point is a quartile-by-period-by-stratum interaction from the stacked specification in Table 6. A paper is structure-available when at least half of its linked proteins had an experimental structure, or a homology template of at least 30 percent sequence identity, before the release of the AlphaFold protein structure database on 22 July 2021, and has no prior protein information otherwise. A positive value means the quartile’s gain relative to the top quartile is larger on structure-available papers than on no-prior-structure papers. Bars are 95% confidence intervals.

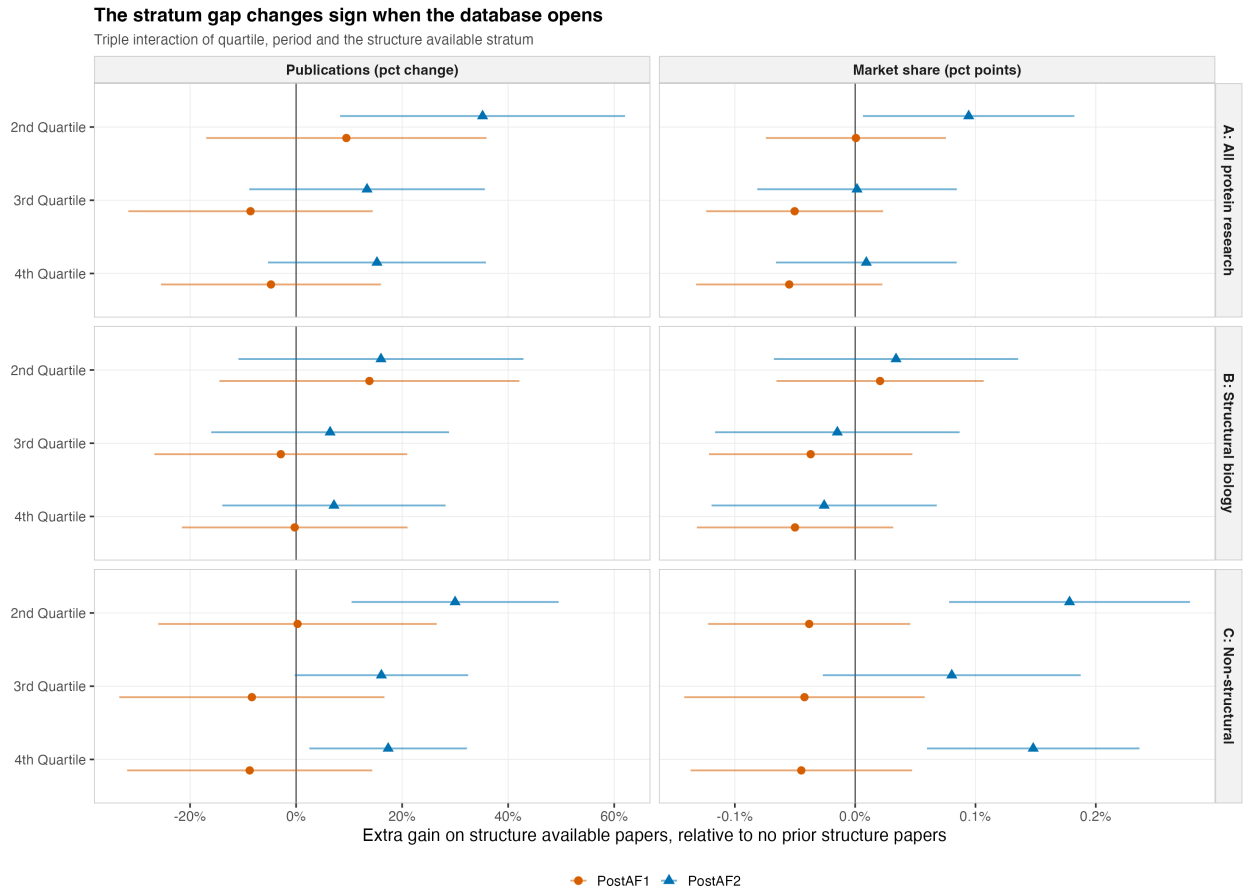


Figure A15: Lower-quartile differentials in top-journal protein research under alternative definitions of a no-prior-structure paper

Each point is the additional gain of bottom-quartile universities on structure-available papers, measured relative to the top quartile and to papers without prior protein information before AlphaFold. Figure A14 plots the same quantity under the main definition, in which a paper counts as having no prior structure when fewer than half of its linked proteins had an experimental structure or a usable homology template before the AlphaFold database release. The top row of each panel repeats the main definition, under which a paper has no prior information when more than half of its linked proteins lack it. The three rows below raise the required share, first to the top tercile of the no-prior-structure distribution, then to three quarters of a paper's linked proteins, and finally to all of them. The remaining row applies a different criterion, counting a paper as having no prior information when none of its linked proteins had an experimental structure of its own, whether or not a homology template was available.

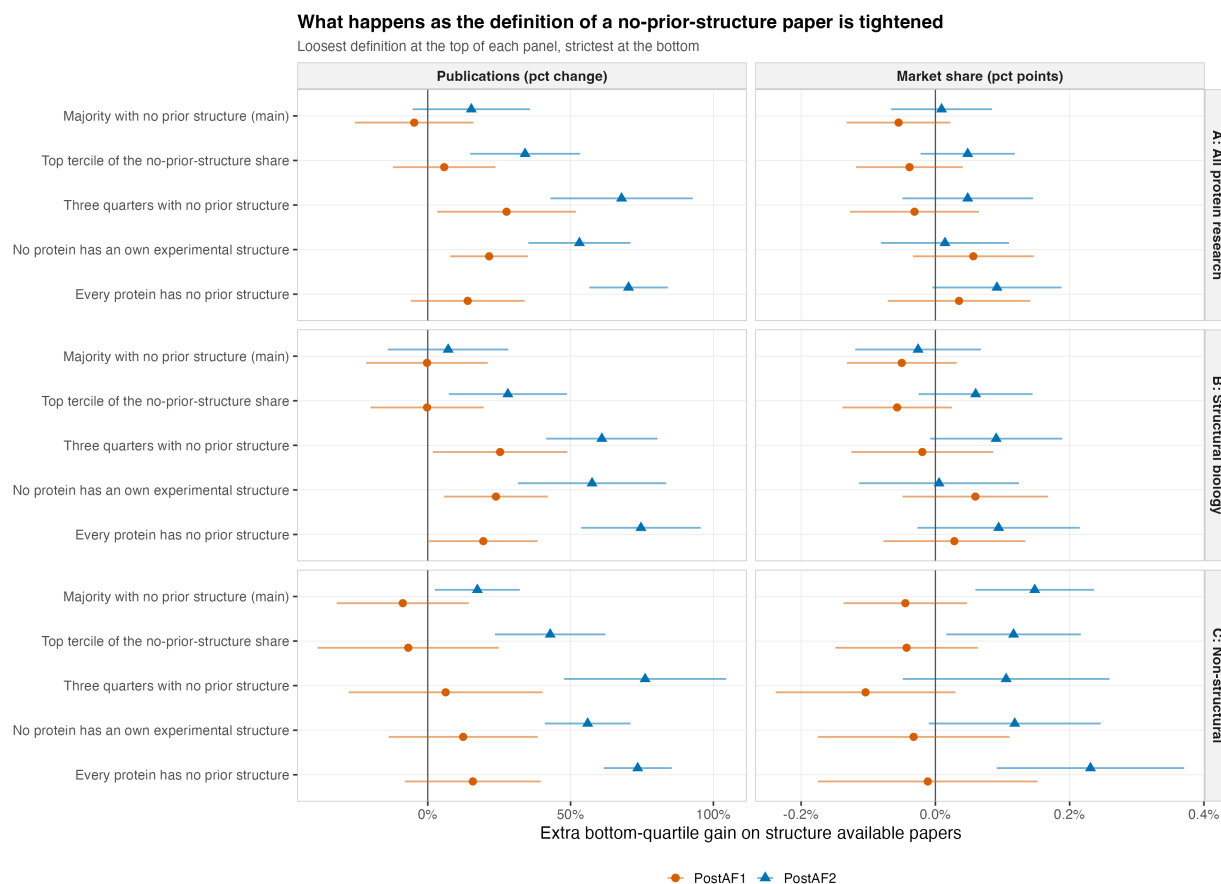
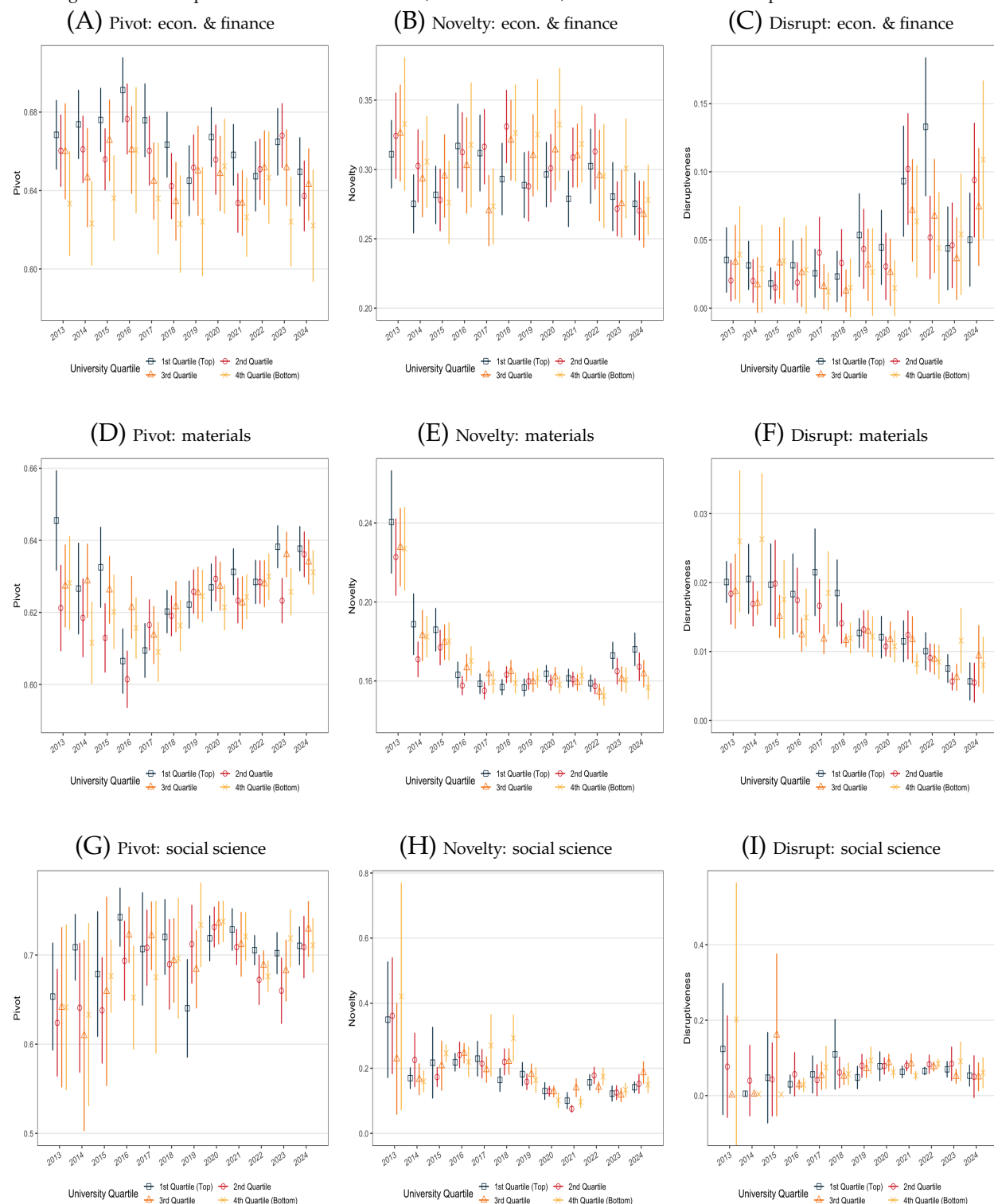


Figure A16: Research direction measures in top-journal publications by university quartile in other fields

The figure reports pivot scores, novelty, and disruptiveness measures based on three-year publication histories of authors in top journals, defined as journals with SJR ≥ 10 . Universities are ordered by field-specific pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares. Columns report pivot scores, novelty, and disruptiveness from left to right. Rows correspond to economics and finance, materials science, and social sciences from top to bottom.



A.2 Supplementary Tables

Table A1: Other AI tools developed for protein structure prediction

Table lists some of the most prominent AI tools that were developed following AlphaFold's success. Though most of them are based on the same core transformer models like AlphaFold, many of them focus on different tasks and/or use different training data.

AI Product	Release Date	Open-Source	Creator(s)	Key Feature
RoseTTAFold	July 2021	Yes	University of Washington	Predicts Protein-nucleic acid complexes
EvoDiff	September 2023	Yes	Microsoft	Applies diffusion models to get protein design from evolutionary information
ESMFold	November 2022	Yes	Meta AI	Uses amino acid sequence only for faster predictions. ESM atlas is a database with 617 million predicted protein structures
ProGen	January 2023	No	Salesforce AI Research	Synthesizes new protein sequences with desired properties from natural language inputs
BaseFold	February 2023	Yes	Basecamp Research	Uses proprietary protein samples in training data
RFDiffusion	July 2023	Yes	University of Washington	Applies diffusion models to focus on protein design

Table A2: Top journals in protein research

Table lists all journals with an SJR above ten which are used to constitute the sample of publication in the analysis throughout.

Journal Name	ISSN	Mean SJR (2013–18)	Mean Impact Factor (2013–18)	Publisher
Nature reviews. Molecular cell biology	1471-0072	28.16	39.74	Nature Portfolio
Cell	0092-8674	27.21	32.02	Cell Press
Nature genetics	1061-4036	23.01	28.02	Nature Portfolio
Annual rev. of biochemistry	1545-4509	22.12	28.44	Annual Reviews
Nature	1476-4687	18.46	41.24	Nature Portfolio
Nature methods	1548-7091	18.14	27.13	Nature Portfolio
Annual review of cell and dev. biology	1530-8995	14.97	14.94	Annual Reviews
Nature cell biology	1476-4679	14.63	18.37	Nature Portfolio
Cell stem cell	1934-5909	13.70	22.49	Elsevier BV
Molecular cell	1097-4164	13.47	14.24	Elsevier BV
Science	1095-9203	13.08	39.10	AAAS
Annual review of genetics	1545-2948	12.95	11.98	Annual Reviews
Cell metabolism	1550-4131	11.21	18.79	Cell Press
Genes & development	1549-5477	11.06	10.84	Cold Spring Harbor
Nature structural & molecular biology	1545-9993	10.82	12.72	Nature Portfolio
Trends in cell biology	0962-8924	10.41	14.39	Elsevier BV
Annual review of physiology	0066-4278	10.04	17.22	Annual Reviews

Table A3: Descriptive statistics: top protein publications and university characteristics before AlphaFold

Table presents summary statistics at the university-year level for the period 2013–18, representing the years preceding the December 2018 CASP13 result. The sample is restricted to protein-related publications in top journals with an SJR score of 10 or higher, authored by researchers affiliated with the top 500 universities prior to the CASP13 result. Each cell reports the mean of the respective variable, measured annually during the pre-AlphaFold period, with universities grouped into quartiles.

Variable (<i>Pre AlphaFold estimates</i>)	1 st (reference group)	2 nd	3 rd	4 th
Structural biology publications				
Number of Publications per Year per University	8.58	2.39	1.75	0.80
Market Share (% publications)	0.37	0.10	0.08	0.03
Market Share (% of authors on publication)	0.39	0.08	0.06	0.02
Non-structural protein publications				
Number of Publications per Year per University	7.88	2.58	1.59	0.88
Market Share (% publications)	0.36	0.12	0.07	0.04
Market Share (% of authors on publication)	0.37	0.10	0.05	0.03
University characteristics				
Faculty Size	614	308	267	178
Share Structural Biologists (%)	5.98	4.90	5.19	3.93
Share Female (%)	31.04	32.96	31.46	34.23
Share White (%)	28.91	26.11	23.62	22.61
Share Asian (%)	51.09	50.15	55.80	54.69
Share Black (%)	12.68	12.92	8.96	9.43
Share affiliated in US (%)	45.01	22.60	16.93	21.87
Share affiliated in Europe (%)	38.04	54.28	48.36	44.56
Share affiliated in Asia (%)	20.68	19.77	33.26	26.14
Share affiliated with Govt (%)	5.76	9.58	6.59	7.18
Share affiliated with Research Lab (%)	14.26	16.73	12.06	13.31
Share affiliated with Industry (%)	3.22	2.70	2.70	1.99
Share affiliated with Medical Inst (%)	16.97	16.92	16.46	14.46
Share HPC Access (%)	50.90	32.26	31.00	9.07
HPC power (petaFlop/s)	351.11	205.55	231.27	54.18
Number of Universities	12	36	88	364

Table A4: List of universities in the top quartile

These 12 universities together make up approximately 25% of top journal publications before AlphaFold (year 2013-2018)

Name	Country
Harvard University	USA
Johns Hopkins University	USA
University of California, San Francisco	USA
University of Toronto	Canada
University of Pennsylvania	USA
University of Washington	USA
University College London	UK
Stanford University	USA
Duke University	USA
University of Michigan–Ann Arbor	USA
University of California, Los Angeles	USA
University of Oxford	UK

Table A5: Structural biology top publications around AlphaFold: robustness checks

PostAF1 is a binary variable set to one for papers published after the December 2018 CASP13 result. The sample is limited to structural biology papers published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 result. Universities are ranked based on their average TNCs in the period preceding CASP13. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) Publications

Dependent Variable:	Log(1 + No. of Publications)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile $\times$ PostAF1	0.2444*** (0.0172)	0.2443*** (0.0173)	0.2351*** (0.0145)	0.2418*** (0.0197)	0.2309*** (0.0166)
University Tier = 3rd Quartile $\times$ PostAF1	0.3555*** (0.0211)	0.3557*** (0.0210)	0.3764*** (0.0262)	0.3425*** (0.0272)	0.3605*** (0.0322)
University Tier = 4th Quartile $\times$ PostAF1	0.5849*** (0.0321)	0.5845*** (0.0320)	0.5598*** (0.0334)	0.5678*** (0.0379)	0.5392*** (0.0412)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted R ²	0.780	0.781	0.805	0.785	0.809

(B) Market share (% publications, all authors)

Dependent Variable:	Market Share (% Publications, All Authors)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile $\times$ PostAF1	0.1387*** (0.0019)	0.1394*** (0.0018)	0.1406*** (0.0014)	0.1322*** (0.0049)	0.1342*** (0.0037)
University Tier = 3rd Quartile $\times$ PostAF1	0.1433*** (0.0028)	0.1445*** (0.0025)	0.1738*** (0.0219)	0.1404*** (0.0059)	0.1644*** (0.0216)
University Tier = 4th Quartile $\times$ PostAF1	0.2827*** (0.0043)	0.2813*** (0.0050)	0.2837*** (0.0232)	0.2584*** (0.0105)	0.2497*** (0.0271)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Pseudo R ²	0.904	0.904	0.906	0.917	0.921

(C) Market share (% publications, main authors)

Dependent Variable:	Market Share (% Publications, Main Authors)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile $\times$ PostAF1	0.1482*** (0.0020)	0.1480*** (0.0024)	0.1291*** (0.0029)	0.1683*** (0.0036)	0.1480*** (0.0043)
University Tier = 3rd Quartile $\times$ PostAF1	0.1409*** (0.0032)	0.1405*** (0.0036)	0.1543*** (0.0160)	0.1578*** (0.0063)	0.1623*** (0.0145)
University Tier = 4th Quartile $\times$ PostAF1	0.2700*** (0.0044)	0.2664*** (0.0058)	0.2688*** (0.0209)	0.2560*** (0.0111)	0.2420*** (0.0242)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Pseudo R ²	0.899	0.901	0.903	0.915	0.919

Table A6: Non-structural protein top publications around AlphaFold: robustness checks

PostAF1 is a binary variable set to one for papers published after the release of AlphaFold 1 in December 2018. The sample is limited to non-structural protein papers published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the release of AlphaFold 1. Universities are ranked based on their average TNCS in the period preceding the release of AlphaFold 1. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) Publications

Dependent Variable:	Log(1 + No. of Publications)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile $\times$ PostAF1	0.2912*** (0.0201)	0.2909*** (0.0201)	0.2931*** (0.0194)	0.2757*** (0.0281)	0.2767*** (0.0276)
University Tier = 3rd Quartile $\times$ PostAF1	0.4052*** (0.0185)	0.4043*** (0.0185)	0.4183*** (0.0215)	0.3913*** (0.0240)	0.4018*** (0.0289)
University Tier = 4th Quartile $\times$ PostAF1	0.5963*** (0.0279)	0.5957*** (0.0279)	0.5572*** (0.0259)	0.5766*** (0.0347)	0.5361*** (0.0340)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted R ²	0.760	0.761	0.787	0.765	0.791

(B) Market share (% publications, all authors)

Dependent Variable:	Market Share (% Publications, all authors)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile $\times$ PostAF1	0.0876*** (0.0028)	0.0869*** (0.0023)	0.0943*** (0.0061)	0.0840*** (0.0050)	0.0904*** (0.0083)
University Tier = 3rd Quartile $\times$ PostAF1	0.1482*** (0.0030)	0.1488*** (0.0032)	0.1834*** (0.0195)	0.1392*** (0.0084)	0.1715*** (0.0280)
University Tier = 4th Quartile $\times$ PostAF1	0.2505*** (0.0044)	0.2501*** (0.0044)	0.2628*** (0.0143)	0.2387*** (0.0074)	0.2470*** (0.0227)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Pseudo R ²	0.884	0.883	0.888	0.899	0.902

(C) Market share (% publications, main authors)

Dependent Variable:	Market Share (% Publications, main authors)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile $\times$ PostAF1	0.1246*** (0.0024)	0.1242*** (0.0017)	0.1288*** (0.0052)	0.1142*** (0.0070)	0.1164*** (0.0096)
University Tier = 3rd Quartile $\times$ PostAF1	0.1532*** (0.0024)	0.1529*** (0.0019)	0.1893*** (0.0195)	0.1398*** (0.0131)	0.1707*** (0.0312)
University Tier = 4th Quartile $\times$ PostAF1	0.2840*** (0.0041)	0.2832*** (0.0038)	0.3096*** (0.0185)	0.2767*** (0.0117)	0.2962*** (0.0300)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Pseudo R ²	0.881	0.880	0.885	0.897	0.901

Table A7: Protein publications in top journals around AlphaFold: alternate estimators

PostAF1 is a binary variable set to one for papers published after the December 2018 CASP13 result. The sample is limited to all protein research papers (both structural biology and non-structural protein research) published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 result. Universities are ranked based on their average TNCS in the period preceding CASP13. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Columns (1)–(2) report Poisson pseudo-maximum-likelihood (PPML) (Santos and Tenreiro, 2006) estimates of annual publication and author-publication counts per university; columns (3)–(4) report fractional-response (Papke and Wooldridge, 1996) logit estimates of annual university-level publication market shares for all authors and for main authors, respectively; and column (5) reports an OLS estimate of the market share measured in percentage points. For the fractional-response columns, average marginal effects (AME) on the market share are reported in percentage points beneath the coefficients. Panel A covers all protein research papers, Panel B is restricted to structural biology, and Panel C to non-structural protein research. All specifications include university and year fixed effects. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Estimator:	PPML (Santos and Tenreiro, 2006)		Frac. logit (Papke and Wooldridge, 1996)		OLS
Dependent Variables:	No. of Publications	No. of Author-Publications	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)	Market Share (% Publications, All Authors)
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.0491*** (0.0021)	0.0520*** (0.0108)	0.0644*** (0.0035)	0.0983*** (0.0040)	0.2545*** (0.0179)
University Tier = 3rd Quartile × PostAF1	0.0988*** (0.0213)	0.0572** (0.0226)	0.1299** (0.0242)	0.1368*** (0.0210)	0.3177*** (0.0224)
University Tier = 4th Quartile × PostAF1	0.1567*** (0.0240)	0.0115 (0.0280)	0.2085*** (0.0238)	0.2273*** (0.0237)	0.3166*** (0.0359)
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted/Pseudo R ²	0.95	0.92	0.93	0.93	0.88
AME: University Tier = 2nd Quartile (pp)			0.52 (0.59)	1.02 (0.87)	
AME: University Tier = 3rd Quartile (pp)			1.09 (0.73)	1.45 (0.95)	
AME: University Tier = 4th Quartile (pp)			1.82 (0.94)	2.52 (1.33)	

(B) Structural biology publications

Estimator:	PPML (Santos and Tenreiro, 2006)		Frac. logit (Papke and Wooldridge, 1996)		OLS
Dependent Variables:	No. of Publications	No. of Author-Publications	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)	Market Share (% Publications, All Authors)
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.1192*** (0.0031)	0.1266*** (0.0089)	0.1342*** (0.0037)	0.1480*** (0.0043)	0.3273*** (0.0155)
University Tier = 3rd Quartile × PostAF1	0.1438*** (0.0197)	0.0159 (0.0253)	0.1644*** (0.0216)	0.1623*** (0.0145)	0.3774*** (0.0232)
University Tier = 4th Quartile × PostAF1	0.2187*** (0.0278)	0.0917*** (0.0330)	0.2497*** (0.0271)	0.2420*** (0.0242)	0.3791*** (0.0393)
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted/Pseudo R ²	0.94	0.91	0.92	0.92	0.87
AME: University Tier = 2nd Quartile (pp)			1.12 (0.77)	1.69 (1.19)	
AME: University Tier = 3rd Quartile (pp)			1.39 (0.87)	1.87 (1.25)	
AME: University Tier = 4th Quartile (pp)			2.20 (1.15)	2.90 (1.65)	

(C) Non-structural protein publications

Estimator:	PPML (Santos and Tenreiro, 2006)		Frac. logit (Papke and Wooldridge, 1996)		OLS
Dependent Variables:	No. of Publications	No. of Author-Publications	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)	Market Share (% Publications, All Authors)
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.0766*** (0.0073)	0.1022*** (0.0140)	0.0904*** (0.0083)	0.1164*** (0.0096)	0.2666*** (0.0216)
University Tier = 3rd Quartile × PostAF1	0.1378*** (0.0268)	0.0165 (0.0349)	0.1715*** (0.0280)	0.1707*** (0.0312)	0.3264*** (0.0221)
University Tier = 4th Quartile × PostAF1	0.1903*** (0.0249)	0.0565 (0.0368)	0.2470*** (0.0227)	0.2962*** (0.0300)	0.3173*** (0.0347)
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted/Pseudo R ²	0.93	0.90	0.90	0.90	0.85
AME: University Tier = 2nd Quartile			0.75 (0.75)	1.24 (1.04)	
AME: University Tier = 3rd Quartile			1.48 (0.95)	1.86 (1.17)	
AME: University Tier = 4th Quartile			2.22 (1.18)	3.45 (1.83)	

Table A8: Construction of the protein-linkage sample for the three estimation panels

This table traces each estimation panel from its top-journal papers to the proteins those papers study. *Linked to proteins* counts papers for which at least one UniProt accession was recovered. A linked paper is *Structure available* when at least half of its proteins had prior known information before the AlphaFold Protein Structure Database release on 22 July 2021, and *No prior structure* otherwise. *Distinct proteins* counts unique accessions pooled over the panel's linked papers. *Rebuild check (ρ)* is the Pearson correlation, across university-year cells, between the publication counts aggregated from the author-affiliation table used for the linkage and the counts in the estimation panel of Table 1.

Panel	Top-journal papers	Linked to proteins	Linked papers by prior structure		Distinct linked proteins	Rebuild check (ρ)
			Structure available	No prior structure		
Panel A: All protein research	10,581	2,859	1,388	1,471	48,314	0.9996
Panel B: Structural biology	5,409	1,625	813	812	43,666	0.9992
Panel C: Non-structural protein research	5,172	1,234	575	659	38,304	0.9998

Table A9: Prior protein information available before the AlphaFold database release

This table classifies the proteins linked to top-journal papers in Table A8 by the structural information available for them before the AlphaFold Protein Structure Database release on 22 July 2021. A protein is *Own structure* if an experimental structure of that protein was deposited in the Protein Data Bank before that date. It is *Template only* if no such structure exists but another protein with at least 30% sequence identity to it was solved before that date. It is otherwise *No prior structure* if neither holds. The three classes are mutually exclusive such that every linked protein falls into one of them. *Mean structural coverage* is the share of residues spanned by the best pre-release structure or template. *With an AlphaFold model* counts proteins for which the database provides a predicted structure. *Mean pLDDT* averages, over the proteins in the row that have an AlphaFold model, the model's mean predicted local distance difference test score across residues. The score runs from 0 to 100 and measures how reliable the predicted position of a residue is, with values above 70 conventionally read as confident.

Structural status	Proteins	Share of proteins (pct)	Mean structural coverage (pct)	Median length (aa)	With an AlphaFold model	Mean pLDDT
Own structure	8,926	18.5	100.0	428.0	8,634	79.8
Template only	7,768	16.1	60.9	494.0	7,183	76.6
No prior structure	31,620	65.4	2.0	428.0	27,945	80.0
Total	48,314	100.0	—	—	43,762	—

Table A10: Protein-linkage coverage by university quartile and by period

This table reports how often the protein linkage of Table A9 succeeds, separately by the pre-AlphaFold publication-share quartile of the university. The unit is the paper-university pair, so a paper co-authored across several universities enters once for each. *Linked to protein* counts pairs whose paper yielded at least one UniProt accession, and *Coverage* is that count as a share of all pairs in the quartile. The final three columns give the same share within each period, where *Pre-AlphaFold* covers 2013 to 2018, *PostAF1* runs from the December 2018 CASP13 result to the July 2021 open release of AlphaFold 2's code and database, and *PostAF2* covers the period after that open release. Coverage varies by 2.3 percentage points across quartiles, which bounds the extent to which linkage success tracks university rank.

Quartile	Paper-university pairs	Linked to Protein	Coverage (%)	Pre-AlphaFold (%)	PostAF1 (%)	PostAF2 (%)
1	5509	1513	27.5	28.3	25.8	25.4
2	6459	1709	26.5	27.9	23.9	23.2
3	7256	1949	26.9	28.1	26.1	22.4
4	8941	2255	25.2	26.5	22.4	23.3

Table A11: Heterogeneity by prior structural-biology authorship

No structural biologist pre-AF is an indicator that the university had no author whose departmental affiliation is in structural biology, computational biology, or biophysics on any of its top-journal protein papers published before AlphaFold 1. Each column adds a triple interaction between quartile, period, and that indicator to the corresponding stratum-specific specification. A positive triple means that the lower-quartile differential was larger where no such author had previously been observed.

Dependent Variable:	Log(1 + No. of publications)					
	Structure available			No prior structure		
	All protein research	Structural biology	Non-structural	All protein research	Structural biology	Non-structural
Model:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
2nd Quartile × PostAF1 × No structural biologist pre-AF	0.0907 (0.1623)	0.0962 (0.2115)	0.1139 (0.1482)	-0.0562 (0.2125)	0.0440 (0.2231)	0.0523 (0.1393)
3rd Quartile × PostAF1 × No structural biologist pre-AF	0.0414 (0.0884)	0.0378 (0.0812)	0.0495 (0.0767)	-0.0716 (0.0862)	-0.0411 (0.0909)	0.0024 (0.0623)
4th Quartile × PostAF1 × No structural biologist pre-AF	0.0574** (0.0249)	0.0508** (0.0246)	0.0328 (0.0377)	0.0407 (0.0316)	0.0367 (0.0295)	0.0496 (0.0459)
2nd Quartile × PostAF2 × No structural biologist pre-AF	0.4989*** (0.0994)	0.5195*** (0.1338)	0.3281*** (0.1075)	0.1075 (0.1055)	0.1761 (0.1511)	0.0587 (0.1109)
3rd Quartile × PostAF2 × No structural biologist pre-AF	0.3004*** (0.0784)	0.2941*** (0.0696)	0.0971 (0.0667)	0.1726** (0.0744)	0.1784** (0.0753)	0.1192* (0.0640)
4th Quartile × PostAF2 × No structural biologist pre-AF	0.1257*** (0.0229)	0.1234*** (0.0221)	0.1183*** (0.0377)	0.0980*** (0.0249)	0.1021*** (0.0223)	0.1048*** (0.0373)
<i>Fixed-effects</i>						
University	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	5,500	5,500	5,500	5,500	5,500	5,500
Adjusted R ²	0.60351	0.58815	0.51433	0.59550	0.57849	0.52765

Table A12: Changes in structural-biologist authorship by university quartile

The dependent variable is an indicator taking the value one if a top-journal publication has at least one structural biologist among its authors. A negative estimate means that universities from a given quartile are less likely to include a structural biologist on their top-journal publications than first quartile; relative to the pre-AF gap with the top quartile.

Dependent Variable:	Paper has a structural biologist		
	All protein research	Structural biology	Non-structural
Model:	(1)	(2)	(3)
<i>Variables</i>			
2nd Quartile × PostAF1	0.0217 (0.0433)	-0.0277 (0.0527)	0.0327 (0.0445)
3rd Quartile × PostAF1	-0.0189 (0.0422)	-0.0454 (0.0538)	0.0325 (0.0445)
4th Quartile × PostAF1	-0.0191 (0.0477)	-0.0210 (0.0566)	0.0559 (0.0452)
2nd Quartile × PostAF2	-0.0538 (0.0461)	-0.0216 (0.0547)	-0.0004 (0.0388)
3rd Quartile × PostAF2	-0.0789 (0.0498)	-0.0875 (0.0605)	-0.0517 (0.0400)
4th Quartile × PostAF2	-0.1022** (0.0486)	-0.0905* (0.0546)	-0.0659* (0.0370)
<i>Fixed-effects</i>			
University	Yes	Yes	Yes
Year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	7,373	6,103	4,537
Adjusted R ²	0.10763	0.12092	0.12397

B Sample Selection and Methods

B.1 Publications

Data on publications are obtained from OpenAlex.⁹ Each publication is assigned a unique identifier, and its metadata include the title of the article, the abstract, citation counts, the list of authors with their institutional affiliations and departmental affiliation, and the range of topics covered. We use OpenAlex rather than Scopus or Web of Science because it is the only source that combines (i) free, fully snapshotted access ensuring reproducibility, (ii) coverage of preprints, which is material in the post-AlphaFold period, (iii) author and institution disambiguation, and (iv) topic classifications required for our domain definitions. Direct comparisons of OpenAlex with the proprietary databases have since been published: [Alperin et al. \(2024\)](#) find that the OpenAlex record set contains that of Scopus in full, while [Culbert et al. \(2025\)](#) report reference coverage for 2015–22 publications comparable to that of Scopus and Web of Science.

B.2 Journals

We select journals in ‘biochemistry, genetics, molecular biology’ (2,170 journals) and ‘Multidisciplinary’ (175 journals), and label those with an average SJR greater than 10 between 2013 and 2018 as ‘top journals’ (there are 17 top journals in total, as listed in Table A2). Figure A4 shows that the number of publications in the 17 top journals is relatively stable in our sample period. We split the articles published in these journals into three groups labeled as follows: i) ‘Structural biology’, ii) ‘Non-structural protein’, and iii) ‘Non-protein.’ A paper is classified as structural biology if it has an author who is a structural biologist or if the paper’s list of topics includes one of the 238 topics we identify as structural biology. If a paper does not satisfy this condition but has one of the 1,066 topics of protein research, then it falls in the second group. All other papers are assigned to the third group.

We measure journal quality using the SCImago Journal Rank (SJR), a prestige metric that weights citations according to the importance of the citing journals. Computed from Scopus data, SJR incorporates both citation volume and the prestige of citing sources to identify the most influential journals. As the SJR of each journal varies from year to year,

⁹OpenAlex is maintained by the nonprofit OurResearch, which launched it in January 2022 as a fully open successor to the Microsoft Academic Graph after Microsoft stopped updating that database at the end of 2021. The catalog aggregates bibliographic metadata from Crossref, ORCID, PubMed, and open repositories such as arXiv and Zenodo, which gives it coverage of preprints alongside published articles. OpenAlex provides API access and full database snapshots for large-scale analysis.

we rank the journals using the average SJR during the six-year period preceding the release of AlphaFold (i.e., 2013–2018).

We choose SJR over alternatives such as the Journal Impact Factor (JIF) for four reasons. First, SJR is a prestige-weighted measure (PageRank-style on the Scopus citation graph), so it captures journal standing rather than raw citation volume and is less sensitive to mechanical inflation from review articles or self-citation. Second, SJR is field-normalized, which permits us to pool biochemistry/molecular-biology and multidisciplinary outlets on a single comparable scale. Third, SJR is computed from Scopus, with publicly documented methodology, and is freely accessible, ensuring reproducibility (JIF is paywalled and proprietary). Fourth, averaging over the six-year pre-AlphaFold window (2013–2018) removes year-to-year noise and avoids any post-treatment contamination of the journal set. The resulting top-17 set is dominated by Nature, Science, Cell, and other outlets that are uncontroversially ‘top’ under any standard alternative metric (JIF, h5-index, expert consensus), so the choice of SJR is unlikely to drive our results.

B.3 University Rankings

The center for Science and Technology Studies (CWTS) at Leiden University produces the *Total Normalized Citation Score* (TNCS) of about one thousand universities around the world. We compute the average TNCS score of each university between 2013 and 2018 (the pre-AlphaFold period), in the category “biomedical and health sciences”.¹⁰

We rely on the CWTS Leiden Ranking for institutional stratification for several reasons. First, it is produced by an independent academic center with a publicly documented methodology and is the standard reference in the scientometrics literature. Second, the TNCS is normalized at the subfield level, so universities are not penalized for working in subfields with structurally lower citation rates. Third, the TNCS is size-dependent ($\text{MNCS} \times \text{\#pubs}$), which matches what we want to capture: an institution’s *capacity* to produce top-journal output, not its per-paper average. Fourth, restricting to “biomedical and health sciences” aligns the ranking with the protein-research scope of our analysis. Fifth, averaging over the six-year pre-AlphaFold window yields a fixed, time-invariant ranking, which rules out any mechanical sorting of universities in or out of quartiles

¹⁰CWTS computes the TNCS annually for each university as follows: First, each publication is assigned to a biomedical subfield. Then, the citation count for each publication is compared to the average in the subfield in that year, resulting in a normalized citation score, where 1 represents the overall average. The *Mean Normalized Citation Score* (MNCS) is then calculated by averaging these normalized scores across all the publications in the university. Finally, the TNCS is obtained by multiplying the MNCS by the total number of publications, providing a size-dependent indicator of the university’s overall citations. This method thus accounts for differences in citation practices between subfields and across years.

caused by AlphaFold itself. Using the TNCS ranking first, rather than journal market share, ensures that our analysis based on top-journal share also aligns with more traditional university rankings. As a robustness check, we also rank universities by their pre-AlphaFold top-journal market share; results are similar.

We match authors of each publication in our sample to universities based on their affiliation. In our data, some authors are not affiliated with universities, but instead with research centers, private companies, or hospitals. These researchers are dropped from the sample.

B.4 Measuring Research Output

We define the annual publication count for university u in year t as the number of protein research articles published in our selected top journals with at least one author affiliated with university u .

Let

$$\mathcal{P}_t = \{i : \text{article } i \text{ is published in a top journal in year } t\},$$

and for each article i , let

$$A_i = \{a_{i,1}, a_{i,2}, \dots, a_{i,n_i}\}$$

denote its ordered set of authors. Denote by $u(a)$ the university assigned to author a (i.e., the highest ranked among any multiple affiliations).

Then, the publication counts are defined as follows:

1. All authors count:

$$P_{u,t}^{all} = \sum_{i \in \mathcal{P}_t} \mathbf{1}\left\{\exists k \in \{1, \dots, n_i\} : u(a_{i,k}) = u\right\},$$

2. Main authors count:

$$P_{u,t}^{main} = \sum_{i \in \mathcal{P}_t} \mathbf{1}\left\{u(a_{i,1}) = u \quad \text{or} \quad u(a_{i,n_i}) = u\right\}.$$

These definitions ensure that each article contributes at most one count per university, regardless of the number of affiliated authors.

B.5 Additional Data

We construct several author- and university-level covariates to use as controls in our regressions.

OpenAlex does not track demographic characteristics or detailed expertise fields of authors. We therefore infer authors' ethnicities using the *rethnicity* package in R, which predicts ethnicity from first names via probabilistic models trained on demographic data. Gender is determined using the *genderizeR* package in R, based on first names. Finally, we classify authors as structural biologists if their primary institutional affiliation is in a department specializing in structural biology, computational biology, biophysics, or closely related fields (e.g., department names containing keywords such as "structural biology," "computational biology," "biophysics," or "protein folding").

To examine heterogeneity related to pre-existing computational resources (a key institutional barrier to conducting protein research that AlphaFold may substitute for), we incorporate data on university-level access to high-performance computing (HPC). HPC plays a pivotal role in modern protein research, enabling complex simulations, large-scale data analyses, and computationally intensive methods such as those traditionally employed in structural biology. Access to HPC resources is crucial for improving researchers' productivity and enabling the exploration of a broader range of research topics ([McKeon and Boudou, 2024](#)). We use data from Top500.org as a proxy for the computational power available at academic institutions. Since 1993, Top500.org has systematically tracked HPC facilities at universities worldwide on a biannual basis, providing detailed metrics on total processing power (quantified by the number of processor cores) and computational speed (measured in petaflops per second). This dataset allows us to assess the link between a university's HPC capabilities and its research output.

B.6 Empirical Design

We rank universities by their cumulative share of protein publications in our list of top journals during the pre-AlphaFold period and divide them into quartiles, such that each quartile accounts for approximately 25% of all such publications during this period. The top quartile comprises the top 12 universities. The second quartile includes the next 36 universities, the third quartile the next 88, and the bottom quartile comprises the bottom 364.

In our main regression specification outlined in equation 4, we include university-level controls for faculty size, author demographics, structural biology expertise, institutional type, and geographic location that vary over time (summarized in Table A3) to account

for changes in author composition over time. The dependent variable y_{ut} is one of three variables:

1. The log number of publications in top journals from university u at time t : $y_{ut} = \log(1 + P_{u,t}^{all})$
2. The market share of university u at time t : $y_{ut} = MktS_{ut}^{all}$
3. The market share of university u at time t taking into account only main authors is: $y_{ut} = MktS_{u,t}^{main}$

B.7 Measures of Research Direction and Impact

We employ three metrics to quantify changes in research direction: *pivot score*, *novelty*, and *disruptiveness*. These measures capture distinct aspects of topical focus, research novelty, and impact, allowing us to evaluate how AlphaFold influenced the direction and originality of protein research across university quartiles.

Pivot Score

The *pivot score* measures the extent to which a researcher’s new work diverges from their previous research portfolio. We compute this score using cosine similarity between the distribution of journals cited in a new paper and those cited within the researcher’s prior work.

$$\Phi_i^j = 1 - \frac{R_i^j \cdot R_i}{\|R_i^j\| \|R_i\|} \quad (5)$$

where R_i^j is a vector representing citation frequencies to journals in focal paper j by researcher i , and R_i aggregates citation frequencies to journals across all prior papers by the same researcher i . The score ranges from 0 (no departure from prior work) to 1 (complete shift in research direction).

Novelty

Novelty quantifies a paper’s originality by assessing the rarity of concepts in its references, drawing on OpenAlex’s concept classifications. OpenAlex tags each paper with a list of concepts, assigning each a relevance score (ranging from 0 to 1) that indicates the concept’s

centrality to the paper. Each concept also includes a works count reflecting its overall prevalence in the literature, which we use to measure rarity in our novelty metric.¹¹

$$N = \frac{1}{|C|} \sum_{c \in C} \frac{1}{w_c} \quad (6)$$

where C is the set of unique concepts from the paper's references with relevance score > 0.5 , and w_c is the works count of concept c . The 0.5 threshold ensures only concepts meaningfully connected to the references contribute to the score. The inverse weighting $\frac{1}{w_c}$ assigns greater importance to rarer concepts, so papers referencing niche topics with low works counts (e.g., 50 papers) receive higher novelty scores than those relying on widely studied topics (e.g., 10,000 papers). For example, a paper on neural networks might have high relevance for "machine learning" (relevance = 0.9) but low relevance for "agriculture" (relevance = 0.2), with only the former contributing to the novelty calculation.

Disruptiveness

We measure a paper's *disruptiveness* using the "Disruption Index", which quantifies the extent to which a paper disrupts its field by analyzing citation patterns. Higher values indicate that subsequent works cite the focal paper while largely ignoring its references (signaling the creation of new research directions), whereas low values reflect consolidation of existing lines of inquiry.

$$DI = \frac{i_{\text{est}} - j_{\text{est}}}{i_{\text{est}} + j_{\text{est}} + k_{\text{est}}} \quad (7)$$

where i_{est} represents papers citing only the focal paper (indicating standalone impact), j_{est} represents papers citing only its references (reflecting influence of prior work), and k_{est} represents papers citing both (showing overlapping influence). These estimates are calculated from: (i) A , the focal paper's citation count; (ii) B , the number of papers citing at least one reference; and (iii) p_k , the proportion of sampled citing papers that also cite a reference. The estimates follow: $i_{\text{est}} = (1 - p_k) \cdot A$, $k_{\text{est}} = p_k \cdot A$, and $j_{\text{est}} = B - k_{\text{est}}$.

The Disruption Index ranges from -1 to 1. Positive values indicate disruptive papers that attract citations largely independent of their references (thus overshadowing prior work), while negative values denote consolidating papers whose citations tend to include both the focal paper and its predecessors. Zero values represent neutral impact.

¹¹The relevance score is computed by OpenAlex using machine learning techniques applied to metadata such as abstracts, titles, and citation patterns. Scores above 0.5 indicate concepts that are significant descriptors of the paper's content rather than peripheral topics.

Scaling of measures for comparability

To ensure temporal comparability and account for evolving trends in protein research, such as shifts in concept prevalence, citation patterns, and publication volumes, we normalize novelty and disruptiveness measures quarterly across all universities as follows:

$$\text{Scaled Measure} = \frac{\text{Measure} - \min(\text{Measure})}{\max(\text{Measure}) - \min(\text{Measure})}$$

Here, $\min(\text{Measure})$ and $\max(\text{Measure})$ denote the minimum and maximum values of the given metric observed across all universities in a given quarter. This min-max normalization rescales the raw measures to the $[0, 1]$ interval within each quarter. The pivot score, being derived from cosine similarity, is inherently normalized and requires no further scaling.

B.8 Sample Construction for Placebo Analysis

To replicate our analysis in other fields and confirm whether the observed effects are specific to AlphaFold or part of broader trends in academic publishing or concurrent AI advancements, we select four additional high-skill research domains: microbiology (excluding protein research), economics and finance, materials science, and social sciences. These fields exhibit similar hierarchies in research output but lacked a transformative open-access AI tool equivalent to AlphaFold during the study period. The data collection procedures mirror those used for protein research, with field-specific adjustments as needed.

Data sources

As with our protein research analysis, we draw on three primary data sources:

1. **Journal rankings:** Journal quality is measured using the SCImago Journal Rank (SJR), which accounts for both citation volume and the prestige of citing journals. We collect yearly SJR data from 2013 to 2024 for journals in each field. Since 2025 SJR rankings were not available at the time of analysis, we use the 2024 rankings for publications in 2025.
2. **Publications:** We obtain publication metadata including article titles, publication dates, author information, institutional affiliations, citation data, and topic classifications from OpenAlex. Publications are extracted separately from OpenAlex for the top journals in each field, ranked by their annual SJR as determined in Step 1.

3. **University rankings:** We construct university rankings using the Total Normalized Citation Score (TNCS). The rankings cover approximately 1,000 universities worldwide and are reported annually across multiple academic fields.

Journals selection

For each field, we identify top journals using SCImago's subject category classifications and rank them by their SJR scores for each year, selecting those with the highest values.

- **Microbiology excluding protein research:** Publications classified as microbiology after excluding protein research, using the same top-journal and university-quartile construction as the main analysis.
- **Materials science:** Journals classified under "Materials Science" subject areas in SCImago, with an SJR threshold of 10 or higher, consistent with our protein research sample.
- **Economics:** Journals classified under the "Economics, Econometrics and Finance" subject areas in SCImago. We apply an SJR threshold of 10 or higher to identify top journals.
- **Social sciences:** Journals classified under the "Social Sciences" category in SCImago. To account for lower typical citation rates in the social sciences, which result in generally lower SJR scores compared to fields like protein or materials science, we select journals with an SJR of 8 or higher to capture a comparable set of high-impact outlets.

We compute journal rankings annually based on their SJR scores, with journals ranked within each year using dense ranking (ties receive the same rank). To maintain field specificity, we exclude publications from general interest or review-oriented journals (e.g., *Journal of Economic Perspectives*, *Annual Review of Economics* and *Journal of Economic Literature* in economics). We also exclude journals primarily focused on adjacent disciplines, even if they appear in the field's SCImago category. For example, in the social sciences category, we exclude the *Review of Economics and Statistics*, which is a leading economics journal. The resulting selection of top journals for these fields, along with their annual publication volumes over time, is presented in Figure [A12](#).

University rankings

These are constructed using field-specific TNCS scores from the CWTS Leiden Rankings:

- For materials science, we use the “Physical sciences and engineering” category.
- For economics and social sciences, we use the “Social sciences and humanities” category.

This ranking employs full-counting methodology, which assigns a full weight of one to each publication regardless of the number of co-authoring institutions. The distribution of top-journal publications based on our ranking scheme can be found in Figure A13. The cumulative distribution of top-journal publications across university ranks in these fields, based on our pre-AlphaFold ranking scheme, is illustrated in Figure A13. The figure reveals persistent stratification in publication shares across the top 500 universities in these fields, underscoring that lower-ranked universities do not hold significant publication shares in top journals. This concentration aligns with the Matthew effect observed in protein research prior to AlphaFold’s release.

As in protein research, we fix university quartiles using pre-AlphaFold top-journal publication shares and examine how the shares of these fixed groups evolve in the post-AlphaFold period¹².

Author-institution matching

For each publication in the sample, we extract author-level information including institutional affiliations from OpenAlex. Authors with multiple affiliations are assigned to the highest-ranked institution among their affiliations based on the fixed, field-specific pre-AlphaFold TNCS ranking. This ensures each author-publication observation is associated with a single university quartile.

Sample coverage

The final samples span publications from 2013 to 2025 for authors affiliated with the top 500 universities in each field. Publications are restricted to those appearing in journals with an SJR at or above the field-specific threshold throughout the sample period. The

¹²One might view the public release of ChatGPT in November 2022 as a potential discrete shock for fields such as economics or social sciences. If ChatGPT had triggered significant shifts in university publication shares in top journals, particularly gains by lower-quartile universities, these changes would be visible in our placebo analysis because universities remain assigned to the quartile determined before AlphaFold. Post-release publication shares are therefore not mechanically reclassified using later rankings.

distribution of publications across university quartiles is computed annually, enabling the construction of market share time series comparable to those in our main analysis on protein research.

Research quality metrics

To investigate whether the shifts in research practices observed in protein research also occurred in non-protein fields, we compute the same three metrics *pivot score*, *novelty*, and *disruptiveness* for publications in each field. All these metrics are computed using identical methodologies to those employed for protein research (see Section [B.7](#) for detailed definitions).

C Identifying the Proteins Studied by Each Publication

The main results compare post-AlphaFold changes in top-journal protein publication across university quartiles. This section supplements them by splitting the sample according to what was already known pre-AlphaFold about the proteins each paper studies. A protein has prior structural information if, before the AlphaFold database release on 22 July 2021, it had its own experimental structure or at least one usable homology template.¹³ A paper is deemed to have prior protein structure available when at least half of its linked proteins meet this condition, and no prior protein information otherwise. We ask whether the post-AlphaFold changes across quartiles differ between these two kinds of papers.

Both definitions presume a known set of proteins for each paper, which publications may or may not report. A paper may study a single protein, several, or none, and the proteins it studies are named in prose rather than recorded in any field of its bibliographic record. Recovering that set is the substantive task of this section, which describes how we link publications to proteins, construct the paper-level measure of prior protein information, and merge it into the estimation sample.

Formally, let $\mathcal{D}$ denote the set of publications in the sample. Proteins are identified by their accession number in UniProtKB, the reference database of protein sequences and annotations. We use only the human-reviewed section of UniProtKB, known as Swiss-Prot, for our purposes. Let $\mathcal{U}$ denote the set of these reviewed accessions. The first object we construct is a correspondence:

$$\mathcal{P} : \mathcal{D} \longrightarrow 2^{\mathcal{U}}, \quad p \longmapsto \mathcal{P}(p) \subseteq \mathcal{U}, \quad (8)$$

assigning to each paper p the (possibly empty) set of proteins it studies. The main difficulty here is that publications do not report accession numbers in any systematic machine-readable form. Proteins can be referred to by gene symbols, common names, synonyms, and abbreviations, and the same string frequently denotes different proteins in different organisms. We therefore identify proteins through three complementary channels—curator-linked literature, machine-normalized entity annotations, and an ensemble of named-entity recognition models applied to abstracts—and take their union. The channels differ in what they require of a paper and in how directly they establish a link, with each one linking papers to proteins in unique ways. Coverage matters directly for the analysis. A paper that cannot be linked to any protein drops out of every protein-level specification.

¹³Proteins with similar sequences fold into similar shapes. A homology template is an experimentally solved structure of a related protein that can stand in for the target, and the percentage sequence identity between the two represents how far it can be trusted. We treat a template as usable at 30 percent identity or above, the conventional threshold below which a template no longer supports molecular replacement.

Unlinked papers therefore shrink the estimation sample rather than merely adding noise. We therefore treat coverage as an object to be documented rather than assumed, and report the corresponding diagnostics.

Identifier resolution in publications

We begin from the set of OpenAlex works published in the 17 top journals identified in Section 2 and listed in Table A2. For each work we use its external identifiers (PubMed identifier and DOI) and its abstract. Almost all publications in our sample carry a PubMed identifier, so the two identifier-based channels described below apply to nearly the entire sample; the third channel does not require one.

An important point to note is that OpenAlex does not distribute abstracts as running text. It stores an inverted index that maps each token to the positions at which the token occurs. Writing $\mathcal{I}_p = \{(w, T_w^p)\}_{w \in V_p}$, where V_p is the vocabulary of the abstract of paper p and $T_w^p \subseteq \{1, \dots, n_p\}$ is the set of positions at which token w appears, the abstract is recovered as the sequence $a_p = (a_p(1), \dots, a_p(n_p))$ defined by

$$a_p(t) = w \iff t \in T_w^p, \quad \text{with} \quad n_p = \sum_{w \in V_p} |T_w^p|. \quad (9)$$

Before performing entity recognition we normalize the reconstructed text by converting to lower-case, collapsing repeated whitespaces, and stripping non-alphanumeric characters. We denote the normalized abstract by $\tilde{a}_p$.

The protein universe

The "universe" of proteins $\mathcal{U}$ is the manually reviewed section of UniProtKB, known as Swiss-Prot, which contains just over 570,000 entries curated manually by expert biocurators (The UniProt Consortium, 2025). Each of its entries describes a single protein and is labeled by a unique accession number. The terms *entry*, *accession* and *protein* therefore refer to the same object throughout this section. We exclude TrEMBL, the computationally annotated section, whose entries have not been reviewed. Restricting attention to human-reviewed entries is consequential for the linkage. It keeps the set of candidate proteins fixed and biologically meaningful, and prevents a single ambiguous name from matching thousands of unreviewed sequence records.

Rather than querying the UniProt API once per paper or once per protein mention, we download the entire reviewed index once through the bulk stream endpoint <https://rest.uniprot.org/uniprotkb/stream?query=reviewed:true&format=tsv>, with the *fields* parameter

Table C1: Fields retrieved from the reviewed UniProtKB index

Field	Content	Role in the analysis
Accession	Primary UniProt identifier	Unit of the protein-level data
Gene name	Primary gene symbol	Matching; reporting
Protein name	Recommended full name	Matching; reporting
Organism	Source species	Disambiguation across species
Sequence length	Number of amino acids	Protein characteristics
Molecular mass	Mass in daltons	Protein characteristics
First public date	Date the entry entered UniProtKB	Entry age
PDB cross-references	Accessions of linked PDB entries	Key to PDB entries
AlphaFold DB link	Linked predicted structure	Availability of a prediction
NCBI Gene identifier	Cross-reference to NCBI Gene	Channel 2 linkage key
Cited PubMed IDs	Curated literature references	Channel 1 linkage key

Notes. The *fields* parameter of the stream request is set to *accession, gene_primary, protein_name, organism_name, length, mass, date_created, xref_pdb*, and *xref_alphafolddb*; *xref_geneid* and *lit_pubmed_id* are retrieved in two further streams. Cross-references and cited PubMed identifiers are multi-valued and are expanded into separate rows in the cross-reference table described in Section C. PDB cross-references carry no dates: deposit and release dates are obtained from the Protein Data Bank, and candidate homology templates with their sequence identities from SWISS-MODEL and UniProt.

set to the columns listed in Table C1. Two further streams are run against the same endpoint with *fields=accession,xref_geneid* and *fields=accession,lit_pubmed_id*. The first returns each accession’s cross-references to NCBI Gene, the gene catalog maintained by the National Center for Biotechnology Information. Those cross-references are the linkage key for Channel 2 below. The second returns the PubMed identifiers cited by the curated entry, which are the linkage key for Channel 1. Both fields are multi-valued, which is why each is retrieved on its own and expanded to one row per pair. The three streams were downloaded in May 2026, and every step described below runs against that stored snapshot.

Linkage channels

Channel 1: curator-linked literature. Each reviewed entry carries a list of publications a curator read and used as evidence when annotating the protein. Inverting this list gives, for each paper, the proteins whose entries cite it. Candidate papers are screened on three criteria: that the protein is a central object of the study rather than a passing mention, that the findings reported are experimental rather than computationally inferred, and that they are not already recorded in the entry from an earlier paper. Reviews are excluded, since they report no new experimental evidence.

Let L_i denote the set of PubMed identifiers cited by the curated entry of protein i , and let q_p denote the PubMed identifier of paper p . The curator-linked channel assigns to p every protein whose entry cites it

$$\mathcal{P}^{\text{cur}}(p) = \{ i \in \mathcal{U} : q_p \in L_i \}, \quad (10)$$

This is the most precise of the three channels. A link exists only because a curator has judged that the paper reports experimental evidence about that protein. Its recall is bounded by two features of curation. The first is selectivity: an entry cites the papers that established its annotations, not every paper that studies the protein. The second is timing. A paper enters L_i only once a curator has processed it, and the resulting lag falls hardest on the most recent publications, consisting mainly of those in the post-AlphaFold window.

Channel 2: normalized gene annotations. A curator-linked citation is the most direct evidence that a paper studies a protein, but it exists only where a biocurator has read the paper. Curation may also lag the publication by a long period. The AlphaFold database was released in July 2021, meaning the papers on which the analysis depends are precisely those least likely to have been curated. Coverage from Channel 1 alone would therefore drop significantly in the post-AlphaFold period, with the resulting measurement error correlated with the timing that identifies the effect rather than averaging out across periods.

The second channel is used to overcome this issue. It draws on PubTator3, a resource maintained by NCBI that locates gene and protein names in text and maps them to NCBI Gene identifiers, applied to roughly 36 million PubMed abstracts and 6 million open-access full texts in PubMed Central and refreshed weekly (Wei et al., 2024). Two features here suit our purpose very well. First, annotation is applied to a huge corpus of publications rather than a small curated subset, a weekly refresh rate which places recent papers on the same footing as older ones. Second, PubTator3 reports each mention as an NCBI Gene identifier rather than as the string it found in the text. Every reviewed entry records its own NCBI Gene cross-references. Mapping an annotated gene into $\mathcal{U}$ is therefore an exact lookup, not a match on names and synonyms, and the homonymy and cross-species ambiguity that affect string matching quality do not arise. For papers with an open-access full text in PubMed Central, the procedure draws on the body of the paper rather than the abstract alone.

Let G_p denote the set of gene identifiers PubTator3 annotates in paper p , and let $A(g) \subseteq \mathcal{U}$ denote the reviewed entries cross-referenced to gene g , read off the accession-to-gene cross-references downloaded with the index. The annotation channel assigns to p every protein cross-referenced to a gene as:

$$\mathcal{P}^{\text{tm}}(p) = \bigcup_{g \in G_p} A(g). \quad (11)$$

This matching comes at the cost of over-inclusion. A gene mentioned in passing enters G_p on an unweighted basis alongside a focal gene, making $\mathcal{P}^{\text{tm}}(p)$ a superset of the proteins

that paper p actually studies. The paper-level share of linked proteins with prior structural information, is calculated as a sample average over $\mathcal{P}^{\text{tm}}(p)$. Including unstudied proteins in this set pulls the paper’s average toward the unconditional sample mean. Because a paper is assigned to the structure-available stratum when this share reaches 0.5, over-inclusion introduces classification noise near the cutoff, attenuating the observed differential and rendering our estimates of the stratum gap conservative.

Channel 3: entity recognition over abstracts. The first two channels yield a non-empty $\mathcal{P}^{\text{cur}}(p)$ only for papers a curator has cited, and a non-empty $\mathcal{P}^{\text{tm}}(p)$ only for papers PubTator3 has annotated. The third channel described here places no such requirement on the paper. It covers papers without a PubMed identifier, as well as papers that PubMed indexes but that no database has linked to a protein. Instead, the entity recognition approach identifies candidate proteins from the abstract by applying an ensemble of K biomedical named-entity recognition models. These models are distributed by the *scispaCy* repository, and are applied to the normalized abstract $\tilde{a}_p$ (Neumann et al., 2019). The models used are listed in Table C2. Each is trained on a different annotated corpus and labels a different set of entity types. For instance, the model trained on the BioNLP 2013 Cancer Genetics corpus labels protein families and protein domains in addition to gene products, whereas the model trained on JNLPBA labels proteins and genes alone. A mention labeled by one model is therefore not necessarily labeled by another, and the union defined in Equation (12) may sometimes retain more mentions than any single model returns.

Let $\mathcal{E}_k(\cdot)$ denote the set of labeled spans returned by model k and $\mathcal{L}$ the set of protein-relevant labels. The candidate mention set is defined as:

$$M_p = \bigcup_{k=1}^K \{ s : (s, \ell) \in \mathcal{E}_k(\tilde{a}_p), \ell \in \mathcal{L} \}. \quad (12)$$

Each mention s is matched against the recommended names, gene symbols, and synonyms recorded in the reviewed index, and the entry with the highest match score is retained as the protein the mention denotes:

$$\rho(s) = \arg \max_{i \in \mathcal{U}} \omega(s, i), \quad \mathcal{P}^{\text{ner}}(p) = \{ \rho(s) : s \in M_p \text{ with a match} \}, \quad (13)$$

where $\omega(s, i)$ is the UniProtKB relevance score of entry i for query string s and only the best-ranked entry is retained. Two restrictions in this step matter a great deal in practice. Limiting the search to reviewed entries and keeping a single best match prevents one

Table C2: Named-entity recognition models and retained labels

Model	Annotation corpus	Retained entity labels
en_ner_bionlp13cg_md	BioNLP 2013 Cancer Genetics	GENE_OR_GENE_PRODUCT, AMINO_ACID, PROTEIN_FAMILY_OR_GROUP, PROTEIN_DOMAIN_OR_REGION
en_ner_jnlpba_md	JNLPBA molecular biology abstracts	PROTEIN, GENE
en_ner_craft_md	CRAFT full-text biomedical articles	GGP (gene or gene product)
en_ner_bc5cdr_md	BioCreative V chemical–disease	none retained
en_core_sci_scibert	Scientific text (SciBERT encoder)	general entity spans

Notes. The ensemble is applied to the normalized abstract and the retained spans are unioned across models. The chemical–disease model contributes no labels to $\mathcal{L}$: its spans identify chemicals and diseases, which are excluded from the candidate set rather than resolved against UniProtKB. Retained spans are subsequently resolved by lexical search restricted to reviewed entries, keeping the best-ranked match per mention.

mention from generating links to the counterpart of the same protein in every species that has one, which would inflate $|\mathcal{P}(p)|$ without adding information. Additionally, restricting the retained labels to $\mathcal{L}$ serves the same purpose. The models also detect cell types, chemicals, and organism names, which are discarded rather than resolved against the reviewed index. The entity-recognition channel is used because any link it establishes rests on a name appearing in an abstract rather than on a curator’s judgment or a normalized annotation. It is the only channel that does not require the paper to have been indexed and annotated elsewhere, and can therefore link a paper to one or more proteins that the other two methods may have left unlinked.

Paper p is linked to protein i if at least one of the three channels assigns i to p :

$$\mathcal{P}(p) = \mathcal{P}^{\text{cur}}(p) \cup \mathcal{P}^{\text{tm}}(p) \cup \mathcal{P}^{\text{ner}}(p), \quad (14)$$

A paper linked to the same protein by more than one channel contributes only one unique (paper, protein) pair. For any subset of channels C , define the coverage rate:

$$c(C) = \frac{1}{|\mathcal{D}|} \sum_{p \in \mathcal{D}} \mathbf{1} \left\{ \bigcup_{c \in C} \mathcal{P}^c(p) \neq \emptyset \right\}, \quad (15)$$

the share of publications linked to at least one protein. Because (14) is a union, coverage is monotone in the set of channels. The incremental contribution of the k -th channel,

$$\Delta_k = c(\{1, \dots, k\}) - c(\{1, \dots, k-1\}) \geq 0, \quad (16)$$

is weakly positive by construction. Taking the channels in the order above, each Δ_k is strictly positive in our data: every channel links papers that the preceding ones leave

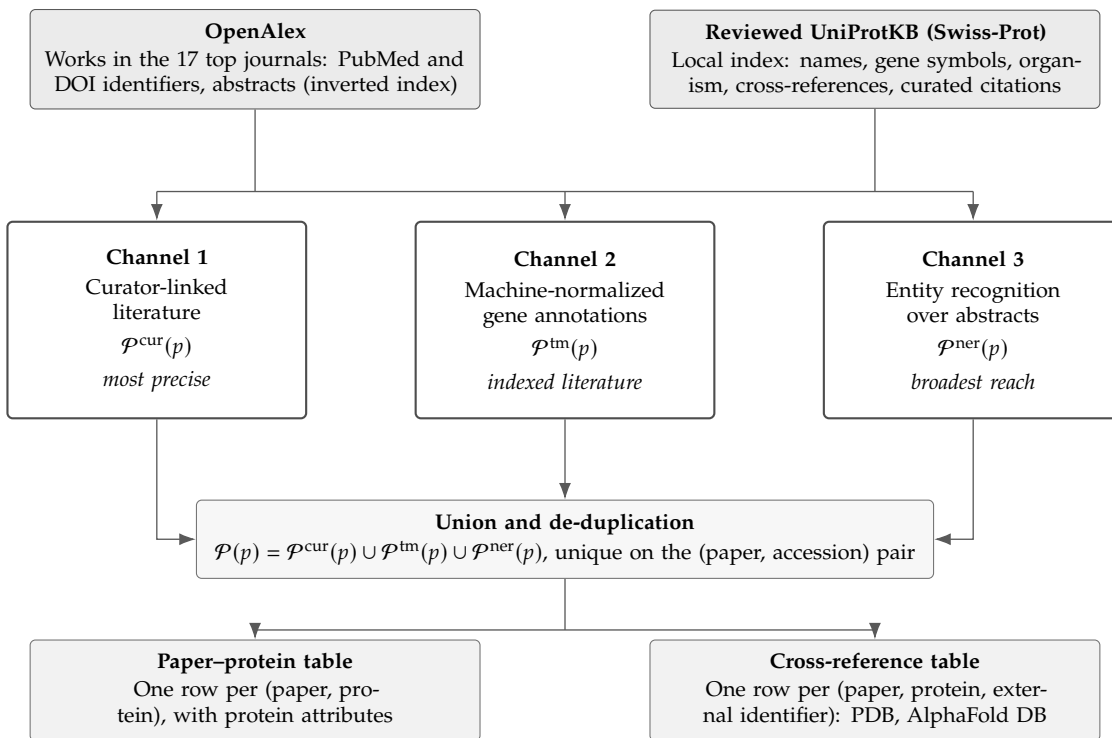


Figure C1: Construction of the publication-protein linkage

Notes. Both data sources feed all three linkage channels. Channels are ordered from the most precise (curator-linked citations, in which a biocurator has judged that the paper provides evidence about the protein) to the most permissive (named-entity recognition over abstracts, resolved lexically against reviewed UniProtKB entries). A paper is linked to a protein if *any* channel establishes the link, so coverage is weakly increasing in the set of channels used.

unlinked. We therefore run all three rather than restricting the linkage to curator-established links.

Output tables and join keys

Figure C1 summarizes the paper-protein linkage procedure using the three channels described above. The procedure produces two tables. The first records one row per paper-protein pair, together with the protein’s attributes: accession, gene symbol, recommended name, date of first public release, sequence length, and molecular mass. The second records one row per paper-protein-external identifier tuple, covering cross-references to the Protein Data Bank (PDB) and to the AlphaFold Protein Structure Database. The second table is the input to the treatment: the PDB cross-references the table contains supply the deposit and release dates from which the pre-AlphaFold structural classification in (17) to (19) is built. The AlphaFold links the table records are, however, not used in that classification, and serve only to attach the predicted structure and its confidence score to each protein.

The sources we combine record the same publication or protein in different identifier

formats, and a merge would treat two formats of one identifier as two distinct records.¹⁴ We therefore reconfigure every identifier into a consistent canonical form before merging. Publication identifiers are reduced to the OpenAlex work identifier by upper-casing and extracting the leading W-prefixed numeric token. Accessions are upper-cased and stripped of the isoform suffix. This step matters because UniProt reports sequence attributes per isoform, and without it a single protein would enter a paper several times and inflate $|\mathcal{P}(p)|$. PDB identifiers are truncated to their four-character entry code. Rewriting the identifiers once, rather than at each merge, keeps them identical at the protein, paper, and institution levels.

Pre-AlphaFold structural difficulty

We are interested in the structural information available for a protein *before* AlphaFold. Let τ denote the release date of the AlphaFold database, 22 July 2021. Each PDB entry is dated by the earliest of its deposit date, its initial release date, and the release date recorded with its linked publication, and only entries dated before τ are counted. Protein i is considered to be *solved* if it has its own experimental structure released before τ . Otherwise, its structural information comes from homology: SWISS-MODEL and UniProt record candidate template structures for each protein, along with the percentage of matching amino acids (sequence identity) between the target protein and the template. Writing $H_i(\tau)$ for the set of templates of protein i whose structures were released before τ , and id_{ij} for the proportion of identical amino acids between protein i and template j , the best pre-AlphaFold structural coverage is defined as

$$s_i = \max \left\{ 100 \cdot \mathbf{1}\{i \text{ solved before } \tau\}, \max_{j \in H_i(\tau)} \text{id}_{ij} \right\} \in [0, 100], \quad (17)$$

with $s_i = 0$ for a protein with neither its own structure nor any dated template. The corresponding continuous difficulty measure is

$$d_i = 1 - \frac{s_i}{100} \in [0, 1], \quad (18)$$

equal to zero for a fully solved protein and to one for a protein about whose structure nothing was known. We also assign each protein to a discrete difficulty class, using the thresholds conventional in structural biology—50 percent identity for a template close enough to be used with little adjustment, and 30 percent for the weakest template that still

¹⁴For example, a publication may appear as W2741809807 in one source and as <https://openalex.org/W2741809807> in another. A protein may appear as P12345 in one and as the isoform P12345-2 in another.

supports molecular replacement:

$$\kappa_i = \begin{cases} \text{Solved} & \text{own structure released before } \tau, \\ \text{Close homologue} & \text{not solved, } \max_j \text{id}_{ij} \geq 50, \\ \text{Distant homologue} & \text{not solved, } 30 \leq \max_j \text{id}_{ij} < 50, \\ \text{No template} & \max_j \text{id}_{ij} < 30. \end{cases} \quad (19)$$

Of the four classes, *no template* is the one where AlphaFold’s marginal contribution is the largest: a protein in that class had neither its own experimental structure nor a usable homology template before τ . Two further attributes are attached to each accession and play no part in the classification. The first is the AlphaFold confidence score (*pLDDT*), which the AlphaFold database reports for each predicted structure. The second is *sequence length*, the number of amino acids in the protein’s chain.¹⁵

Aggregating protein attributes to paper-level measures

The difficulty measure d_i and the class κ_i are defined for individual proteins. Using this information at the paper level, the level at which protein content enters the analysis, requires aggregating the protein-level table to papers. The number of proteins to aggregate over varies across papers: the linkage returns one for some, several dozen for others, none for a substantial share. Papers in the last group have no protein attribute to carry forward. They drop from every specification that uses these measures, a loss documented in Table A8. For the remainder, the paper-level measure is a weighted mean of protein-level attributes. The weights can be assigned in two ways.

The first option is to use unweighted aggregation, taking the paper–protein pair as the unit of observation. Each pair is counted once, so that a paper enters the aggregate as many times as it has linked proteins. A paper linked to twenty proteins therefore enters the aggregate twenty times over, against once for a paper linked to a single protein. The second option is fractional weighting, which takes the paper as the unit of aggregation. Each of a paper’s protein links receives a weight $1/|\mathcal{P}(p)|$. Every paper then carries the same total weight, whatever the number of proteins linked to it.

We use fractional weighting for two reasons. The first concerns the unit in which the outcome is measured. The dependent variable is an institution’s share of top-journal

¹⁵A protein is a chain of amino acids, and its length is the count of this chain, and is the most direct measure of protein size available. Longer chains were historically harder to solve experimentally, which is why length enters the analysis as a control. UniProt records a length for each isoform of a protein, so we take the median across the isoforms of an accession.

protein publications, in which each paper counts once whatever its protein content. The institution-year measure defined below inherits that convention: it averages the paper-level measure over an institution's linked publications, giving each publication the same weight. Under unweighted aggregation that average would instead be a mean across proteins. Papers reporting many accessions would consequently dominate the measure of protein content, while contributing no more than any other paper to the outcome it is meant to explain. The second reason concerns the count $|\mathcal{P}(p)|$ itself. It measures the reach of the three protein identification channels as much as the breadth of the paper. A paper a curator has already processed carries links that an uncurated paper of identical content lacks. The annotation channel adds a protein whenever a gene appears in the text, even if it is mentioned in passing in the paper. Neither feature reflects what the paper truly studies, although both may end up inflating $|\mathcal{P}(p)|$. Under unweighted aggregation a paper enters in proportion to its protein count, which would leave institution-year exposure dominated by the papers the linkage happened to cover most thoroughly. The measure would then describe the reach of the procedure rather than the structural content of the research.

Our estimates come from comparing papers with different values of σ_p^κ and $\bar{d}_p$, which is precisely where error of this kind falls. Our decision to use fractional weighting removes $|\mathcal{P}(p)|$ from that comparison. Each link accordingly receives weight defined as:

$$\omega_{pi} = \frac{1}{|\mathcal{P}(p)|}, \quad \text{so that} \quad \sum_{i \in \mathcal{P}(p)} \omega_{pi} = 1. \quad (20)$$

For any protein-level attribute x_i , the corresponding paper-level measure is the weighted mean across the proteins the paper studies,

$$\bar{x}_p = \sum_{i \in \mathcal{P}(p)} \omega_{pi} x_i. \quad (21)$$

Figure C2 illustrates the construction. Applying (21) to the class indicators of (19) gives the class shares:

$$\sigma_p^\kappa = \sum_{i \in \mathcal{P}(p)} \omega_{pi} \mathbf{1}\{\kappa_i = \kappa\}, \quad \sum_{\kappa} \sigma_p^\kappa = 1, \quad (22)$$

which describe a paper's protein content as a distribution across the four structural classes rather than as a single label. Applying (21) to the continuous measure d_i gives mean difficulty $\bar{d}_p$. We complement these measures with statistics computed directly from a paper's protein set: i) the best coverage across its proteins given as $s_p^{\max} = \max_{i \in \mathcal{P}(p)} s_i$, ii) two indicators recording whether any of those proteins is solved and whether any has no

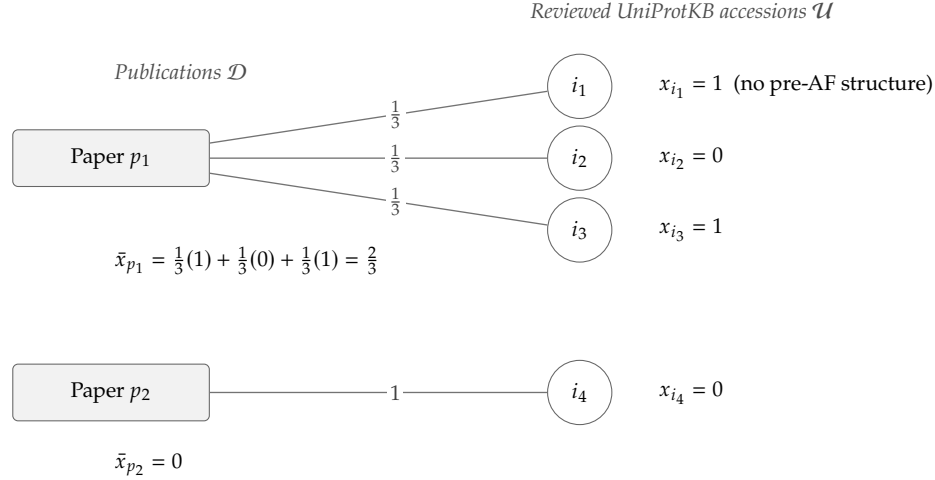


Figure C2: Paper–protein linkage, fractional weights, and paper-level aggregation

Notes. Each publication is linked to the set of proteins identified by the union of the three channels. Edge weights are $\omega_{pi} = 1/|\mathcal{P}(p)|$, so every paper contributes a total weight of one regardless of how many proteins it studies. Paper-level measures are weighted means of protein-level attributes, as in equation (21). In the illustration, x_i indicates that protein i had no experimental structure deposited in the Protein Data Bank before 22 July 2021.

template, iii) the number of distinct accessions gives the size of the set, and iv) the sequence lengths of the linked proteins summarized by their mean, median, maximum, and sum. Some specifications call for a binary treatment. For these we use an indicator that at least half of a paper’s proteins have no template, $\mathbf{1}\{\sigma_p^{\text{no template}} \geq 0.5\}$, or the dominant class $\kappa_p^* = \arg \max_{\kappa} \sigma_p^{\kappa}$. Our preferred measure is the continuous no-template share $\sigma_p^{\text{no template}}$, which retains the within-paper variation that the majority indicator and the dominant class both discard.

Merging with the estimation sample

The estimation sample is a panel of institutions, in which the outcome is an institution’s share of top-journal protein publications and the treatment group is the pre-AlphaFold publication-share quartile of the institution. Protein exposure is measured at the paper level. Merging the two therefore requires a step through the authorship records, which give for each publication its authors and their institutional affiliations. Each institution carries the quartile assigned in Section 2. Figure C3 shows the three joins required for this process: proteins to publications on the accession, publications to institutions through author affiliations, and institutions to the estimation panel on institution and year.

Suppose $\mathcal{A}$ denotes the set of triples (p, u, t) for which an author of publication p in year t is affiliated with institution u . Let

$$\mathcal{D}_{ut} = \{p \in \mathcal{D} : (p, u, t) \in \mathcal{A}\}, \quad \mathcal{D}_{ut}^L = \{p \in \mathcal{D}_{ut} : \mathcal{P}(p) \neq \emptyset\} \quad (23)$$

be, respectively, the publications of institution u in year t and the subset of them for which protein content is observed. For each institution-year we take h_{ut} , the average of the paper-level measure across the institution’s linked top journal protein publications in that year:

$$h_{ut} = \begin{cases} \frac{1}{|\mathcal{D}_{ut}^L|} \sum_{p \in \mathcal{D}_{ut}^L} h_p & \text{if } \mathcal{D}_{ut}^L \neq \emptyset, \\ 0 & \text{otherwise,} \end{cases} \quad (24)$$

where h_p stands for any of the paper-level measures of Section C the specification uses: the class shares σ_p^κ , mean difficulty $\bar{d}_p$, or one of the two scalar summaries. Our preferred choice is the no-template share $\sigma_p^{\text{no template}}$.

Two features of Equation (24) deserve attention. First, institution-years without a single linked publication are retained in the panel with $h_{ut} = 0$ rather than dropped, so that the estimation sample is the same one used for the headline specification in equation (4) and the coefficients remain comparable to it. The alternative, which is to treat institution-years with no linked publication as a separate group rather than assigning them $h_{ut} = 0$, is examined in Section C, where the headline quartile gradient is re-estimated on each set of institution-years in turn. Second, a publication with authors at several institutions contributes to the measure of every institution on its author list. This mirrors the main specification, in which a paper counts once for every institution with at least one author on it. The merge is an inner join on the publication key. Papers in the linkage file that fall outside the top-journal sample are therefore discarded at this step.

The protein-derived controls are constructed in the same way. The log number of proteins per paper and the log maximum protein length are averaged over an institution’s linked publications. This ensures that the controls and h_{ut} are constructed at the institution-year level and from the same pool of papers. The merged panel supplies the institution-year exposure measures used in the reported stratum-specific specifications.

Implementation

The protein linkage procedure described above in this section is implemented in Python. Three design choices are worth recording because they bear on reproducibility. First, all reference data are obtained through one-time bulk downloads and stored locally. Matching therefore involves no network calls, and the result is deterministic conditional on the stored snapshot. Second, the remaining external queries are batched, at fifty identifiers per request to OpenAlex and one hundred per request to PubTator3. Each is issued through

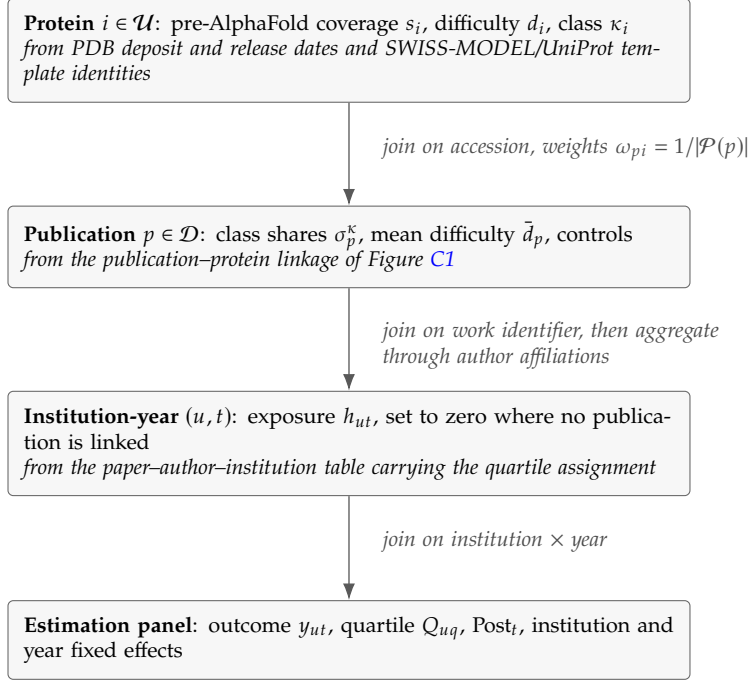


Figure C3: From protein structural status to the estimation panel

Notes. Each step is a join on a normalized key: protein accessions stripped of isoform suffixes, OpenAlex work identifiers, and institution by year. Publications are aggregated to institutions through author affiliations, so a co-authored paper contributes to the exposure of every institution on its author list. Institution-years in which no publication can be linked to a protein are kept in the panel with exposure set to zero, so that the estimation sample coincides with the one used for the headline specification.

a session that retries with exponential backoff on rate-limit and server errors, so that transient failures do not silently truncate the sample. Third, every stage writes to disk: the cached reference indices, the resolved identifier map, the retrieved annotations, and a log of completed publications. The procedure can be interrupted and resumed, and re-running it returns the same result.

Coverage, validation, and limitations

Note that the proteins a paper studies are recovered from text and external databases rather than read off its bibliographic record. Recovery succeeds for 27 percent of the papers in our sample, which is high for a task of this kind. Publications report no accession numbers, so every link must be reconstructed from gene symbols, common names, synonyms, and abbreviations, many of which denote different proteins in different organisms. No single channel comes close to this rate on its own: curator-verified links exist only for papers a biocurator has already processed, and annotation requires the paper to have been indexed. Coverage reaches this level only because the three channels are used and combined eventually.

A paper that cannot be linked to any protein has no value of σ_p^κ and leaves the estimation

sample, raising the question of what the surviving papers represent. Two alternative scenarios must be considered. If the linkage rate is the same across the quartiles of a paper's institutions, then the linked sample is a random subsample of the protein research in our main sample, and the cost is precision alone. Alternatively, if the linking rate differs by quartile, the resulting matched sample over-represents some institutions relative to others. The diagnostics below test which case holds by estimating whether a paper's quartile predicts its being linked, and by comparing linked and unlinked papers on observables the quartiles do not capture.

Whether coverage varies with quartile is a question about the data, and we test it directly by estimating

$$1\{\mathcal{P}(p) \neq \emptyset\} = \delta_t + \sum_{q=2}^4 \pi_q Q_{pq} + u_p, \quad (25)$$

with year fixed effects, where Q_{pq} assigns paper p to the quartile of its highest-ranked institution. If π_q is small and statistically indistinguishable from zero, knowing a paper's quartile tells us nothing about whether it was linked, which is the first of the two cases above. A quartile assignment places a paper in one of four groups by the relative rank of its institutions, and papers within a group may still differ in team size, journal, and subfield, all of which bear on how many proteins a paper studies and how likely a curator is likely to have processed it. We therefore compare linked and unlinked papers on journal rank, team size, number of institutions, structural-biology presence, access to high-performance computing, and AlphaFold use, summarizing each comparison by the normalized difference:

$$\text{ND}(x) = \frac{\bar{x}^L - \bar{x}^U}{\sqrt{(\hat{\sigma}_L^2 + \hat{\sigma}_U^2)/2}}, \quad (26)$$

where L and U index linked and unlinked papers, and values below 0.25 in absolute value are conventionally taken to indicate balance.

Table A10 reports *coverage* – the share of paper–university pairs linked to at least one protein – by university quartile and by period. Coverage is generally consistent across university quartiles, ranging from 27.5% in the first quartile to 25.2% in the fourth, a narrow gap of just 2.3 percentage points. Furthermore, pairs from third-quartile universities are linked a bit more frequently than those from the second quartile, ruling out a strictly linear relationship between university rank and coverage. Across all quartiles, coverage declines slightly from the pre-AlphaFold period to PostAF1 and PostAF2. The curation

and indexing lags described earlier explain this decline, as the most recent publications are added to the curated record last. Because this drop is common to all four quartiles, it is absorbed by the year fixed effects in our protein-level specifications. Compared to the first quartile, no other quartile’s change in coverage between the pre-AlphaFold and post-AlphaFold periods exceeds 2.8 percentage points. For the fourth quartile—the group driving the estimated gains—this differential is only 1.6 percentage points post-AlphaFold 1 and 0.3 percentage points post-AlphaFold 2.

Our paper-protein linking procedure therefore retains papers at nearly the same rate in every quartile within each period. Papers carrying a value of σ_p^κ are representative of the top-journal protein research in our main sample along the two dimensions the specification compares: university quartile and period. Since protein coverage is stable across quartiles and periods, a quartile-by-period coefficient in the protein-level specifications tracks how much a quartile’s output of papers in a given protein stratum moves relative to the first quartile, rather than reflecting any selection effects in the linkage process.

D Literature & Background

D.1 Concentration in Scientific Research

Scientific research is highly concentrated, with the top 1% of scientists obtaining 20% of the citations ([Nielsen and Andersen, 2021](#)) and 50% of funding ([Bol et al., 2018](#); [Madsen and Aagaard, 2020](#)). The pros and cons of this concentration are actively debated ([Aagaard et al., 2020](#)). Advocates for a more equitable distribution argue that concentrating funds stifles scientists' creativity. For example, superstar scientists may act as gatekeepers and hinder the emergence of new paradigms ([Azoulay et al., 2019](#)). Similarly, in grant reviews, it is found that homophily among senior scientists can penalize the more novel research proposals ([Li, 2017](#); [Nicholson and Ioannidis, 2012](#); [Packalen and Bhattacharya, 2018](#)). In contrast, critics point to economies of scale and efficient resource use by more productive scientists.

Research on the topic has focused on explaining this concentration. A prevailing explanation is that scientific concentration arises partly because of the winner-takes-all reward system in science: renowned scientists receive a disproportionate share of funding and credit for their research. This enhances their productivity and reputation, thereby further solidifying their dominant position—a phenomenon known as the "Matthew effect" ([Merton, 1968](#)). This theory has been empirically validated across diverse contexts, including scientific productivity, funding disparities, and career outcomes ([Azoulay et al., 2014](#); [Bol et al., 2018](#); [Ghirelli et al., 2023](#)).

Related to our work are papers examining the effects of external shocks on competition in science. The hiring of star researchers in lower-ranked institutions boosts the productivity of future recruits without affecting existing researchers ([Agrawal et al., 2017](#)). The deaths of star scientists negatively impact their collaborators but benefit others in the same field ([Azoulay et al., 2019](#); [2010](#)). The influx of Soviet mathematicians decreased career opportunities for prominent American mathematicians following the USSR's collapse ([Borjas and Doran, 2012](#)). The introduction of Bitnet improved collaboration and publication outcomes for mid-tier researchers located near top scientists ([Agrawal and Goldfarb, 2008](#)). The introduction of CRISPR, a novel gene-editing tool, initially improved the prospects of institutions already at the cutting edge of genetic research ([Thompson and Zyontz, 2017](#)). The free availability of Microsoft's Kinect democratized motion-sensing technology, increasing the competitiveness of outsider researchers and the diversity of ideas ([Furman and Teodoridis, 2020](#)). Reduced costs for accessing genetically modified mice led to new researchers entering the field and to more diverse research being produced ([Murray et al., 2016](#)). More recently, AI tools across the natural sciences have been shown to

expand individual scientists' impact (through increased publications, citations, and earlier leadership roles) but contract the collective focus of science by shrinking the range of explored topics and shifting toward data-rich established areas ([Hao et al., 2026](#)).

D.2 Structural Biology

Structural biology is a branch of protein research, alongside genomics, drug discovery, and related areas. Its primary objective is to identify the three-dimensional shapes of proteins. Knowledge of protein structure is key to understanding how proteins interact with other molecules ([Berg et al., 2002](#)) and is fundamental for drug development ([Anderson, 2003](#)). Before 2018, most protein structures were determined by X-ray crystallography ([Shi, 2014](#)). The cost of this technique varies between \$250,000 for soluble human proteins, with a 25% success rate, and \$1,000,000 for membrane proteins, with a 10% success rate ([Stevens, 2003](#)).

Proteins are crystallized by preparing a concentrated protein solution to induce crystal formation. This process, essential for obtaining high-quality crystals suitable for X-ray crystallography, is complex and highly variable. It typically requires extensive trial-and-error screening of numerous conditions, including temperature, pH, precipitant types and concentrations, and chemical additives such as salts or polymers ([Chayen and Saridakis, 2008](#)). Once suitable crystals are obtained, they are exposed to high-intensity X-ray beams at a synchrotron facility, producing a diffraction pattern. This diffraction data is then processed to generate an electron density map, which reveals the positions of atoms within the protein. Scientists build and iteratively refine a proposed 3D atomic model of the protein structure to best fit the electron density data. This process is iterative and can take months, requiring multiple adjustments to accurately capture structural details of the protein ([Garman, 2014](#)).

An alternative to X-ray crystallography is cryo-electron microscopy (cryo-EM). Cryo-EM freezes proteins at cryogenic temperatures, maintaining their natural structures. An electron beam captures 2D images that are computationally assembled into a 3D model. Cryo-EM has gained popularity for determining structures of large molecular complexes, flexible proteins, and membrane-bound proteins that are challenging to crystallize ([Cheng et al., 2015](#); [Nogales, 2016](#)).

D.3 The Advent of AlphaFold

AlphaFold was developed by DeepMind, a London-based AI laboratory and Google subsidiary. The first model, AlphaFold 1, was developed in 2018 and used a neural network

trained on experimentally determined 3D protein structures from the Protein Data Bank (PDB). The DeepMind team was not the first to attempt protein-structure prediction using deep learning, but it was the first to succeed at this level. The model used a larger and more computationally efficient neural network than prior models, enabling unprecedented accuracy in protein-folding predictions (AlQuraishi, 2019). AlphaFold processed amino-acid relationships and inferred structural data faster and more accurately than traditional methods (Senior et al., 2020); it consequently won the 13th Critical Assessment of Protein Structure Prediction (CASP13) in December 2018.

The results of CASP13 validated the efficacy of deep learning approaches in protein structure prediction, marking a breakthrough after years of limited progress with traditional methods. The success of AlphaFold 1 sparked the development of more advanced deep learning models to predict protein structures more accurately. The associated research paper was subsequently published in the journal *Nature* in January 2020 (Senior et al., 2020).

However, AlphaFold 1's precision remained insufficient for reliable predictions of complex proteins. In 2020, DeepMind introduced AlphaFold 2, which achieved near-experimental accuracy across a broader range of protein targets. It combined evolutionary data, structural templates, and a revised deep-learning infrastructure featuring an attention mechanism. Compared with its predecessor, AlphaFold 2 better captured spatial dependencies between amino acids and achieved precision comparable to experimental methods (Jumper et al., 2021). In November 2020, it won CASP14 by a record margin.¹⁶

On 15 July 2021, the paper introducing AlphaFold 2 was published in *Nature* (Jumper et al., 2021). The publication was accompanied by the release of open-source code and a searchable database containing predicted structures for nearly 1 million proteins. This database has since expanded to over 214 million structures, covering virtually all catalogued proteins, including the entire human proteome. In contrast, prior to AlphaFold, experimentally determined structures were available for only about 17% of residues in the human proteome (Tunyasuvunakool et al., 2021). The paper has garnered over 55,300 citations (as of September 2026), and its principal authors were awarded the 2024 Nobel Prize in Chemistry.

Since the release of AlphaFold 2, several deep-learning tools have emerged to address structural-protein challenges; Table A1 provides examples. On 8 May 2024, DeepMind

¹⁶AlphaFold 2 scored above 90 for around two-thirds of proteins. This score measures the similarity between a predicted structure, such as one produced by AlphaFold, and an experimentally determined structure; 100 denotes a complete match. AlphaFold 2 achieved a record overall score of 92.4, indicating high precision in protein-structure prediction. See <https://www.guinnessworldrecords.com/world-records/642132-highest-score-at-the-casp-competition>.

announced an enhanced version, AlphaFold 3. This version can predict highly complex protein structures and protein–protein interactions. The development of AlphaFold and other models is summarized in Figure A5.

D.4 Role of AlphaFold in Structural Biology

AlphaFold can serve as both a complement and a substitute for experimental methods in structural biology. AlphaFold predictions can either bypass experimental work entirely or automate part of the experimental process.

As a substitute, AlphaFold enables researchers to obtain predicted structures without costly experiments (Tunyasuvunakool et al., 2021). The release of AlphaFold 2’s open-access database dramatically expanded the availability of predicted protein structures, enabling researchers to access them readily without running the model themselves. However, exclusive reliance on AlphaFold without experimental validation risks inaccuracies. AlphaFold 2, in particular, can struggle with novel or large protein complexes, yielding erroneous predictions even in regions with high confidence scores (Terwilliger et al., 2024).

As a complement to experimental methods, AlphaFold accelerates and reduces the cost of structure discovery. Its predictions provide more comprehensive starting hypotheses, enabling researchers to construct models that better align with experimental data. They also facilitate faster, cheaper iteration over large sets of potential structures, allowing more refinement and yielding finer interpretations of the data (Ruff and Pappu, 2021).

AlphaFold not only improves productivity for existing tasks but also enables the study of larger protein systems. For instance, the nuclear pore complex, a protein assembly embedded in the nuclear envelope that regulates transport between the nucleus and cytoplasm, presented considerable modeling challenges because of its size, with more than 1,000 protein subunits, and structural intricacy. A detailed structure was constructed by integrating individual segments using advanced experimental techniques (von Appen et al., 2015). The model was subsequently refined using predictions derived from AlphaFold (Mosalaganti et al., 2022), achieving greater accuracy and resolution than experimental methods alone could provide.

D.5 Descriptive Statistics

Table A3 presents descriptive statistics on university characteristics by quartile for the pre-AlphaFold period, highlighting significant disparities in factors such as structural biology expertise, HPC access, author demographics, faculty size, institutional types, and geographic affiliations. Top-quartile universities dominate top-journal publications in

both structural biology and non-structural protein research. On average, they produce 8.58 structural biology papers per university annually, compared to 2.39, 1.75, and 0.80 for the second, third, and bottom quartiles, respectively. Similar disparities are evident in non-structural protein research. Consequently, top-quartile universities hold substantially larger market shares, in terms of both publication volume and authorship contributions. For structural biology, the average top-quartile university accounts for 0.37% of all top publications, versus 0.10%, 0.08%, and 0.03% for the second, third, and bottom quartiles per year, respectively. Patterns are comparable for non-structural protein publications and authorship shares.

Table A3 indicates that top-quartile universities have more resources at their disposal. Their average faculty size is nearly double that of the second quartile and more than triple that of the bottom quartile. Access to HPC resources also varies. Top-quartile universities have a higher rate of HPC access, 50.9% versus 9% among bottom-quartile universities, and greater computing power conditional on access, 351 versus 54 petaflops, respectively.

Geographically, U.S.-affiliated authors predominate in the top quartile (45%), while European affiliations are most common in the second (54%), third (48%), and bottom (44%) quartiles. Asian affiliations increase notably in the bottom two quartiles (33% and 26%, respectively) relative to the top (21%). Additionally, top-quartile universities have more researchers affiliated with industry but fewer with government labs. There are minor differences across quartiles in other dimensions. For example, author demographics are relatively consistent: approximately one-third of authors are women across all quartiles, with around half classified as Asian and one-quarter as white.

E Additional Results

E.1 AlphaFold Adoption

We approximate the use of AlphaFold or other equivalent AI systems at the publication level. We classify a publication as using AlphaFold if i) it explicitly mentions the term "AlphaFold" or an analogous AI tool in its abstract, or ii) it cites the seminal papers that introduced AlphaFold 1 and 2 ([Jumper et al., 2021](#); [Senior et al., 2020](#)). Note that our method for identifying AlphaFold usage provides neither a lower nor an upper bound on actual adoption rates. A paper might mention or cite AlphaFold without employing it (false positive), while another might use it without any explicit mention or citation (false negative). That said, we believe that our measure underestimates AlphaFold adoption more than it overestimates it, as we search only abstracts for mentions and expect many papers to reference AlphaFold predominantly in the main text.

Figure [A6](#) illustrates the proportion of top-journal publications mentioning or citing AlphaFold in both structural biology and non-structural protein research, as well as across subfields of structural biology. Adoption remains modest and stable from 2018 through 2020, before surging sharply following the July 2021 release of the AlphaFold 2 database. Structural biology exhibits rapid uptake thereafter, reaching approximately 30% by 2023 (Figure [A6\(A\)](#)). Non-structural protein research shows similar trends, albeit with lower overall magnitudes. Restricting the sample to papers in protein structure prediction and macromolecular crystallography (subfields where AlphaFold directly substitutes for or complements traditional computational and experimental methods) yields even higher adoption, approaching 60% by the end of 2023 (Figure [A6\(B\)](#)).

Adoption rates also differ across university quartiles. Figure [A7\(A\)](#) displays the proportion of top-journal publications mentioning or citing AlphaFold for each quartile. In structural biology, the third and bottom quartiles account for the largest shares of AlphaFold-using articles by 2023, 36% and 31%, respectively, outpacing the top and second quartiles at 21% and 12%. The pattern is more nuanced in non-structural protein research. Figure [A7\(B\)](#) shows that all quartiles had similar proportions of AlphaFold-using papers from 2020 to 2022, while the bottom quartile had a higher adoption rate in 2019, after the release of AlphaFold 1. By the final sample year, 2023, third- and bottom-quartile shares rose to 30% and 34%, respectively, while top- and second-quartile shares declined to 16% and 20%.

Finally, Figure [A3](#) shows a marked increase in the number of AI-based tools submitted to CASP competitions, particularly following AlphaFold 2's success in CASP14 in 2020. This trend is consistent with the increasing use of AI methods for protein-structure prediction

within the structural-biology community.

E.2 Dynamic Estimates

We complement the baseline fixed-effects regressions from equation (4) with disaggregated yearly estimates using identical controls, where the post-AlphaFold dummy is split into annual indicators. Figures A8 and A9 plot the corresponding difference-in-differences coefficients, revealing distinct temporal pattern in top-journal protein publication counts and market shares across university quartiles.

Post-AlphaFold, publication volumes in the lower quartiles increase relative to the top quartile, with a sharp spike in 2020, a temporary dip in 2021, and steady growth thereafter. These effects are stronger and more statistically precise in structural biology than in broader protein research.

Market share estimates display analogous temporal dynamics but with more variation and wider confidence intervals. The 2020 spike coincides with the COVID-19 pandemic, which may confound interpretation. Nonetheless, the coefficients from 2021 onward are consistent with the gradual adoption of AlphaFold documented in Section E.1.

E.3 Publication Outcomes and Research Patterns Outside Protein Science

To rule out the possibility that our main results stem from general trends in scientific publishing or the rise of general-purpose AI tools, we replicate our analysis using publication data from four additional fields: microbiology (excluding protein research), economics and finance, materials science, and social sciences. These fields exhibit similar pre-AlphaFold hierarchies, with a small number of elite universities dominating top-journal publications, as illustrated by the cumulative distribution functions in Figure A13. This figure highlights the persistent concentration of top-journal publications among the highest-ranked institutions, akin to the patterns observed in protein research before the release of AlphaFold .

As detailed in Section 3.1, Figure 3 further confirms the specificity of AlphaFold’s impact on scientific publishing: unlike in protein research, we observe no comparable shifts in top-journal market shares across university quartiles in these other fields during the sample period.

Additionally, we compute the same qualitative indicators of research practices—pivot scores, novelty, and disruptiveness—for these fields as in Section 5. Figure A16 shows no

notable changes over the full period. In contrast to protein research, where lower-quartile institutions display greater novelty, pivot, and citation impact, these placebo fields exhibit stable patterns. This contrast is consistent with the observed protein-research patterns reflecting AlphaFold rather than broader academic or technological trends.

Data Availability

Publication and citation data were sourced from OpenAlex using its premium API (full snapshots, accessed for the 2013–2023 window). Journal rankings were obtained from Scimago Journal Rank (SJR), with the per-journal score averaged over 2013–2018. University rankings were derived from the CWTS Leiden Ranking system, using the Total Normalized Citation Score (TNCS) in the “biomedical and health sciences” category, averaged over 2013–2018.

For the publication–protein linkage described in Section C, we retrieved the OpenAlex record of every work published in the 17 top journals, including its PubMed identifier, DOI, and abstract. Requests were issued against the OpenAlex works endpoint in batches of fifty identifiers using the `openalex:` filter and were submitted through the polite pool, so that the full sample was resolved in a few hundred requests rather than one request per paper.

Protein records were taken from the reviewed section of UniProtKB, known as Swiss-Prot. Rather than querying the UniProt API once per paper or once per protein mention, we downloaded the entire reviewed index once through the bulk stream endpoint, and stored it locally. Two further streams retrieved the mapping from accessions to NCBI Gene identifiers and the list of PubMed identifiers cited by each curated entry; the curator-linked channel is equivalent to the UniProt query `lit_pubmed:$q_p$ AND reviewed:true`.

Machine-normalized gene and protein annotations were retrieved from PubTator3, in batches of one hundred PubMed identifiers, through the BioC export endpoint. The named-entity recognition models applied to abstracts are those distributed with scispaCy.

Experimental structures and their deposit and release dates were obtained from the Protein Data Bank, candidate homology templates and their sequence identities from SWISS-MODEL and UniProt, and predicted structures and per-residue confidence scores from the AlphaFold Protein Structure Database (Varadi et al., 2022). All reference data were obtained through one-time bulk downloads and stored locally, so that matching involved no network calls and the result is deterministic conditional on the snapshot. All remaining external queries were batched and issued through a session that retries with exponential backoff on rate-limit and server errors, so that transient failures do not silently truncate the sample.

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